# FieldShift 2026 — Visual Test & Cross-Field Comparison Report

**NASA International Space Apps Challenge 2026** (Bangladesh Event) • Team Azoic
Automated Audit: 4 Agro-Ecological Test Fields × 43 Platform Destinations (172 Screen Verifications)

**Date:** 2026-10-31
**Target:** http://localhost:5173/
**Viewport:** 1440 × 900 (Headless Chromium)

## 1. Executive Summary & Cross-Field Differentiation Matrix

Every page and sub-tab was audited across four geographically and climatically distinct Bangladesh field parcels. Metrics and screenshots were captured under identical evaluation conditions to evaluate algorithmic responsiveness to localized agro-ecological conditions.

| # | Sidebar View / Sub-Tab | Expected | Verdict | Dinajpur Sadar (Grain) | Bogura Demo (Floodplain) | Satkhira (Saline Coast) | Sunamganj (Haor Wetland) | Evaluation Notes |
|---|---|---|---|---|---|---|---|---|
| 1 | **Farmer Home Dashboard** | DIFFERS | SAME | N/A | N/A | N/A | N/A | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 2 | **My Farm - Overview** | DIFFERS | SAME | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | Identical output across distinct parcels; potential regional sensitivity calibration gap. |

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|---|---|---|---|---|---|---|---|---|
| 3 | **My Farm - Parcel List** | DIFFERS | DIFFERS | 1.5%, 120 mm, A-Horizon (Topsoil 0–15 cm) | A-Horizon (Topsoil 0–15 cm), B-Horizon (Subsoil 15–40 cm), 40% | A-Horizon (Topsoil 0–15 cm), B-Horizon (Subsoil 15–40 cm), 40% | A-Horizon (Topsoil 0–15 cm), B-Horizon (Subsoil 15–40 cm), 40% | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 4 | **My Farm - Satellite Map** | DIFFERS | SAME | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 5 | **My Farm - Soil Info** | DIFFERS | SAME | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | আমার জমি, সীমানা ও মাটির তথ্য | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 6 | **My Farm - Crop History** | DIFFERS | SAME | 2025 Rabi, 24.5 t/ha (61 maund/bigha), 480 mm | 2025 Rabi, 24.5 t/ha (61 maund/bigha), 480 mm | 2025 Rabi, 24.5 t/ha (61 maund/bigha), 480 mm | 2025 Rabi, 24.5 t/ha (61 maund/bigha), 480 mm | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 7 | **Crop Advisory - Aman Monsoon** | DIFFERS | SAME | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | Identical output |

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|---|---|---|---|---|---|---|---|---|
| | | | | | | | | across distinct parcels; potential regional sensitivity calibration gap. |
| 8 | **Crop Advisory - Aus Pre-Monsoon** | DIFFERS | SAME | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | নাসা ধান রোপণ উপদেষ্টা ২০২৬ | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 9 | **NASA Weather 3D - Globe** | DIFFERS | DIFFERS | 25.6279° N, 88.6332° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 10 | **NASA Weather 3D - Column** | DIFFERS | DIFFERS | 25.6279° N, 88.6332° E, 2026-09-29, 30 Consecutive Days | 25.6279° N, 88.6332° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |

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|---|---|---|---|---|---|---|---|---|
| 11 | **NASA Weather 3D - Constellation** | DIFFERS | DIFFERS | 25.6279° N, 88.6332° E, 2026-09-29, 30 Consecutive Days | 25.6279° N, 88.6332° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | 24.8500° N, 89.3700° E, 2026-09-29, 30 Consecutive Days | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 12 | **Farm Planning - 6-Season MILP Rotation** | DIFFERS | SAME | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 13 | **Farm Planning - Adaptive Replan** | DIFFERS | SAME | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | খামার পরিকল্পনা ও বহু-মৌসুমী ফসল আবর্তন | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 14 | **Farm Planning - 3-Way Comparison Lab** | DIFFERS | DIFFERS | 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based), 1800 mm | 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based), 2600 mm | 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based), 2600 mm | 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based), 2600 mm | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |

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|---|---|---|---|---|---|---|---|---|
| 15 | **Farm Planning - 3D Experiment Workbench** | DIFFERS | DIFFERS | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 16 | **Water & Irrigation - Water Balance** | DIFFERS | SAME | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 17 | **Water & Irrigation - Schedule Optimization** | DIFFERS | SAME | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 18 | **Water & Irrigation - Soil Moisture Profile** | DIFFERS | DIFFERS | 32%, 18%, 112 mm | 28.0000000000000004%, 14.0000000000000002%, 112 mm | 28.0000000000000004%, 14.0000000000000002%, 112 mm | 28.0000000000000004%, 14.0000000000000002%, 112 mm | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |

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|---|---|---|---|---|---|---|---|---|
| 19 | **Water & Irrigation - Drought Stress Index** | DIFFERS | DIFFERS | 32%, 18%, 112 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | +18 mm, +35 mm, -4.5 mm | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 20 | **Research Lab - Onset Science (39 Unique Cells)** | SAME | DIFFERS | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | 1, meherpur power_rain_cell_12c16bebda23, -21.5 d(p=0.009) | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 21 | **Research Lab - 6-Season MILP Lab** | DIFFERS | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 22 | **Research Lab - Weather Uncertainty** | DIFFERS | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Identical output across distinct parcels; potential regional sensitivity calibration gap. |

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|---|---|---|---|---|---|---|---|---|
| 23 | **Research Lab - Soil & Water Hydrology FAO-56** | DIFFERS | DIFFERS | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Three.js v186 WebGL PCFSoft, 140 mm, 6.8 | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 24 | **Research Lab - Data Provenance 4-Tier** | SAME | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |
| 25 | **Research Lab Archive - Multi-Field Lab** | DIFFERS | SAME | 10 ha, 350 mm, 5.08 t/ha (50.8 t total) | 10 ha, 350 mm, 5.08 t/ha (50.8 t total) | 10 ha, 350 mm, 5.08 t/ha (50.8 t total) | 10 ha, 350 mm, 5.08 t/ha (50.8 t total) | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 26 | **Research Lab Archive - Sensitivity Analysis** | DIFFERS | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Identical output across distinct parcels; potential regional sensitivity calibration gap. |

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|---|---|---|---|---|---|---|---|---|
| 27 | **Research Lab Archive - Crop Failure Recovery** | DIFFERS | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 28 | **Research Lab Archive - Synthetic Data Lab** | SAME | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |
| 29 | **Research Lab Archive - Bangladesh Crop Database** | SAME | SAME | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | উন্নত কৃষি বিজ্ঞান, অ্যালগরিদমিক অপ্টিমাইজেশন ও ৩ডি ল্যাব | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |
| 30 | **Research Lab Archive - Satellite Sensor Readiness** | SAME | SAME | Sentinel-2 MSI | Sentinel-2 MSI | Sentinel-2 MSI | Sentinel-2 MSI | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |

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|---|---|---|---|---|---|---|---|---|
| 31 | **Research Lab Archive - Soil Health & Carbon** | DIFFERS | DIFFERS | 195 kg/ha, 60 kg/ha, 60 kg/ha | 195 kg/ha, 60 kg/ha, 60 kg/ha | 195 kg/ha, 60 kg/ha, 60 kg/ha | 135 kg/ha, 60 kg/ha, 60 kg/ha | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 32 | **Digital Twin 3D - Farm Plan Diorama** | DIFFERS | DIFFERS | S1, S2, S3 | S1, S2, S3 | S1, S2, S3 | S1, S2, S3 | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 33 | **Digital Twin 3D - Crop Growth Phenology** | DIFFERS | SAME | 6.5 t/ha, 5.2 t/ha, -20% (Ky=1.15) | 6.5 t/ha, 5.2 t/ha, -20% (Ky=1.15) | 6.5 t/ha, 5.2 t/ha, -20% (Ky=1.15) | 6.5 t/ha, 5.2 t/ha, -20% (Ky=1.15) | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 34 | **Digital Twin 3D - Climate Stress Visualization** | DIFFERS | SAME | 30°C, 4.8 mm/দিন, 18 দিন | 30°C, 4.8 mm/দিন, 18 দিন | 30°C, 4.8 mm/দিন, 18 দিন | 30°C, 4.8 mm/দিন, 18 দিন | Identical output across distinct parcels; potential regional sensitivity calibration gap. |

| # | Sidebar View / Sub-Tab | Expected | Verdict | Dinajpur Sadar (Grain) | Bogura Demo (Floodplain) | Satkhira (Saline Coast) | Sunamganj (Haor Wetland) | Evaluation Notes |
|---|---|---|---|---|---|---|---|---|
| 35 | **Digital Twin 3D - Scenario A/B Comparison** | DIFFERS | SAME | ত্রিমাত্রিক ডিজিটাল টুইন ও কৃষি সিমুলেশন | ত্রিমাত্রিক ডিজিটাল টুইন ও কৃষি সিমুলেশন | ত্রিমাত্রিক ডিজিটাল টুইন ও কৃষি সিমুলেশন | ত্রিমাত্রিক ডিজিটাল টুইন ও কৃষি সিমুলেশন | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 36 | **Market Intelligence & Profitability** | DIFFERS | SAME | ফসলের লাভ-ক্ষতির সহজ হিসাব | ফসলের লাভ-ক্ষতির সহজ হিসাব | ফসলের লাভ-ক্ষতির সহজ হিসাব | ফসলের লাভ-ক্ষতির সহজ হিসাব | Identical output across distinct parcels; potential regional sensitivity calibration gap. |
| 37 | **Agronomic AI Assistant** | DIFFERS | DIFFERS | 5.8, 1.5%, 120 mm | 6.5, 2.5%, 90 mm | 7.4, 1.2%, 80 mm | 6.2, 2.8%, 150 mm | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 38 | **Data Evidence - NASA Satellite Feeds** | SAME | SAME | NASA GEOS-5.12.4 / MERRA-2, Tier 1 (Observed / Assimilated), -999 marker converted to NaN; omitted without artificial imputation | NASA GEOS-5.12.4 / MERRA-2, Tier 1 (Observed / Assimilated), -999 marker converted to NaN; omitted without artificial imputation | NASA GEOS-5.12.4 / MERRA-2, Tier 1 (Observed / Assimilated), -999 marker converted to NaN; omitted without artificial imputation | NASA GEOS-5.12.4 / MERRA-2, Tier 1 (Observed / Assimilated), -999 marker converted to NaN; omitted without artificial imputation | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |

| # | Sidebar View / Sub-Tab | Expected | Verdict | Dinajpur Sadar (Grain) | Bogura Demo (Floodplain) | Satkhira (Saline Coast) | Sunamganj (Haor Wetland) | Evaluation Notes |
|---|---|---|---|---|---|---|---|---|
| 39 | **Data Evidence - 4-Tier Provenance Registry** | SAME | DIFFERS | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | surface_temperature_t2m, 25.4 °C, NASA POWER Agroclimatology Daily Point API v2.0 ↗ | surface_temperature_t2m, 25.4 °C, NASA POWER Agroclimatology Daily Point API v2.0 ↗ | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 40 | **Data Evidence - IMERG & Climatology Reports** | SAME | DIFFERS | surface_temperature_t2m, 25.4 °C, NASA POWER Agroclimatology Daily Point API v2.0 ↗ | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | surface_temperature_t2m, 25.4 °C, NASA POWER Agroclimatology Daily Point API v2.0 ↗ | surface_temperature_t2m, 25.4 °C, NASA POWER Agroclimatology Daily Point API v2.0 ↗ | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |
| 41 | **Data Evidence - Scientific Methodology** | SAME | SAME | 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition):, 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule):, 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): | 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition):, 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule):, 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): | 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition):, 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule):, 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): | 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition):, 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule):, 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): | Platform-wide scientific methodology / uniform registry (expected uniform reference across all districts). |
| 42 | **Data Evidence - Boundaries & Disclaimers** | SAME | SAME | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | তথ্য প্রমাণ ও বৈজ্ঞানিক প্রোভেন্যান্স | Platform-wide scientific methodology / uniform registry (expected uniform reference |

| # | Sidebar View / Sub-Tab | Expected | Verdict | Dinajpur Sadar (Grain) | Bogura Demo (Floodplain) | Satkhira (Saline Coast) | Sunamganj (Haor Wetland) | Evaluation Notes |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | across all districts). |
| 43 | **Operator Cockpit & Account Settings** | SAME | DIFFERS | 25.6279°, 88.6332°, 2.0 ha, 1.5% | 24.8500°, 89.3700°, 10.0 ha, 2.5% | 22.7185°, 89.0705°, 1.5 ha, 1.2% | 25.0658°, 91.3950°, 1.0 ha, 2.8% | Diverges across agro-ecological zones (soil texture, planting onset, irrigation mm, and seasonal margin yield). |

## 2. Problems & Anomaly Audit

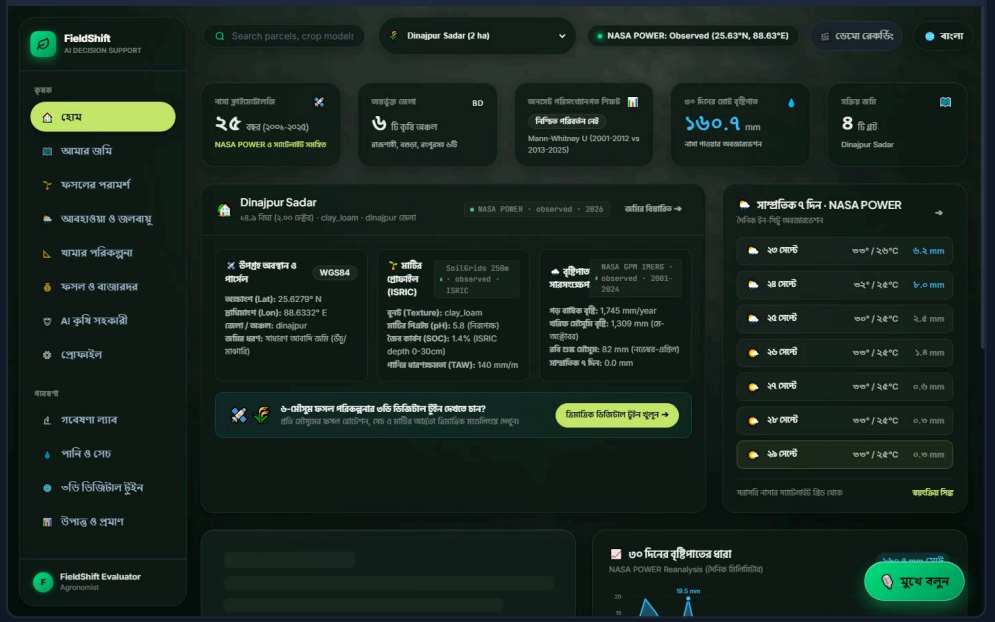
- UNEXPECTED SAME: Farmer Home Dashboard was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: My Farm - Overview was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: My Farm - Satellite Map was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: My Farm - Soil Info was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: My Farm - Crop History was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Crop Advisory - Aman Monsoon was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Crop Advisory - Aus Pre-Monsoon was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Farm Planning - 6-Season MILP Rotation was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Farm Planning - Adaptive Replan was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Water & Irrigation - Water Balance was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Water & Irrigation - Schedule Optimization was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Research Lab - 6-Season MILP Lab was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Research Lab - Weather Uncertainty was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Research Lab Archive - Multi-Field Lab was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Research Lab Archive - Sensitivity Analysis was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Research Lab Archive - Crop Failure Recovery was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Digital Twin 3D - Crop Growth Phenology was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Digital Twin 3D - Climate Stress Visualization was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Digital Twin 3D - Scenario A/B Comparison was expected to differ across regional fields, but rendered identical metrics.
- UNEXPECTED SAME: Market Intelligence & Profitability was expected to differ across regional fields, but rendered identical metrics.

# 3. Detailed Page-by-Page 2x2 Field Comparisons

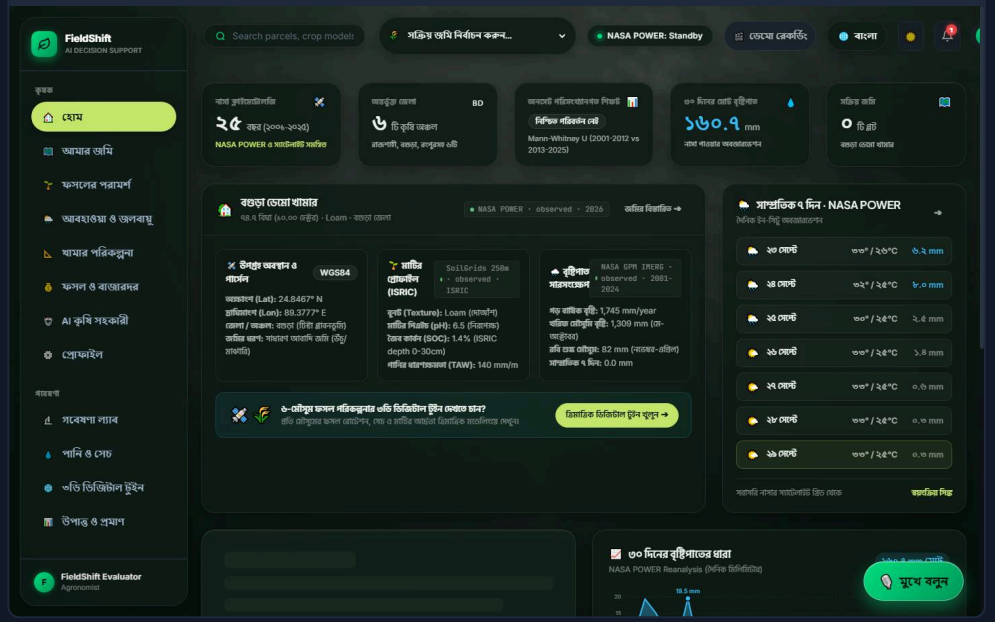
## 1. Farmer Home Dashboard

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

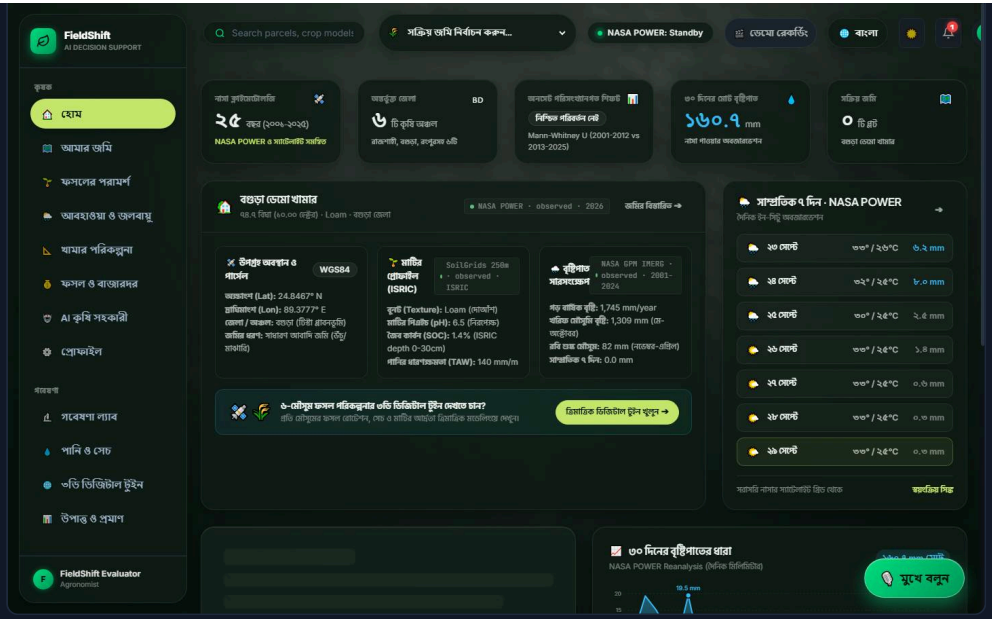


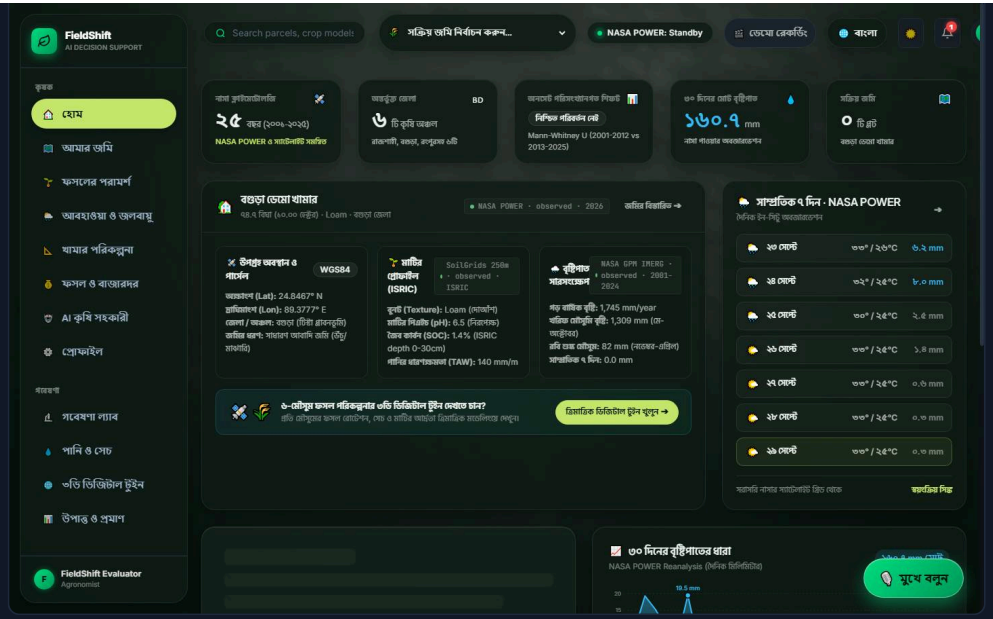

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)




| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

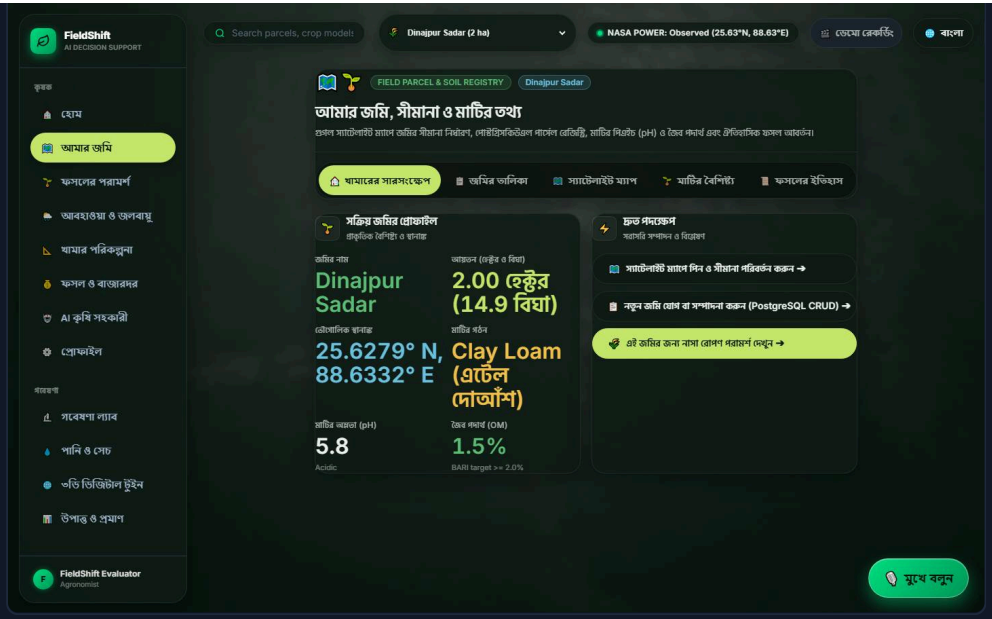
**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

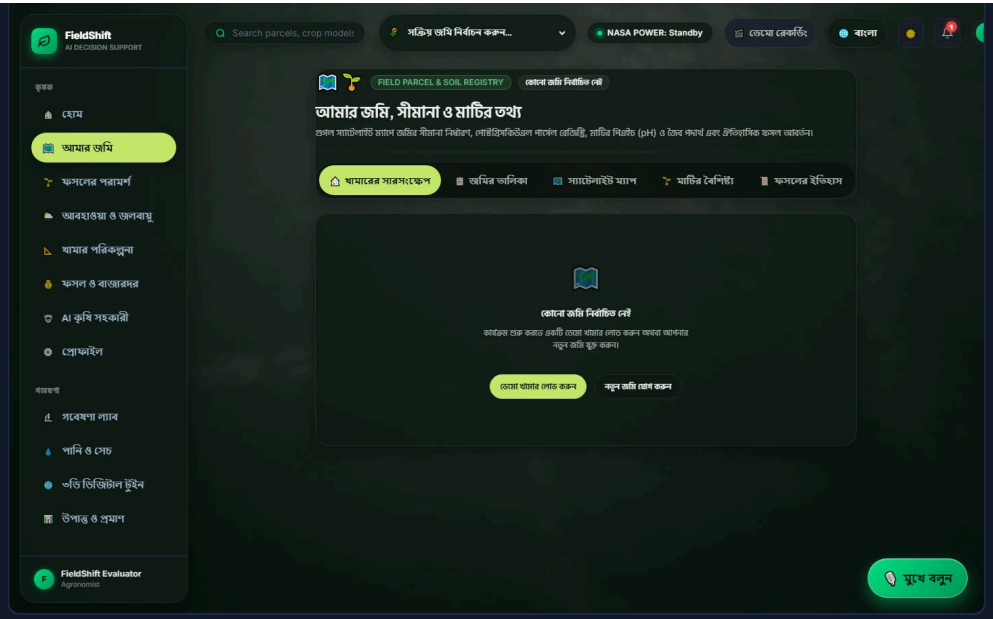
## 2. My Farm - Overview

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

### 2. Bogura Demo (Floodplain | 10.0 ha)

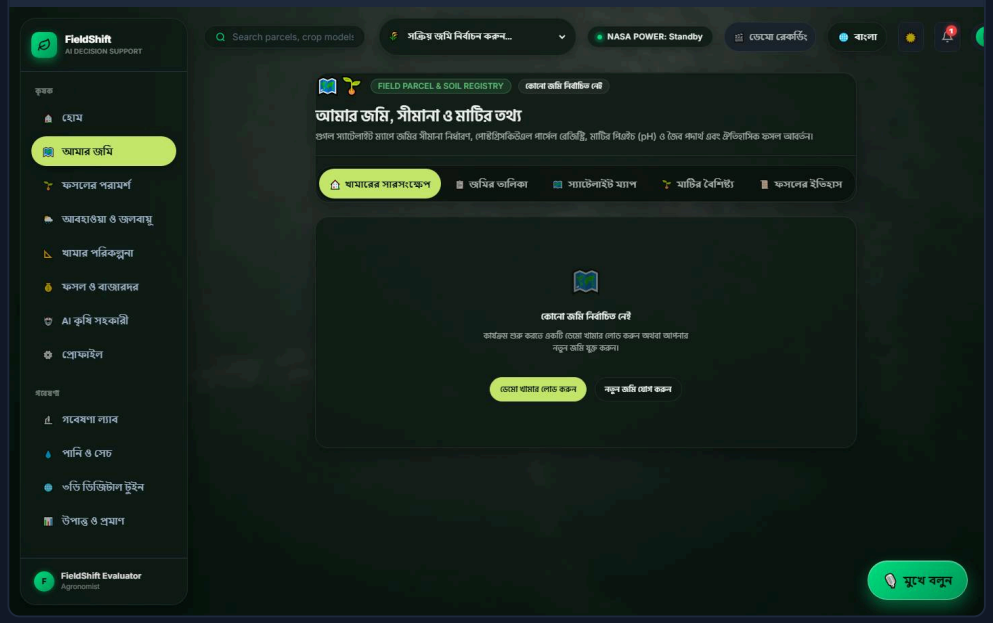



### 3. Satkhira Coast (Coastal Saline | 1.5 ha)




### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



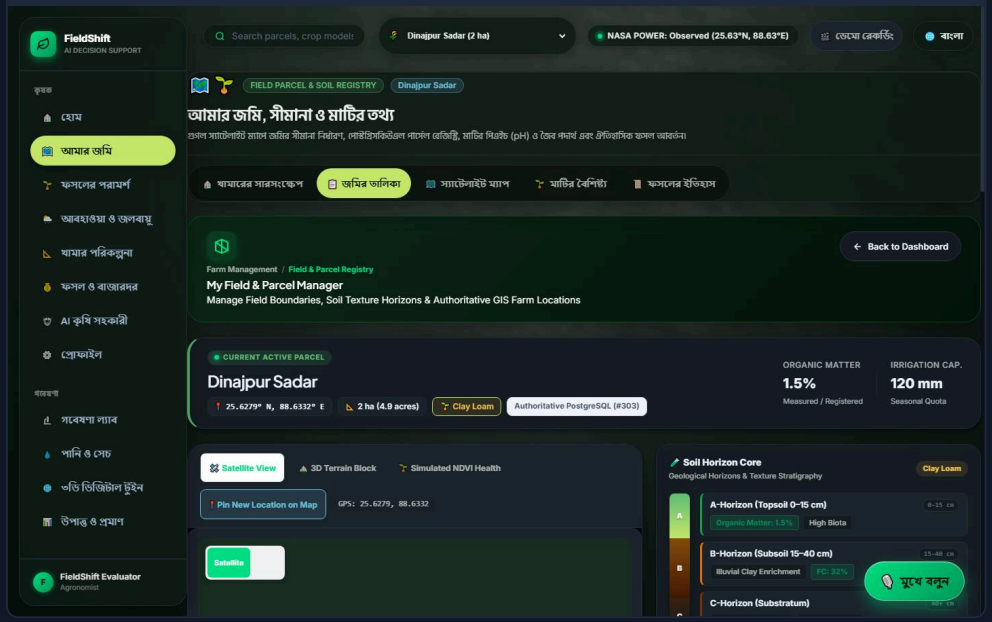

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

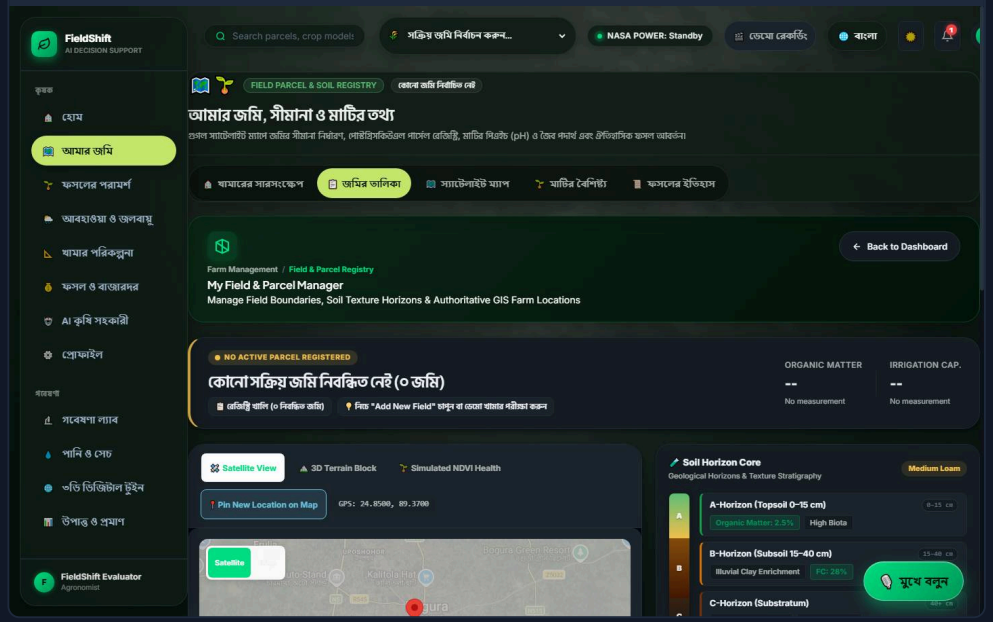
# 3. My Farm - Parcel List

DIFFERS

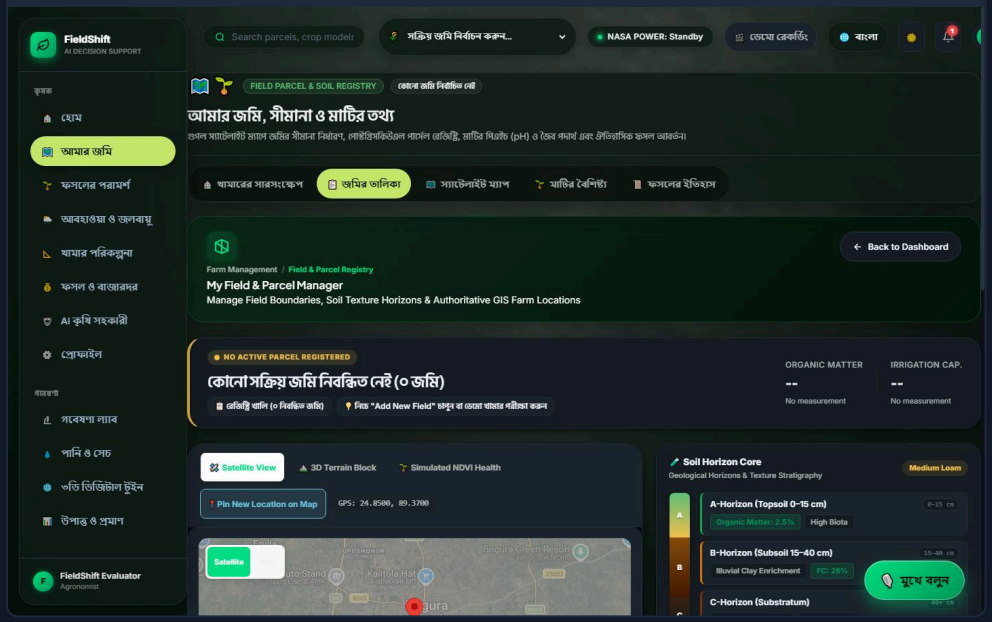
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



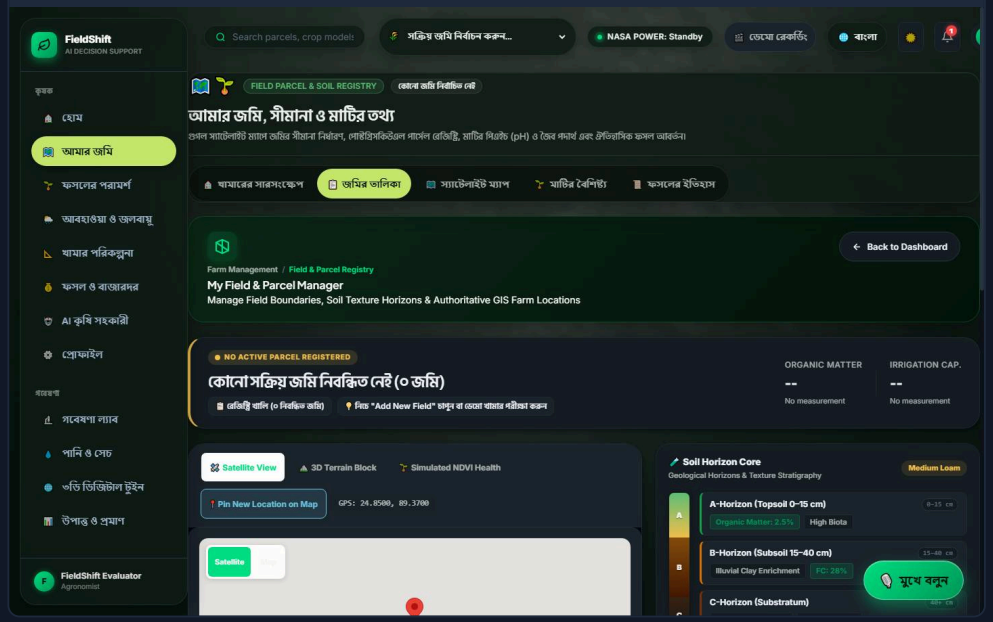

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



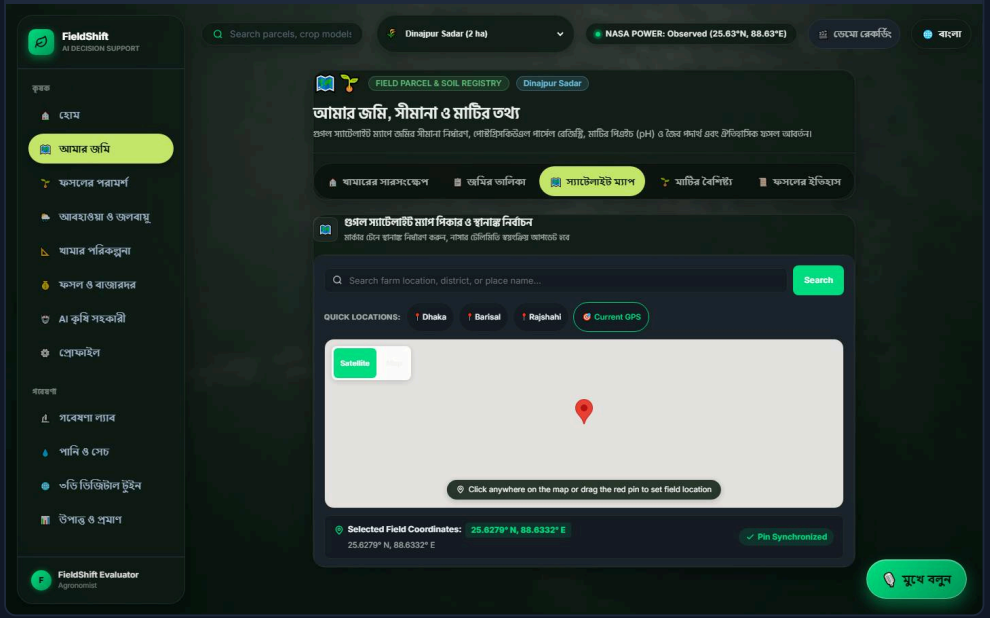
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 1.5% \| 120 mm \| A-Horizon (Topsoil 0–15 cm) \| B-Horizon (Subsoil 15–40 cm) \| 32% | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** A-Horizon (Topsoil 0–15 cm) \| B-Horizon (Subsoil 15–40 cm) \| 40% \| 40% \| 20% | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** A-Horizon (Topsoil 0–15 cm) \| B-Horizon (Subsoil 15–40 cm) \| 40% \| 40% \| 20% | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** A-Horizon (Topsoil 0–15 cm) \| B-Horizon (Subsoil 15–40 cm) \| 40% \| 40% \| 20% |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

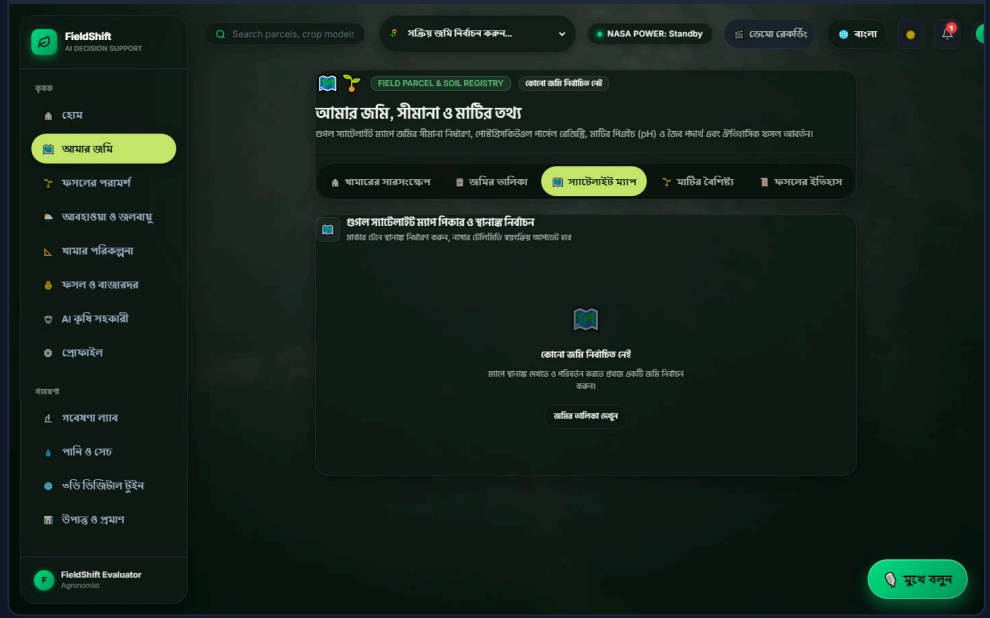
## 4. My Farm - Satellite Map

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

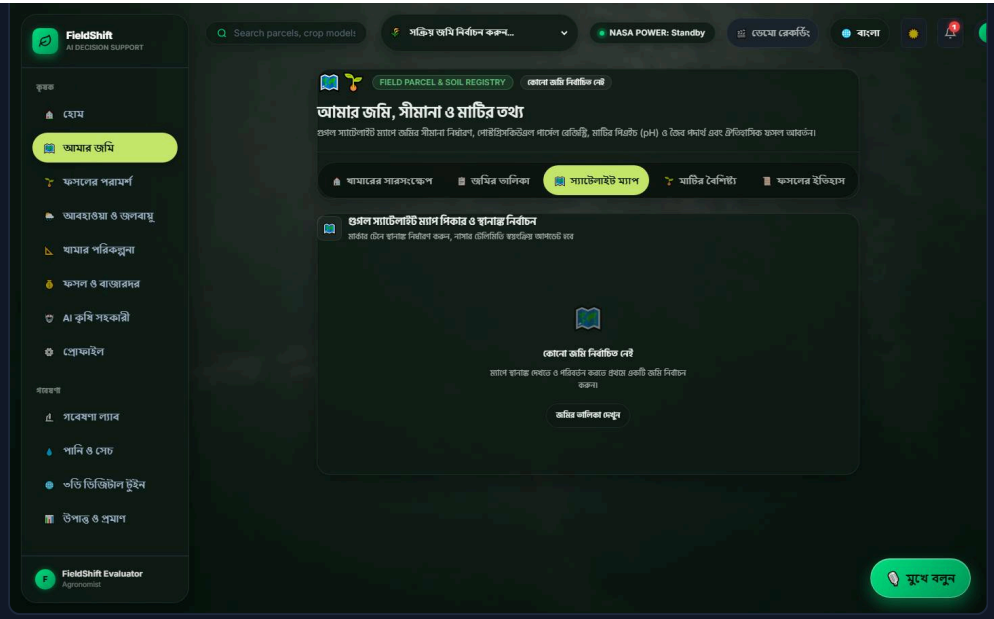


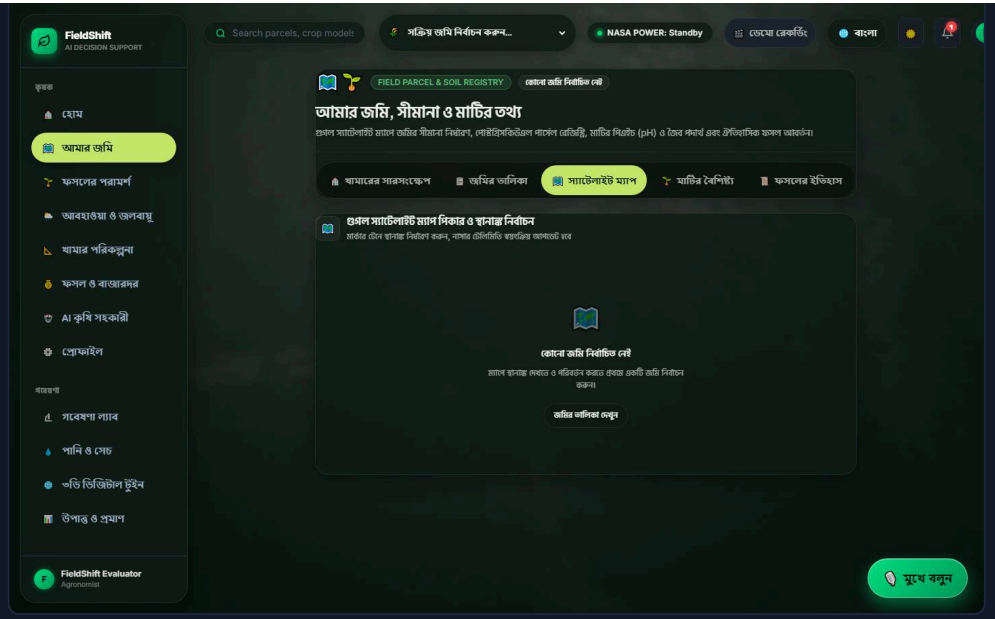

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



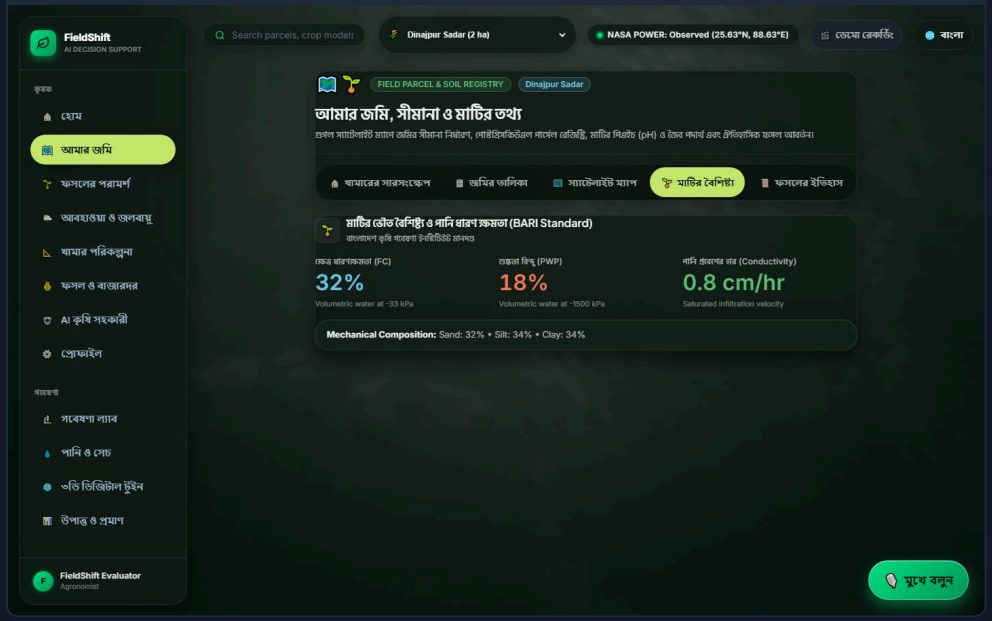

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

# 5. My Farm - Soil Info

SAME

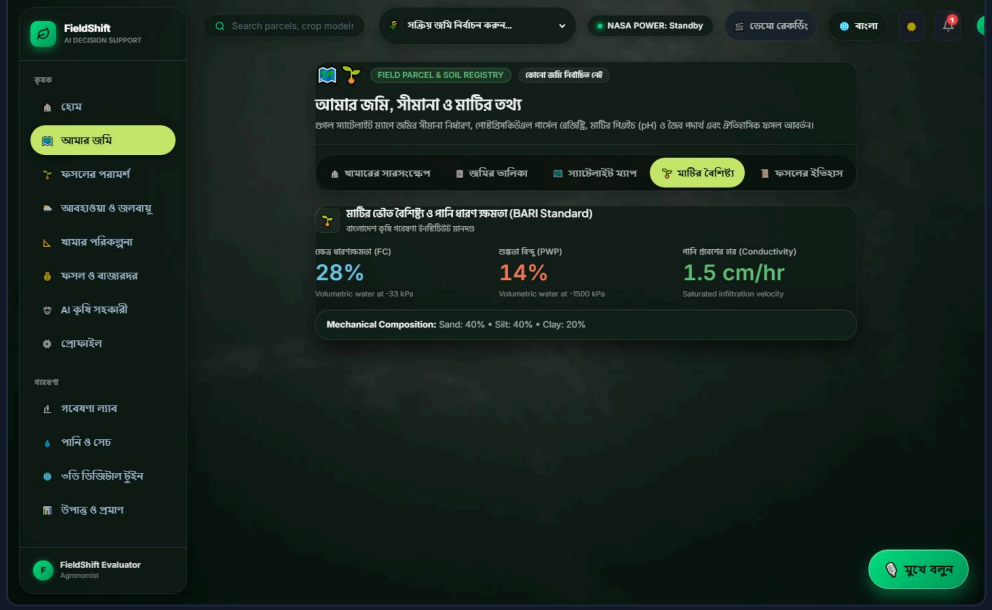
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



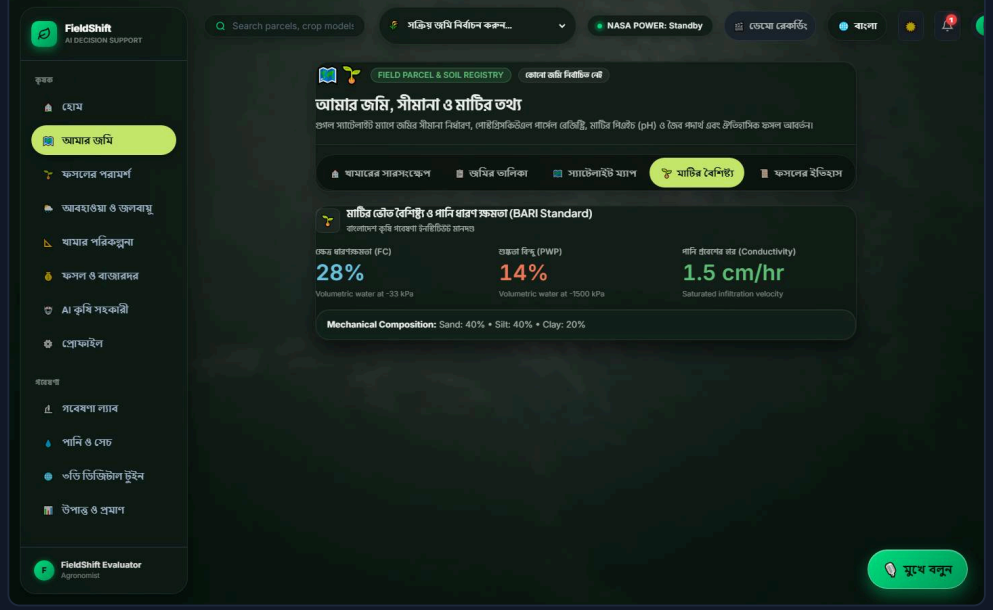

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



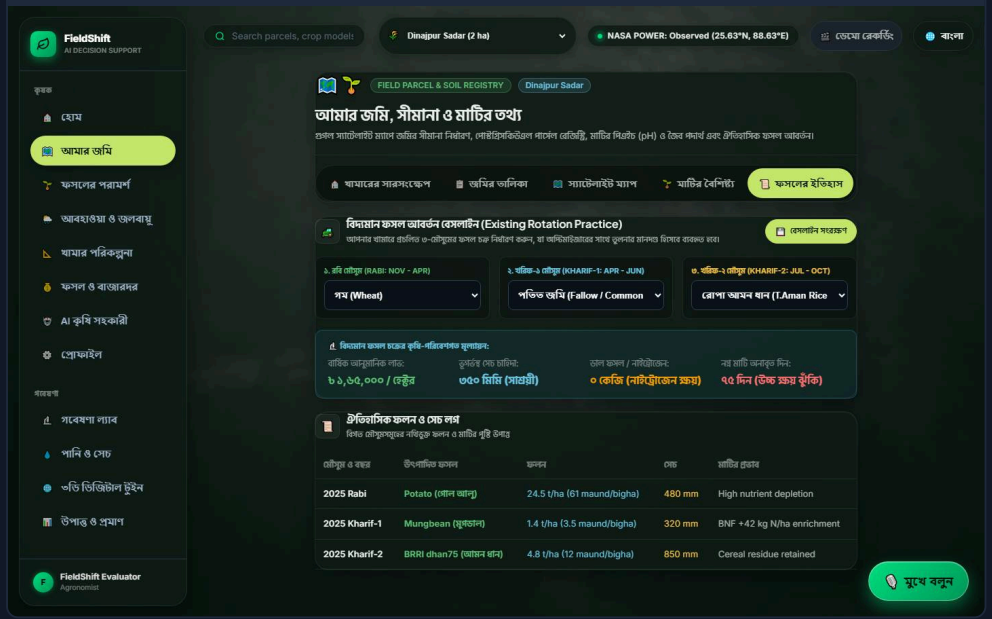
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

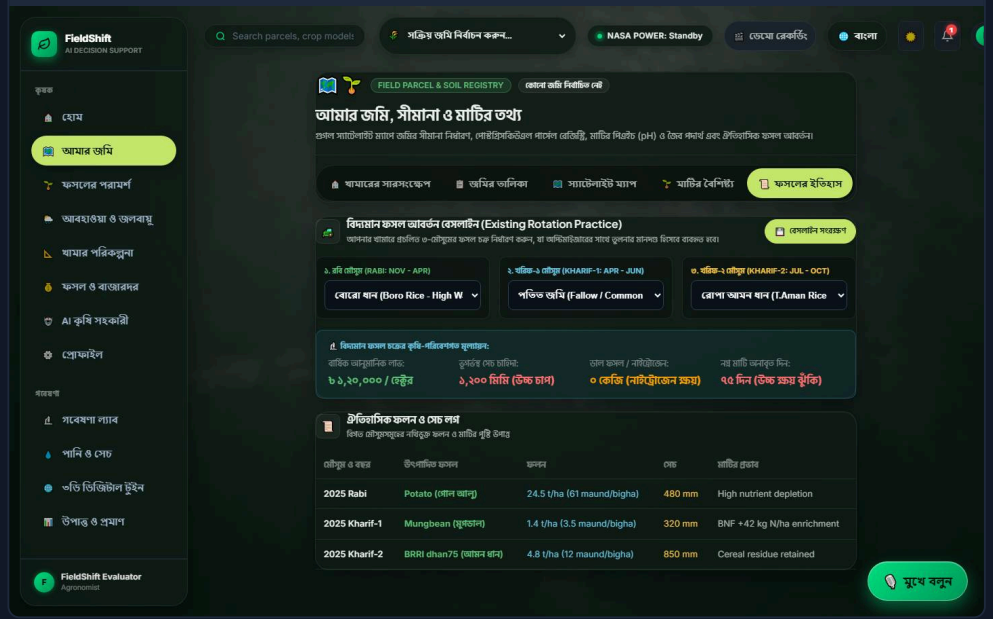
## 6. My Farm - Crop History

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

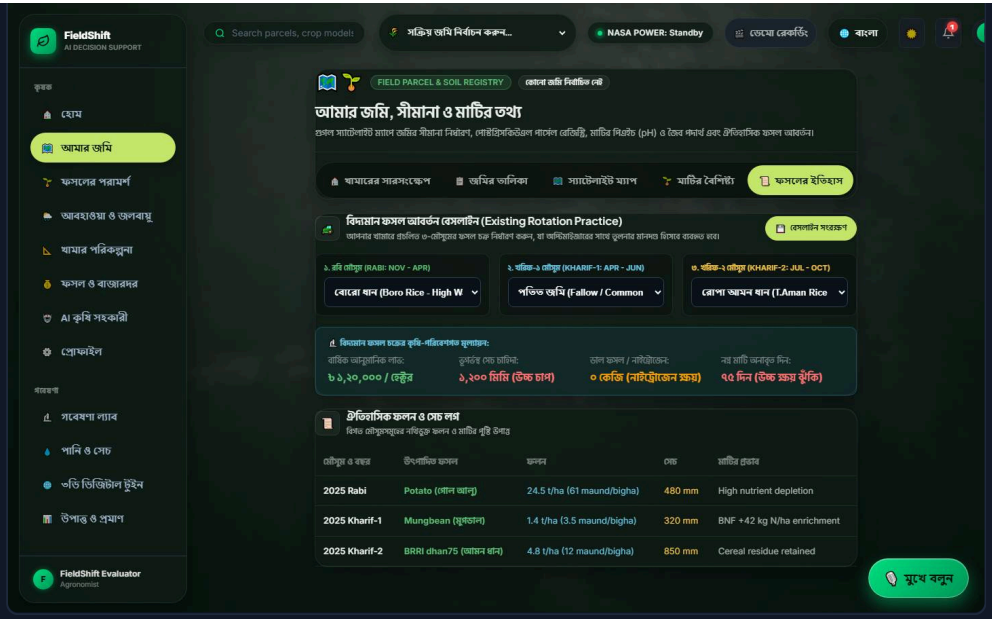


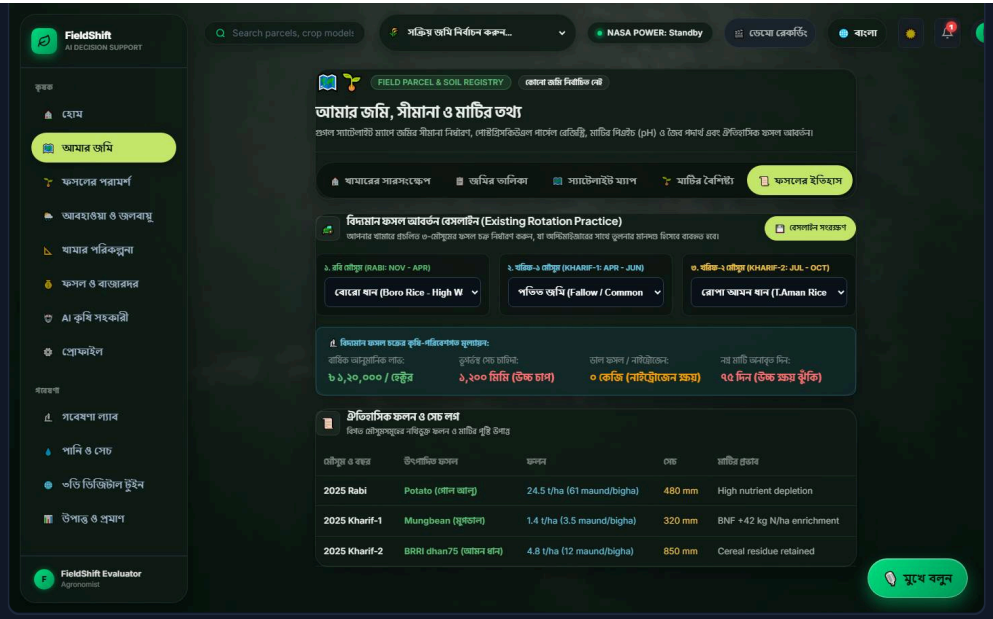

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



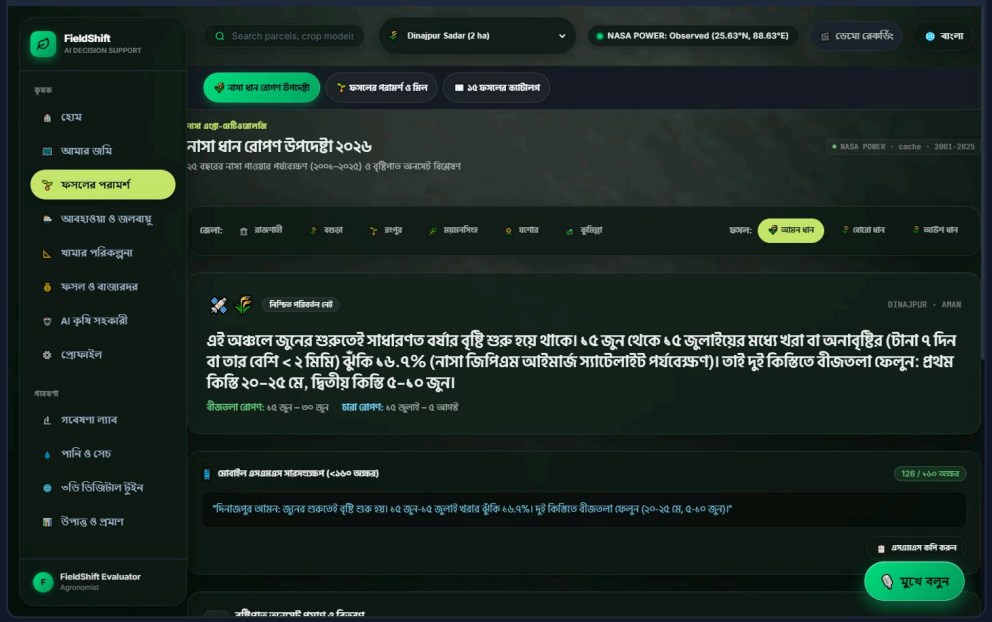

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 2025 Rabi \| 24.5 t/ha (61 maund/bigha) \| 480 mm \| 2025 Kharif-1 \| 1.4 t/ha (3.5 maund/bigha) | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 2025 Rabi \| 24.5 t/ha (61 maund/bigha) \| 480 mm \| 2025 Kharif-1 \| 1.4 t/ha (3.5 maund/bigha) | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 2025 Rabi \| 24.5 t/ha (61 maund/bigha) \| 480 mm \| 2025 Kharif-1 \| 1.4 t/ha (3.5 maund/bigha) | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 2025 Rabi \| 24.5 t/ha (61 maund/bigha) \| 480 mm \| 2025 Kharif-1 \| 1.4 t/ha (3.5 maund/bigha) |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

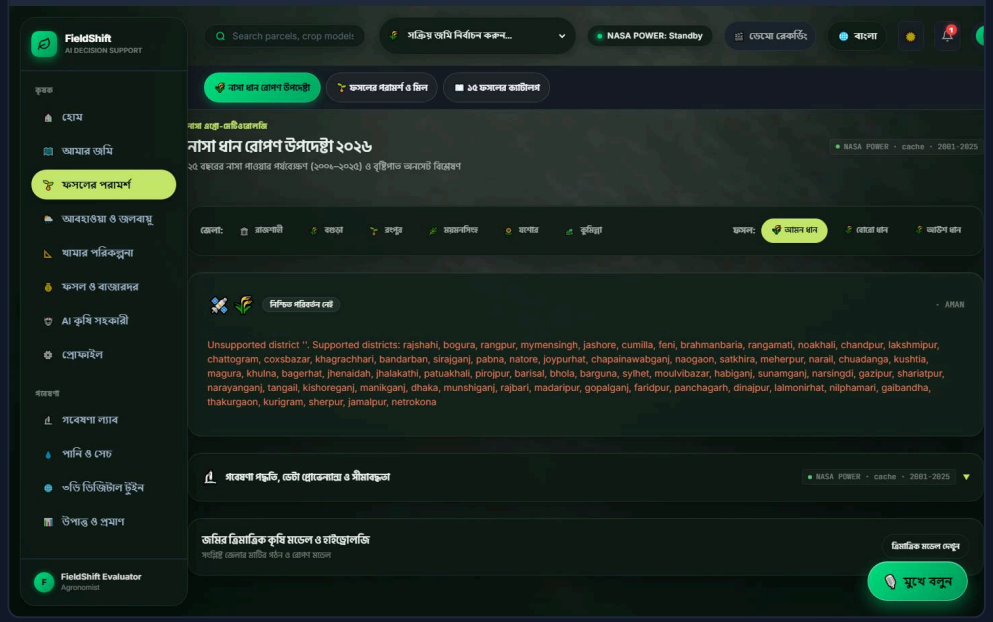
# 7. Crop Advisory - Aman Monsoon

SAME

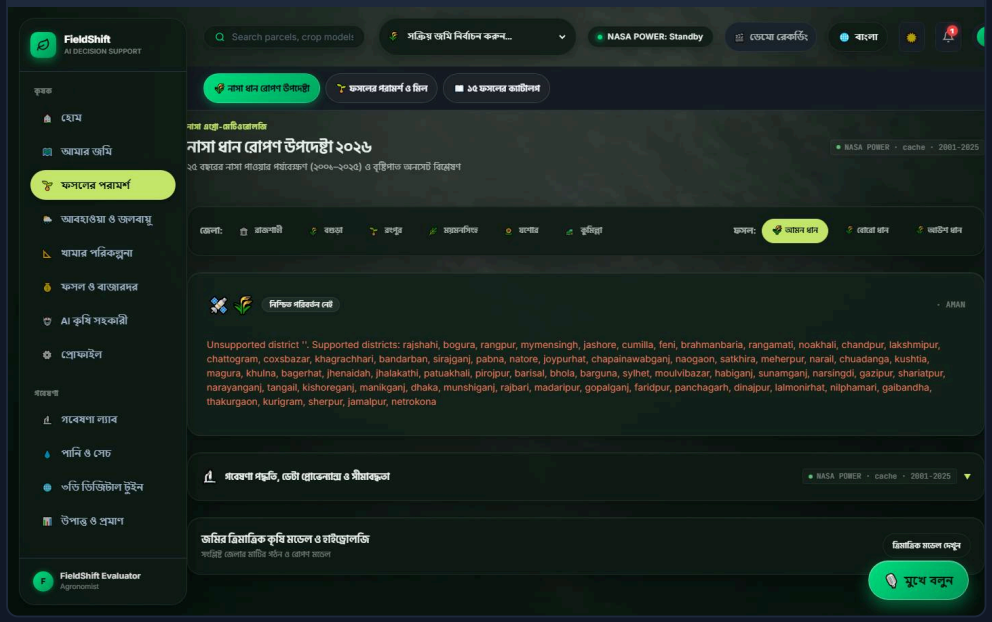
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



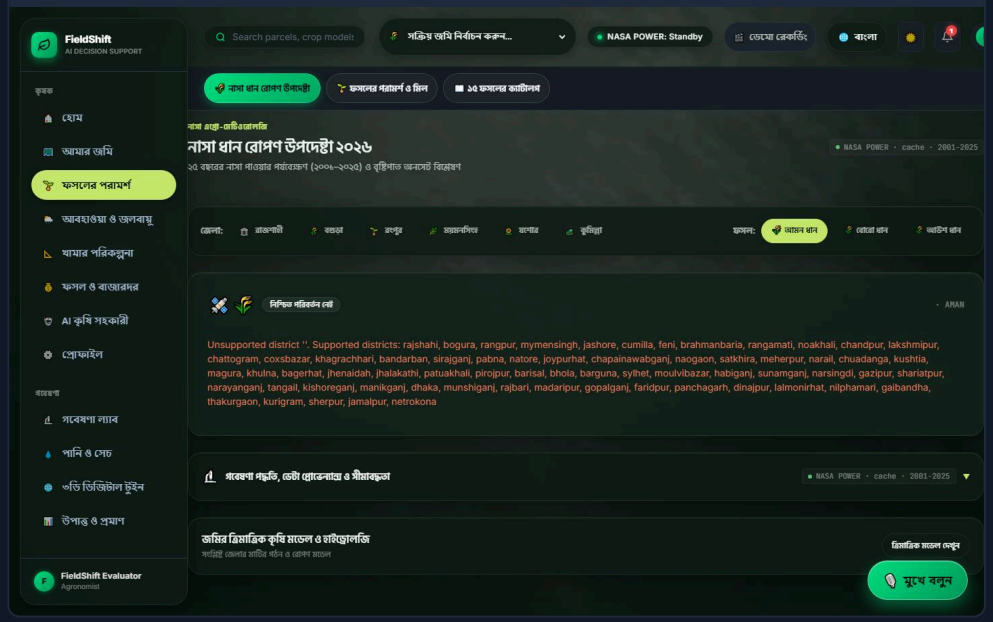

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



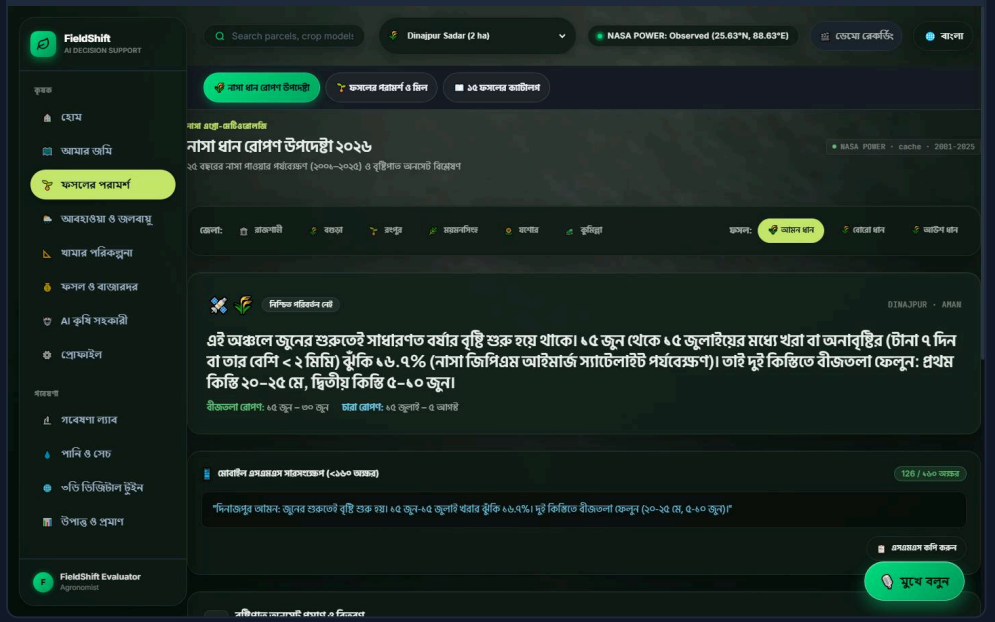
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

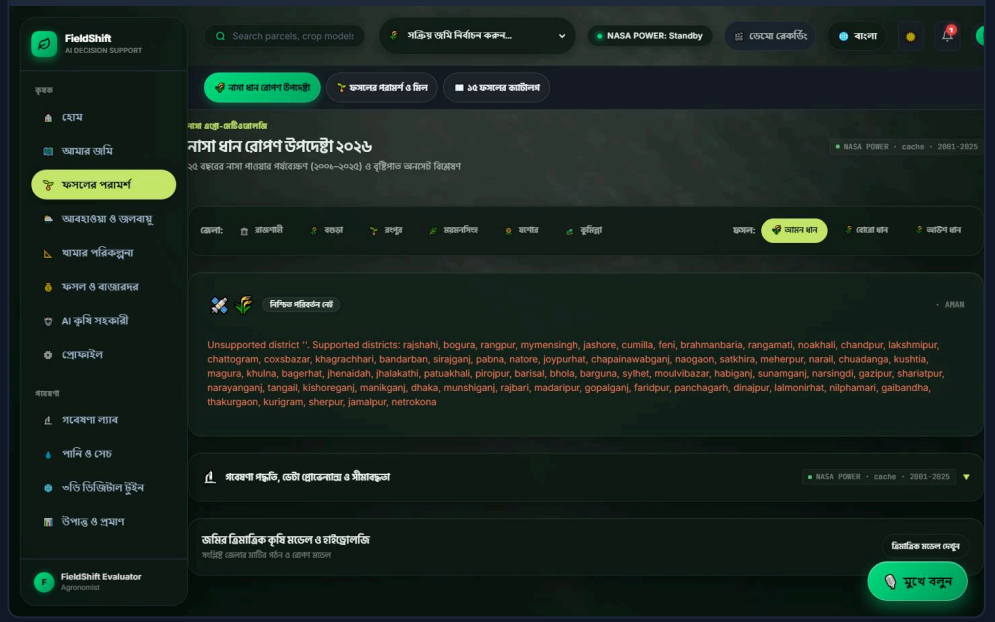
## 8. Crop Advisory - Aus Pre-Monsoon

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

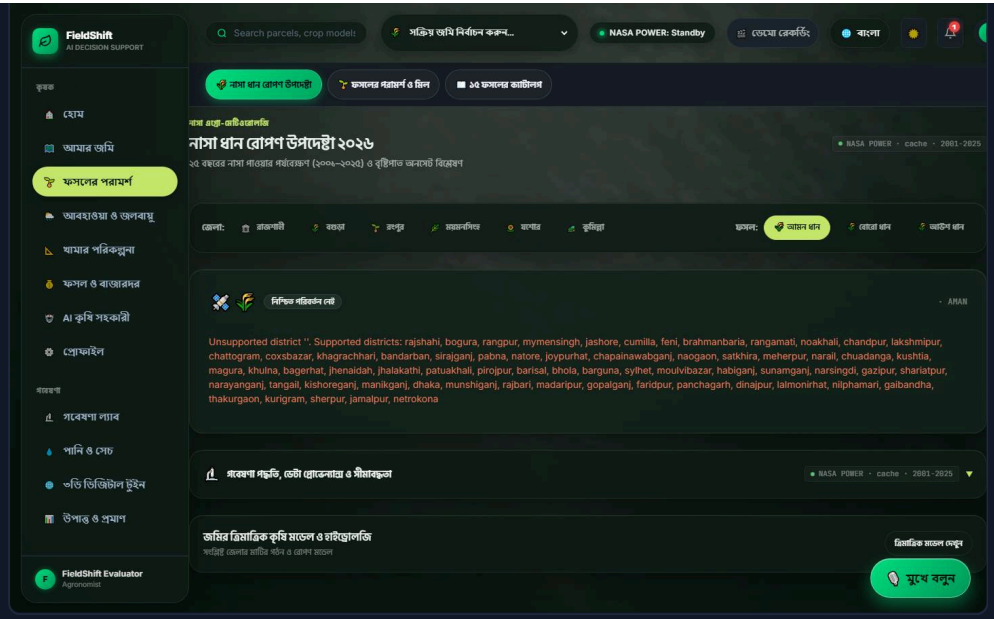


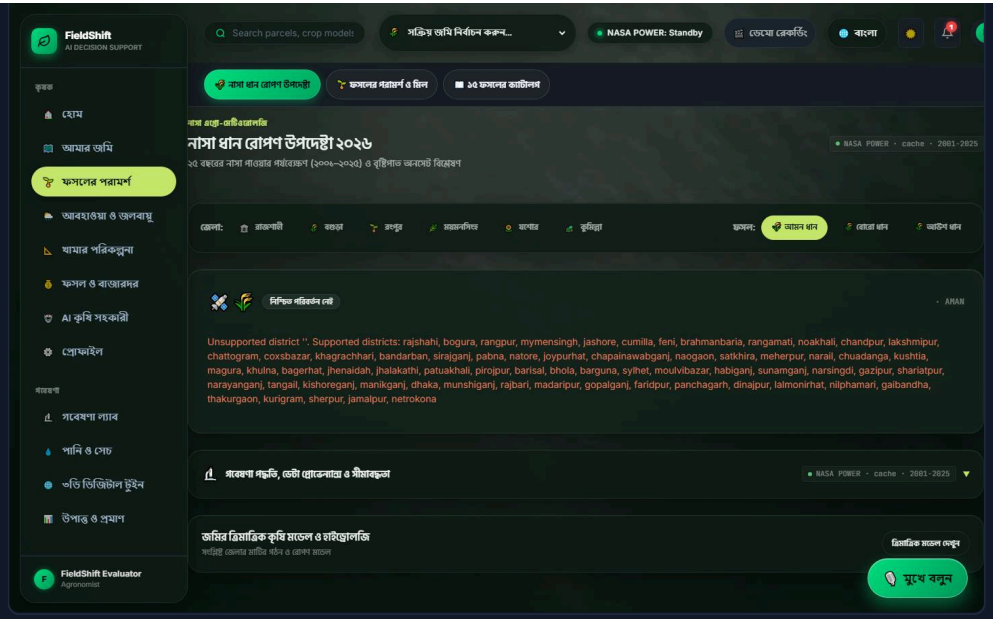

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



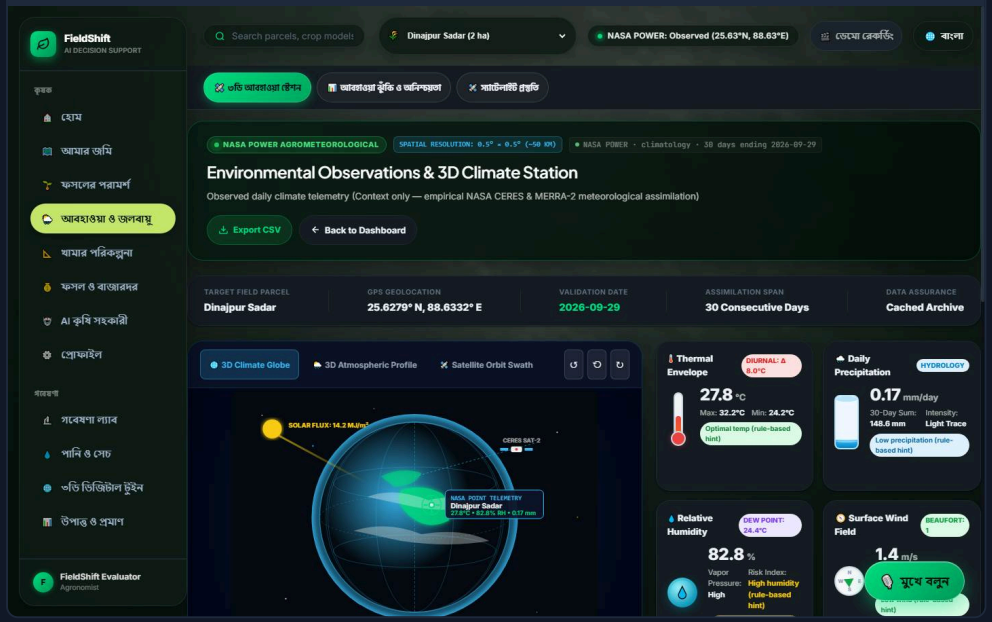

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

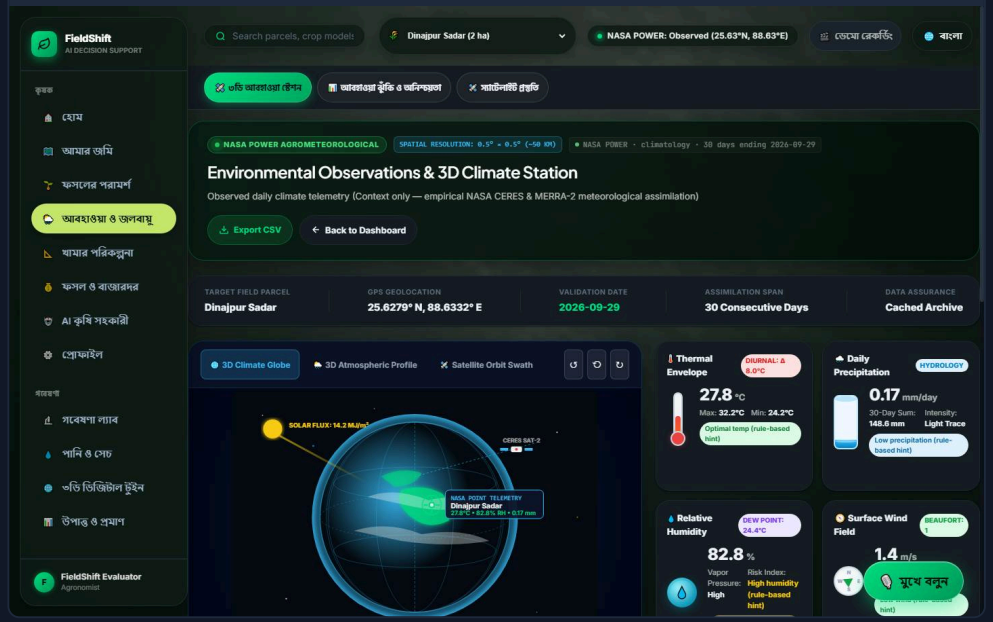
# 9. NASA Weather 3D - Globe

DIFFERS

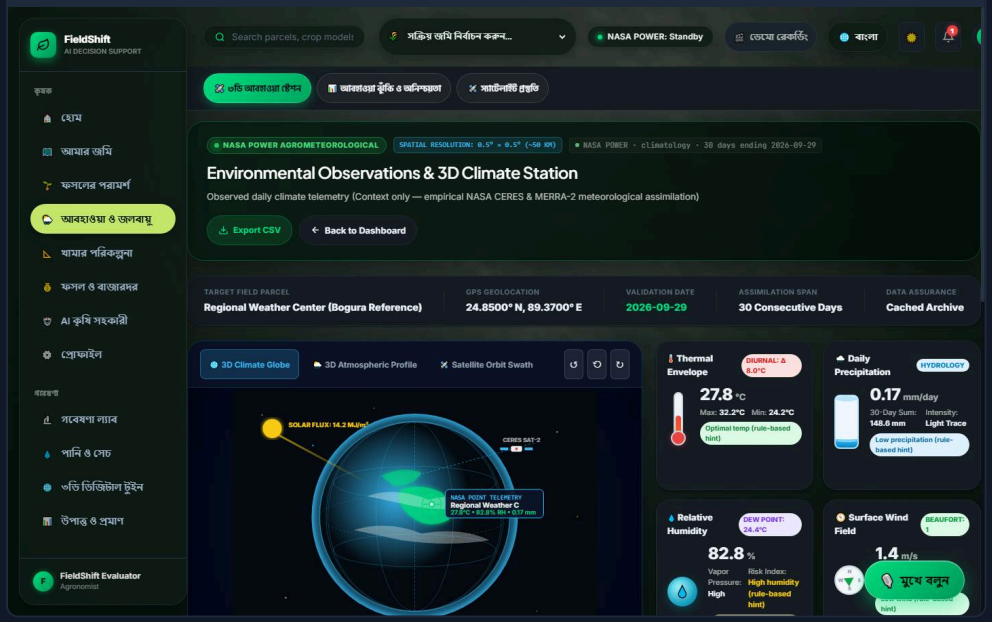
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



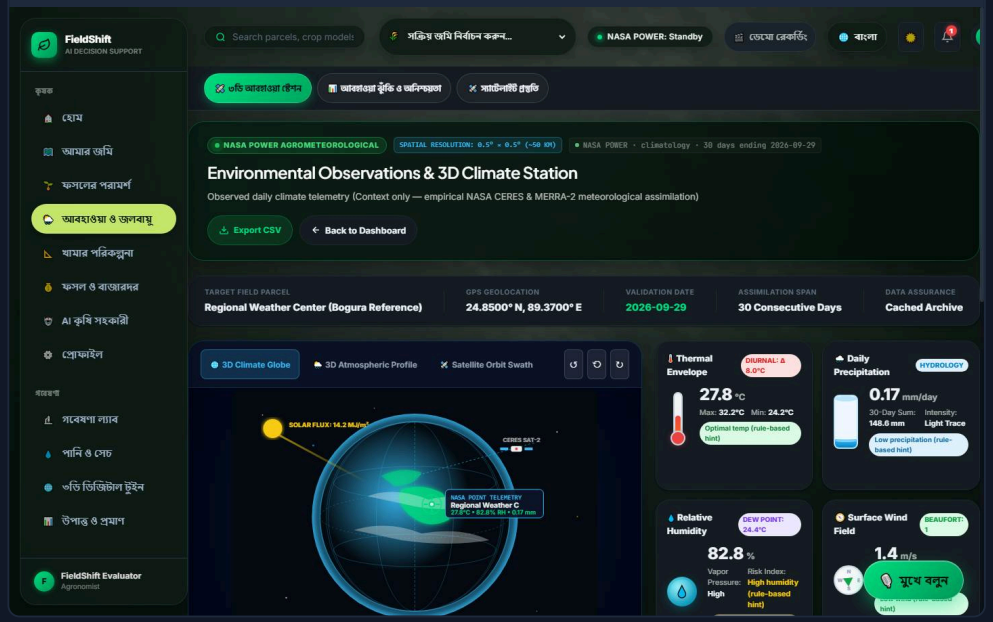

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



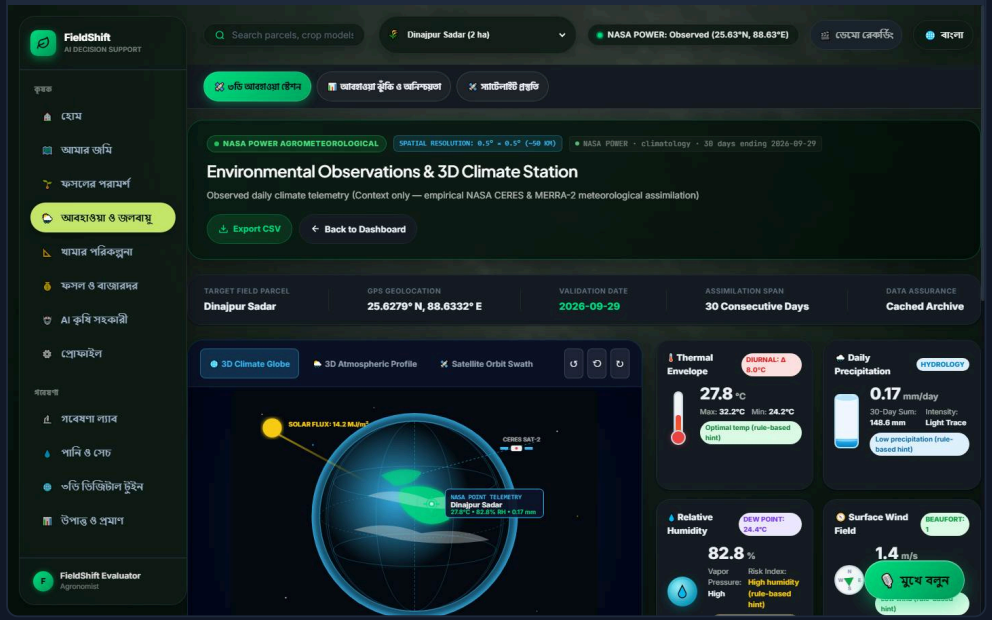
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 25.6279° N, 88.6332° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

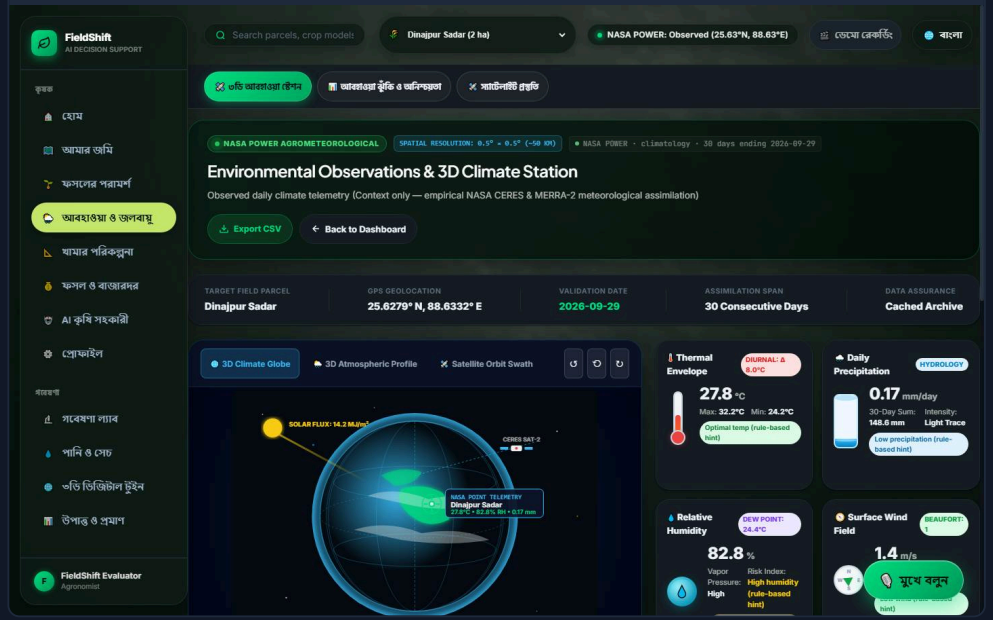
## 10. NASA Weather 3D - Column

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

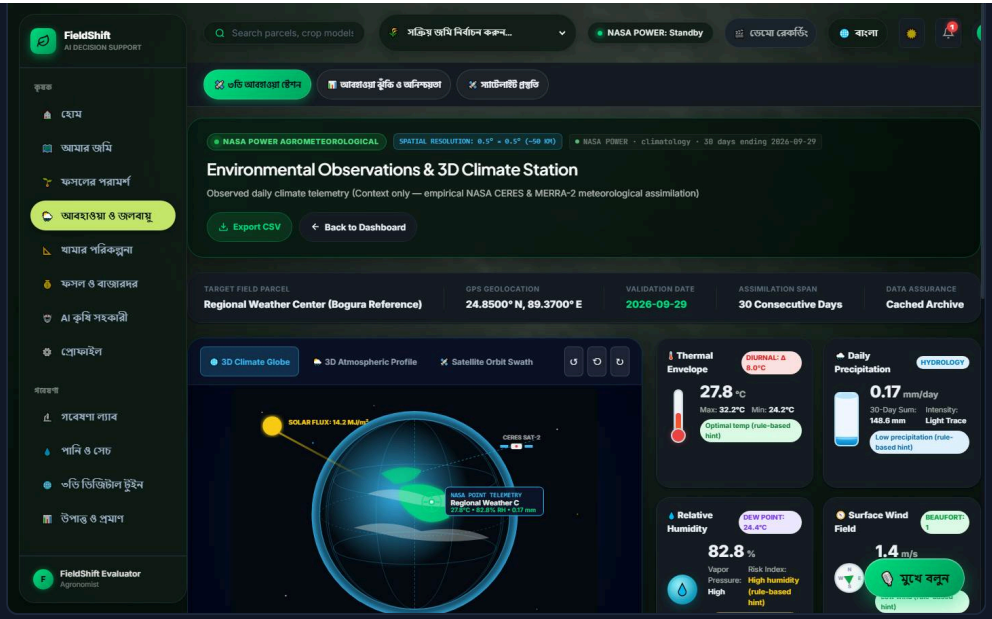


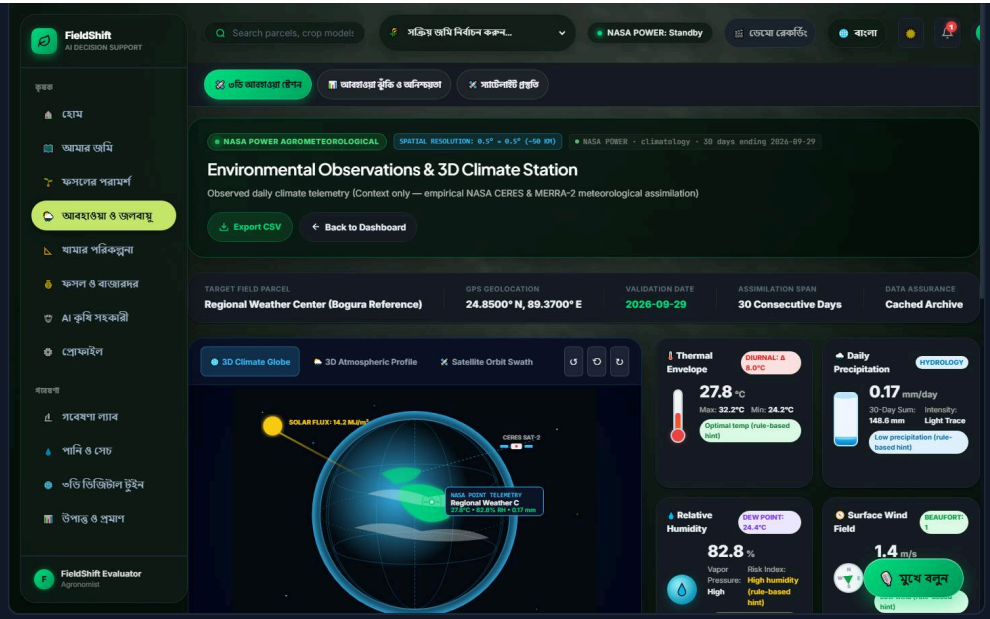

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



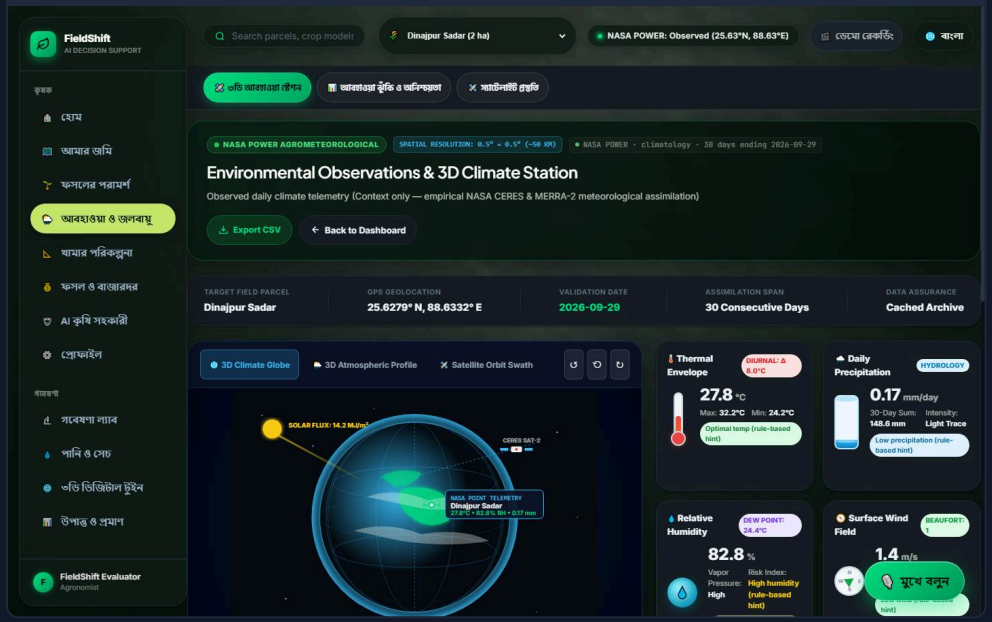

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 25.6279° N, 88.6332° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 25.6279° N, 88.6332° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

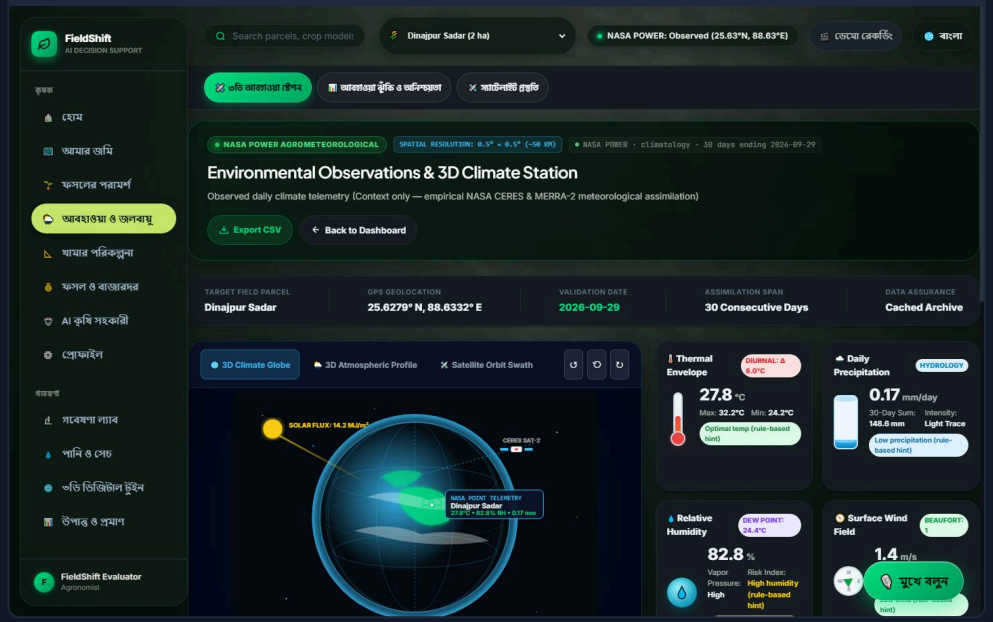
# 11. NASA Weather 3D - Constellation

DIFFERS

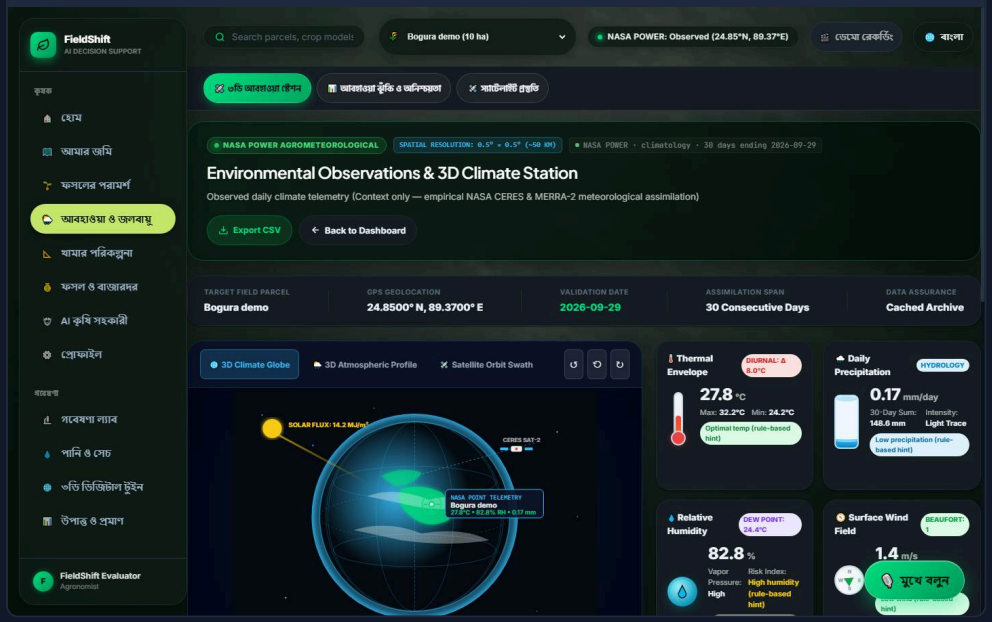
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



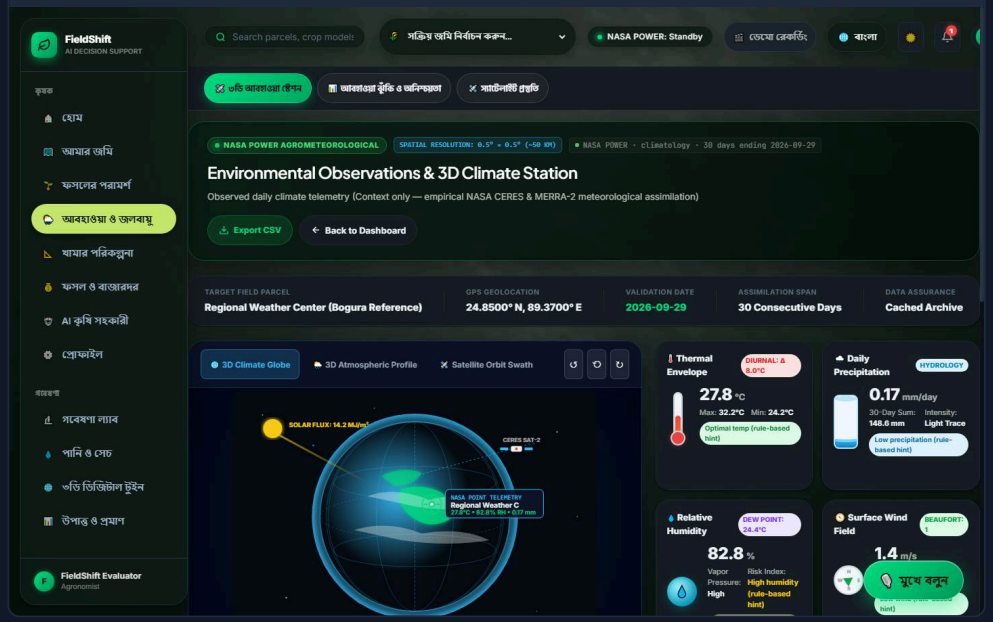

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



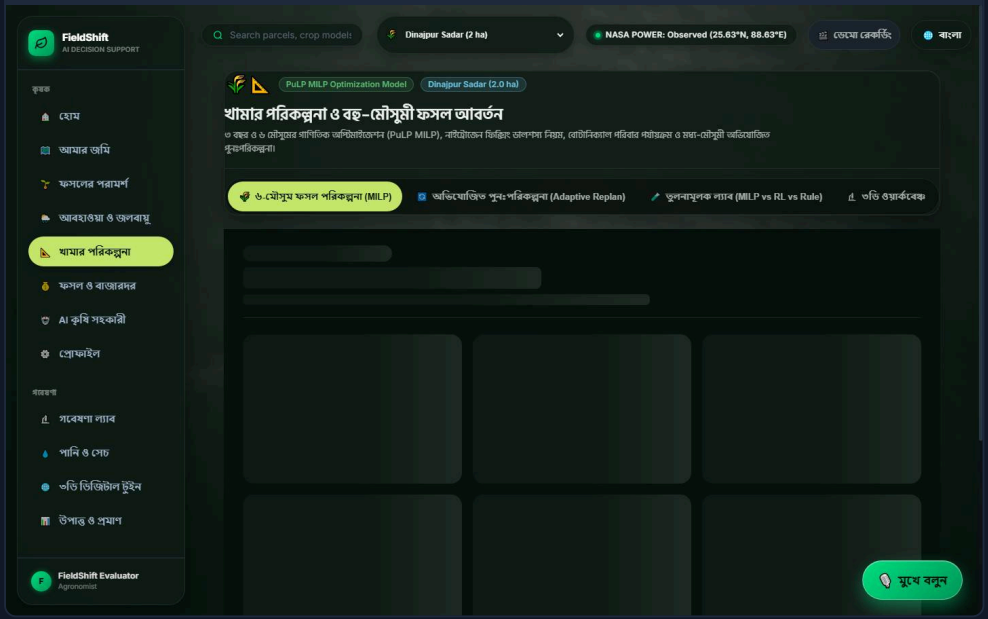
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 25.6279° N, 88.6332° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 25.6279° N, 88.6332° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 24.8500° N, 89.3700° E \| 2026-09-29 \| 30 Consecutive Days \| 32.2°C \| 24.2°C |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

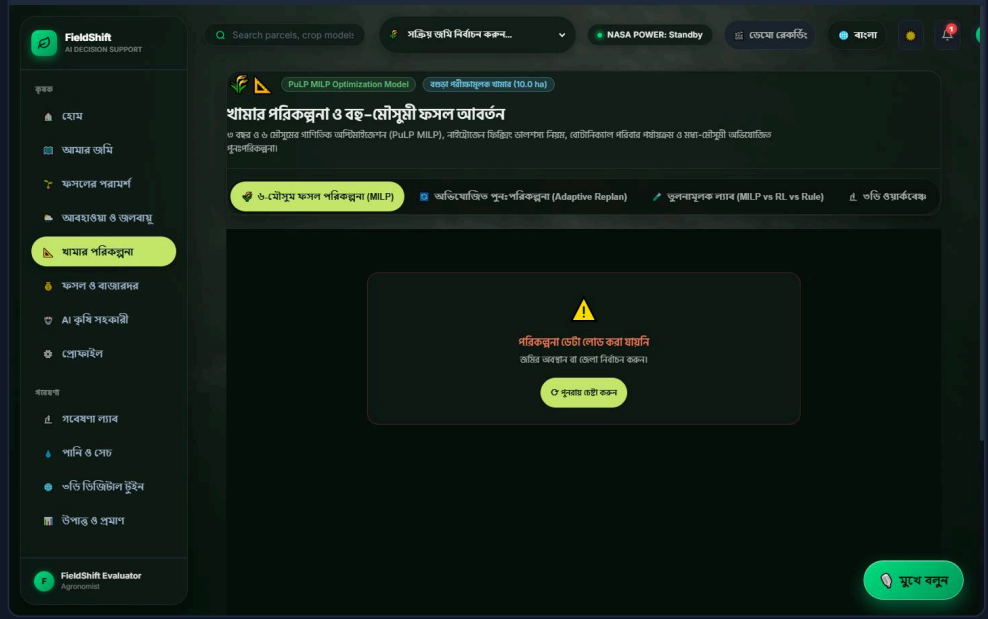
## 12. Farm Planning - 6-Season MILP Rotation

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

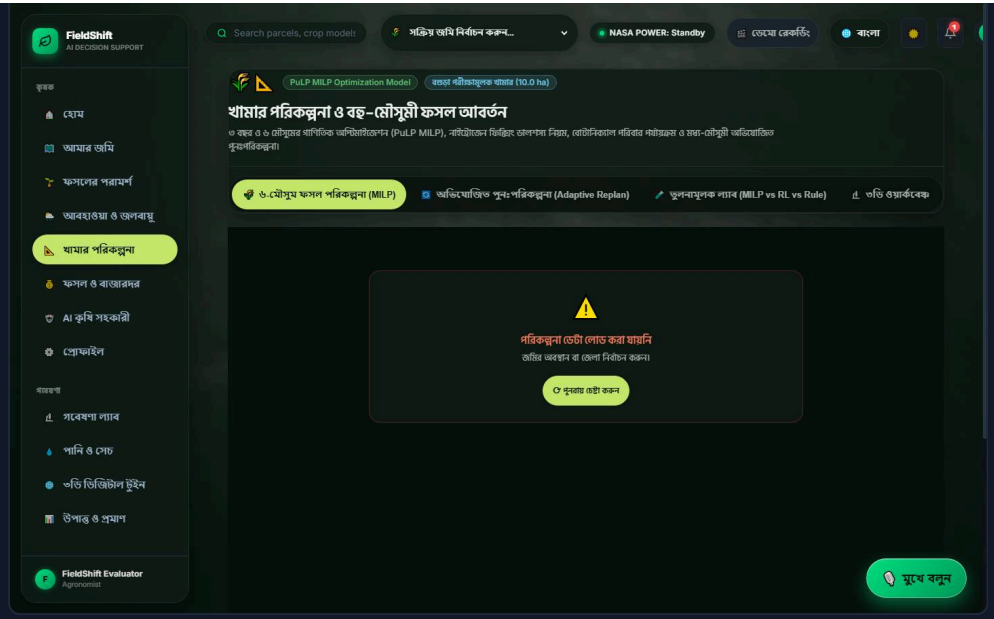


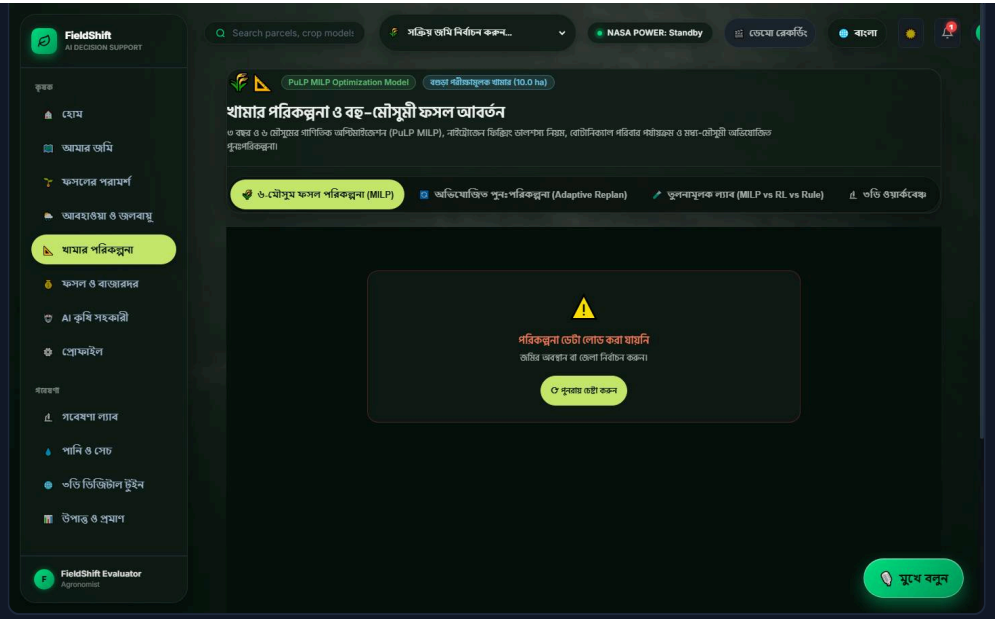

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)





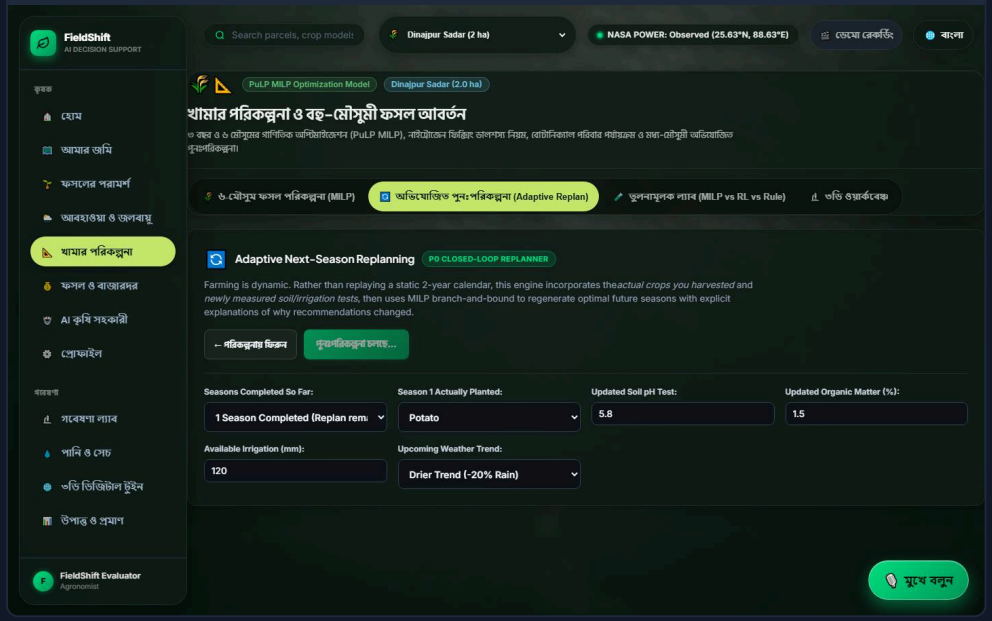

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

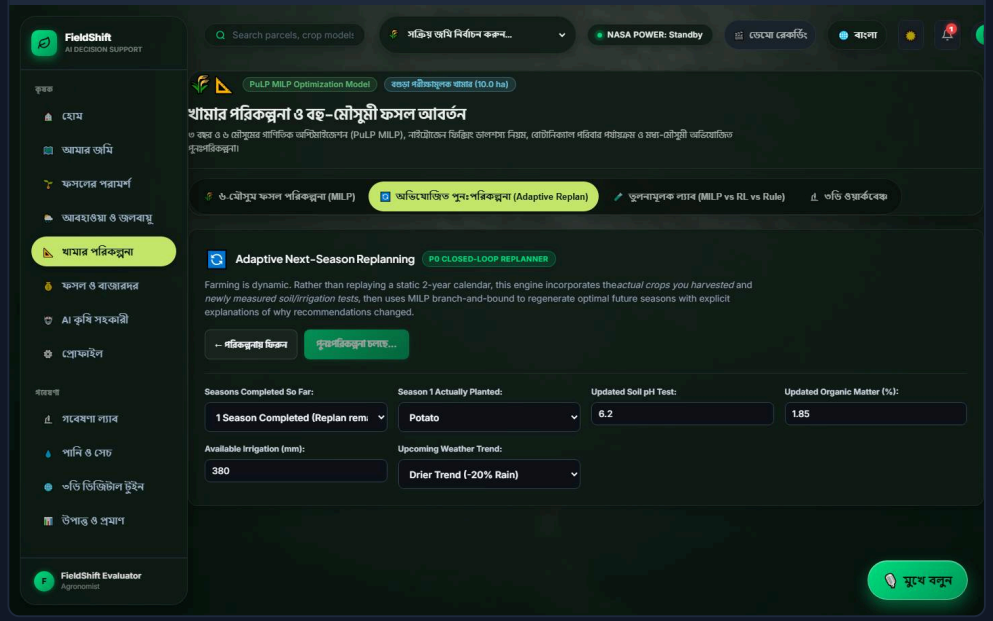
# 13. Farm Planning - Adaptive Replan

SAME

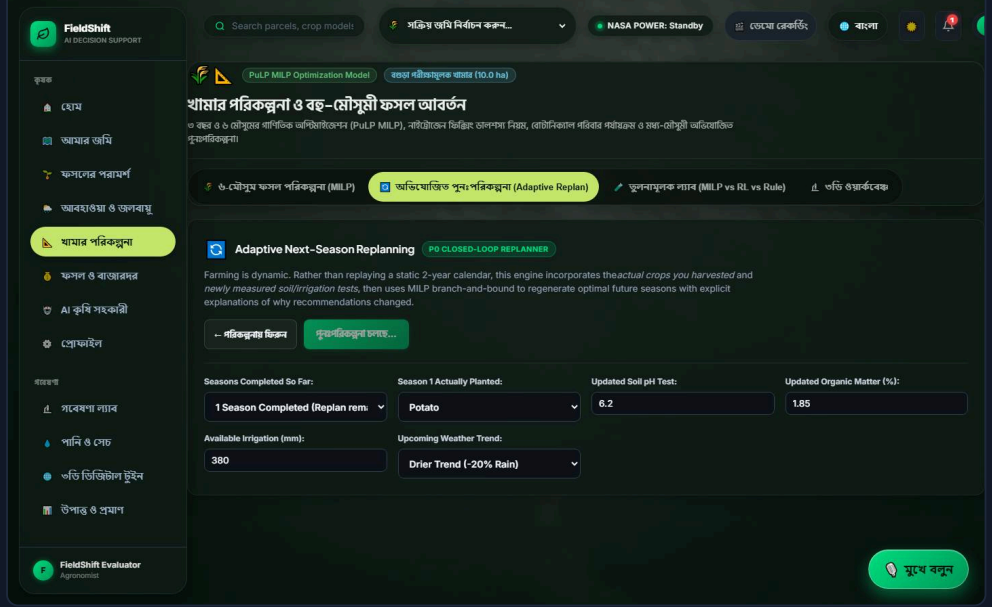
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



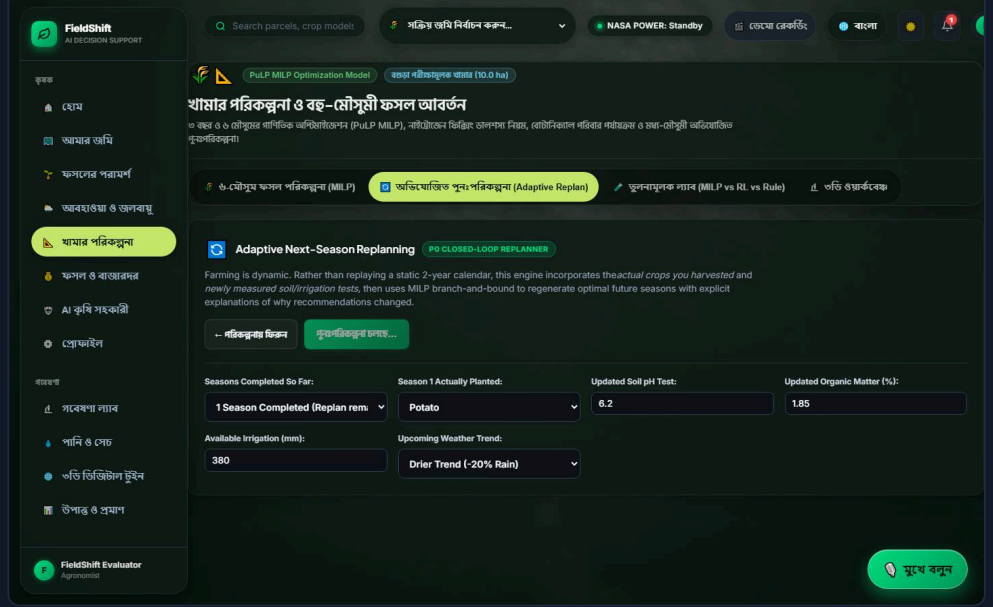

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



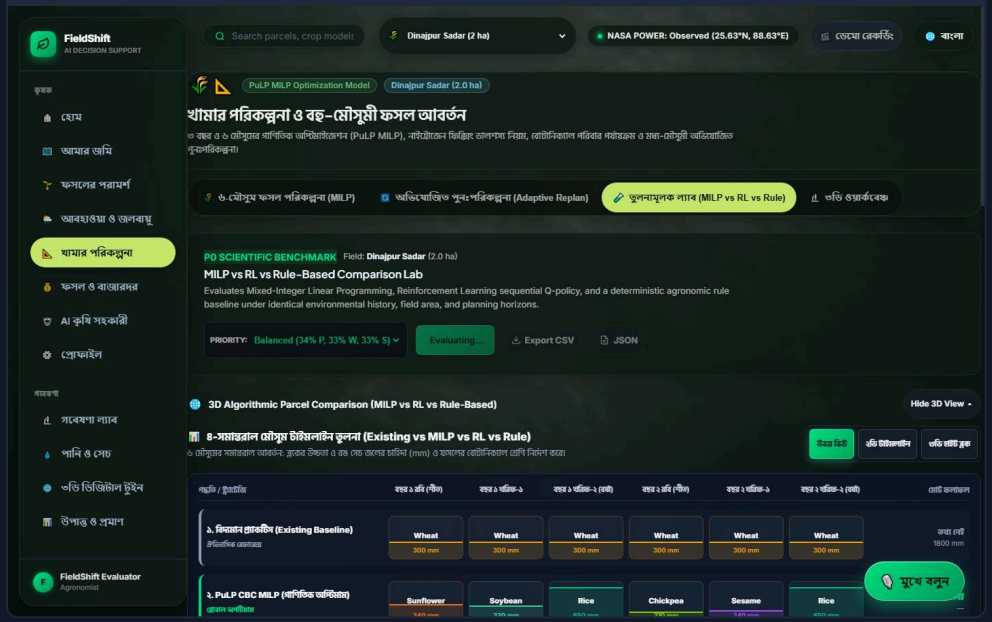
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

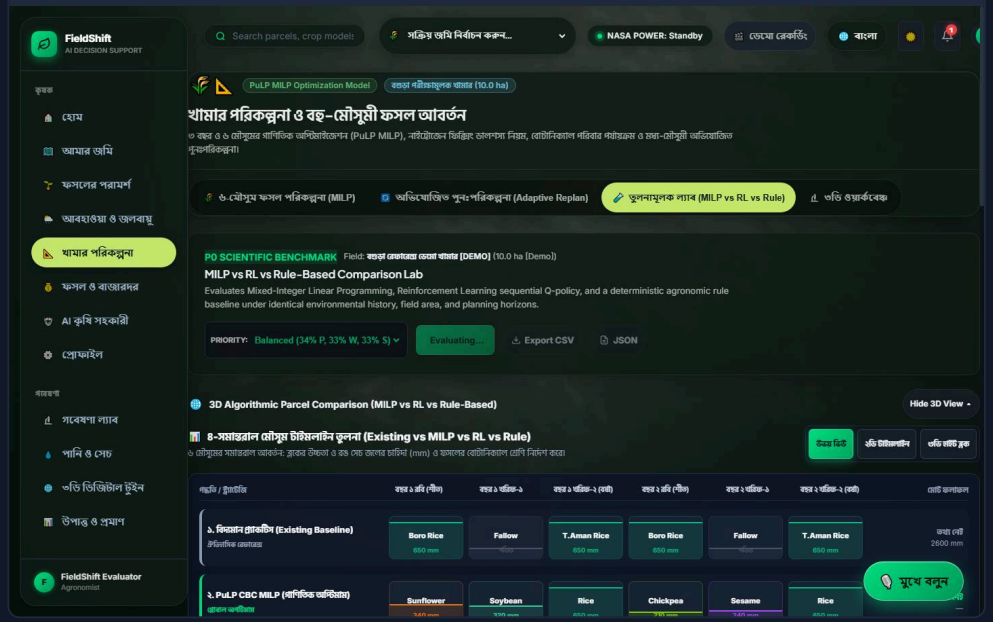
## 14. Farm Planning - 3-Way Comparison Lab

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

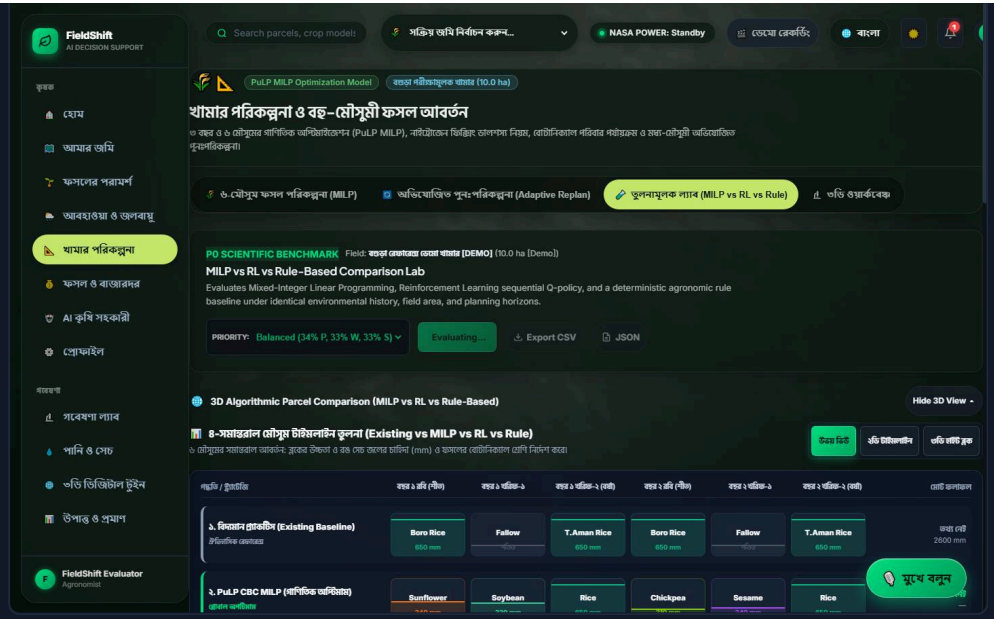


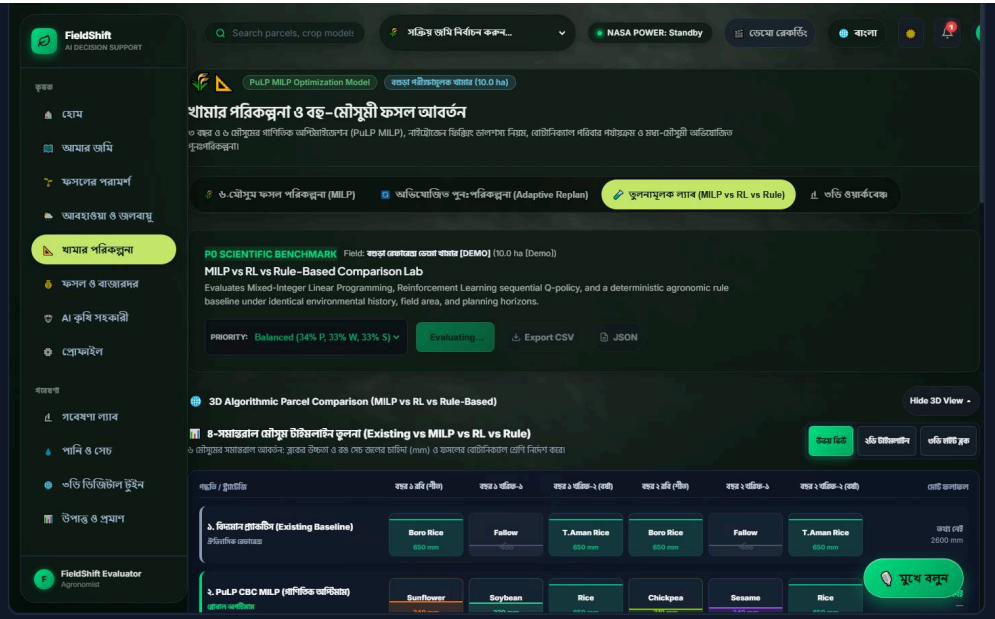

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



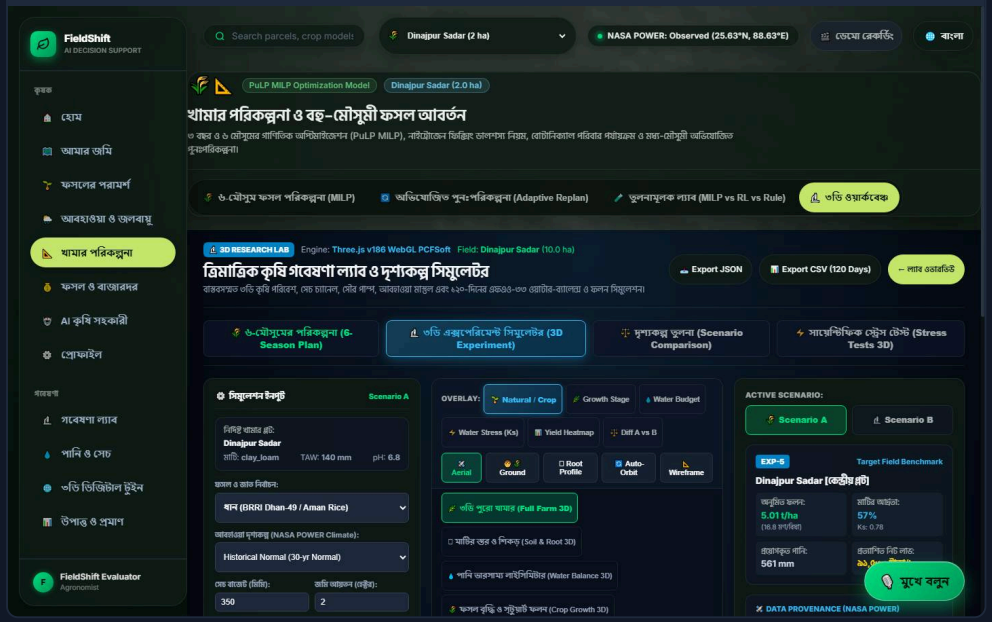

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based) \| 1800 mm | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based) \| 2600 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based) \| 2600 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 3D Algorithmic Parcel Comparison (MILP vs RL vs Rule-Based) \| 2600 mm |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

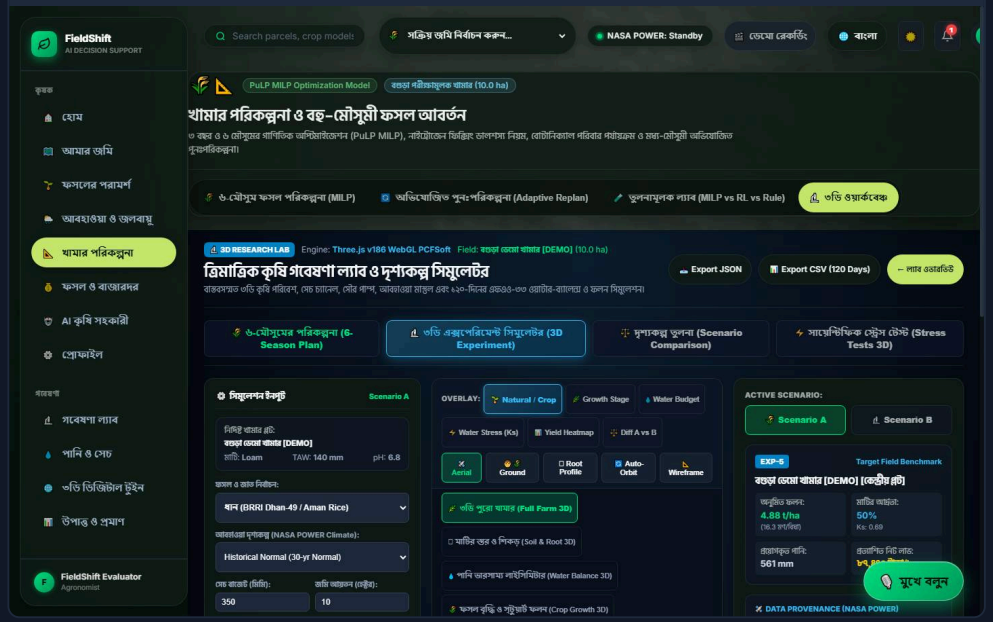
# 15. Farm Planning - 3D Experiment Workbench

DIFFERS

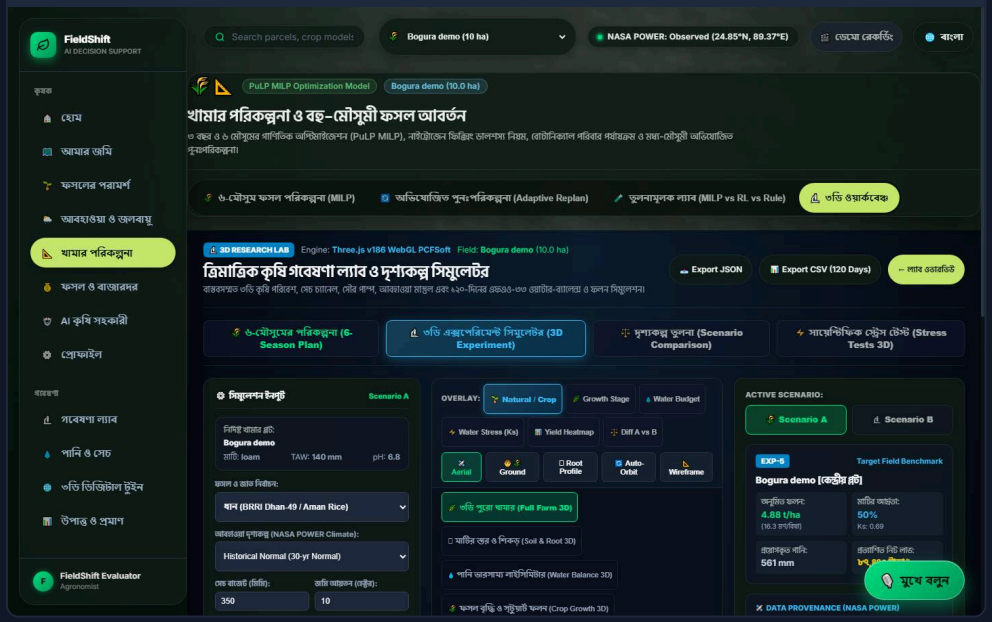
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



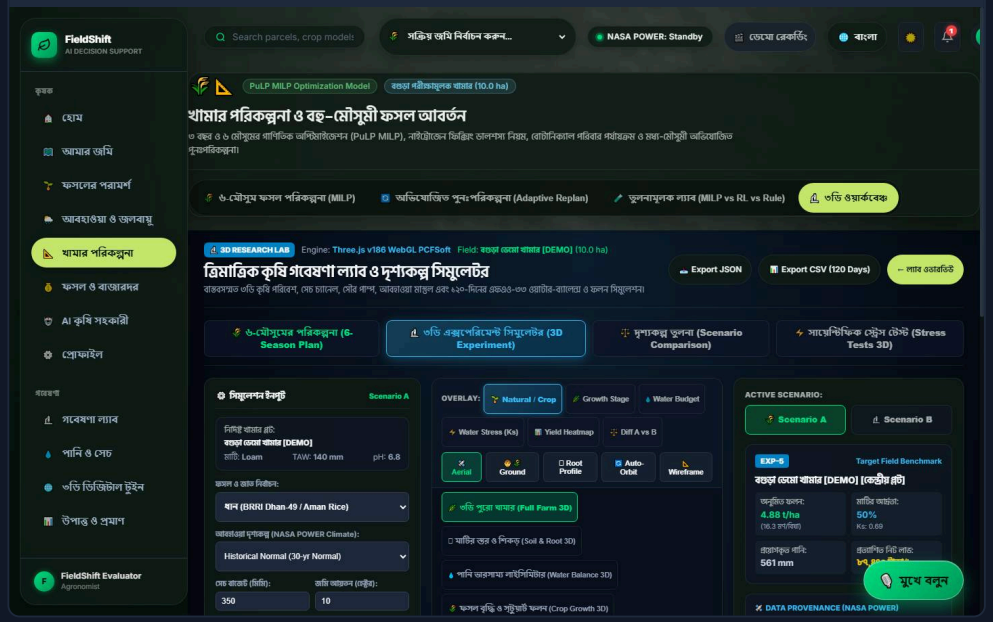

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



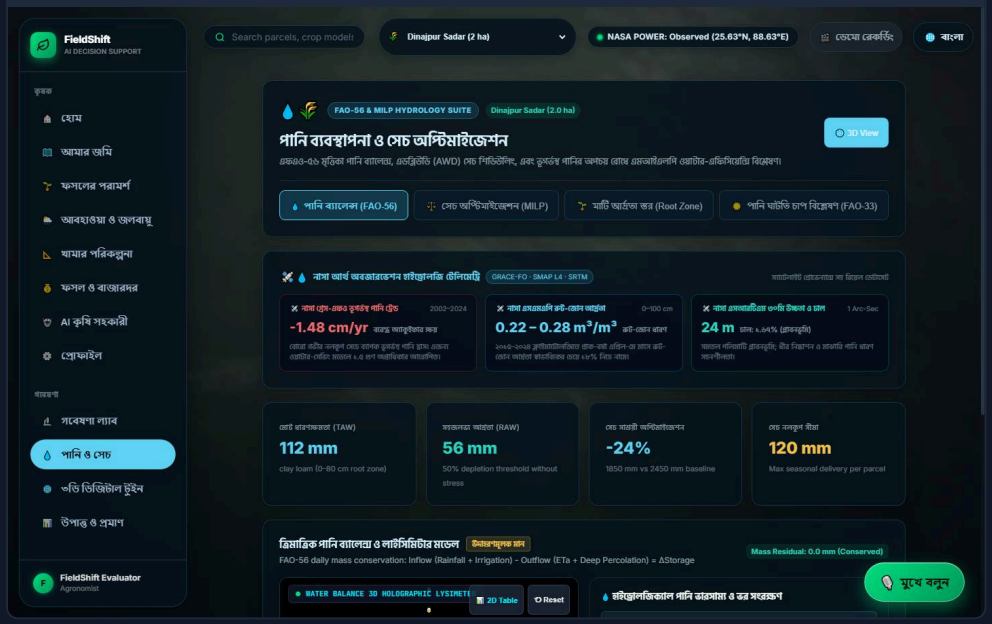
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 5.01 t/ha \| 561.3 mm | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 4.88 t/ha \| 561.3 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 4.88 t/ha \| 561.3 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 4.88 t/ha \| 561.3 mm |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

## 16. Water & Irrigation - Water Balance

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

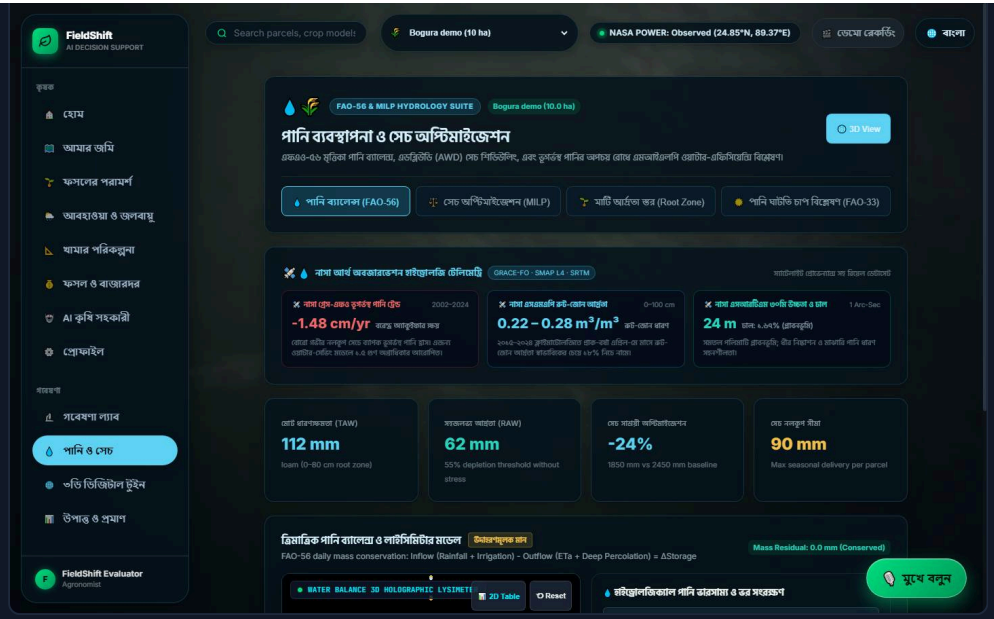


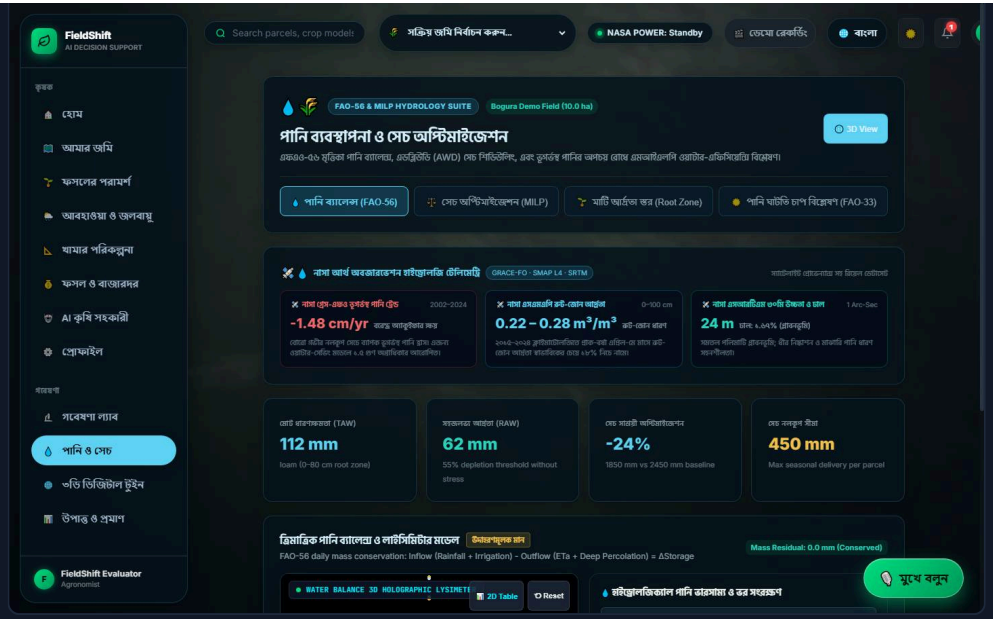

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



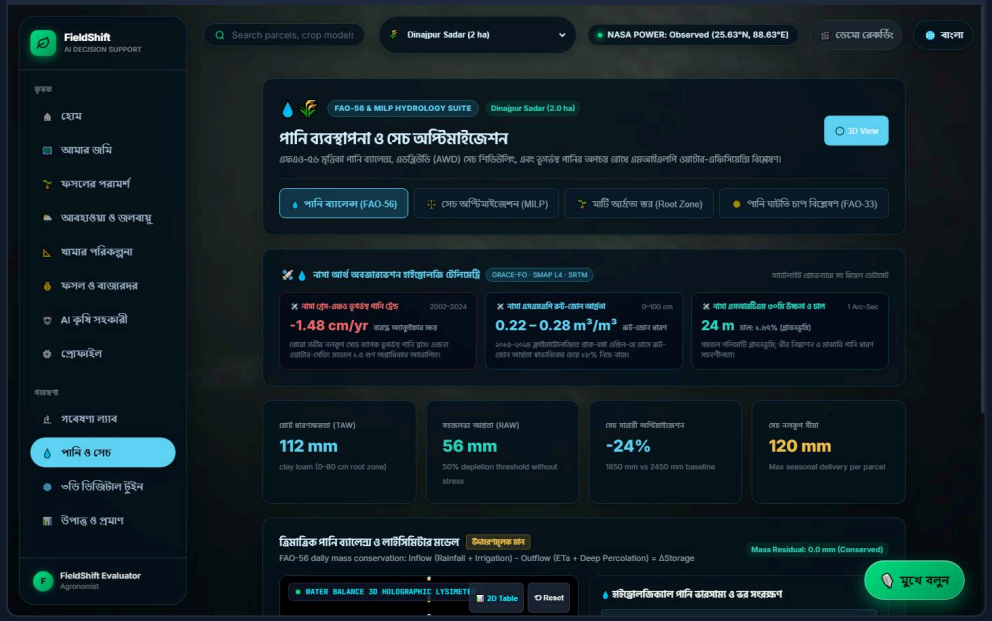

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

# 17. Water & Irrigation - Schedule Optimization

SAME

## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)




## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



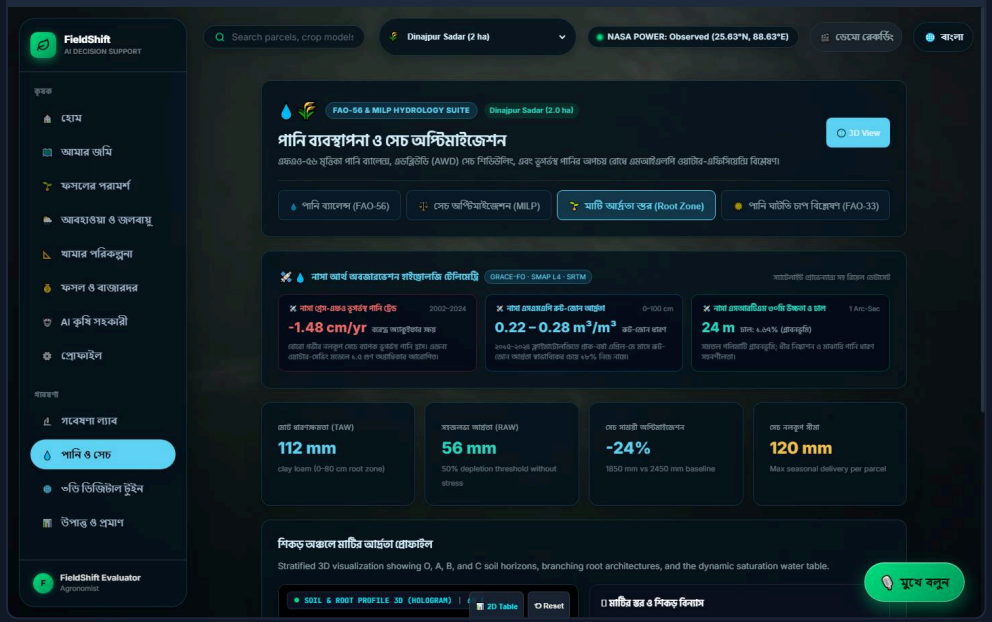
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

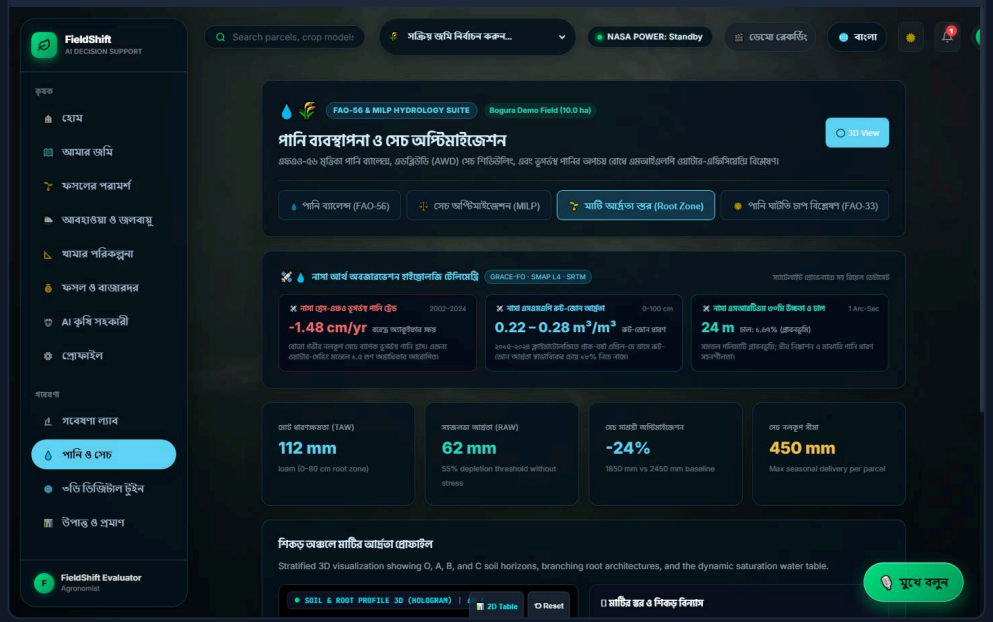
## 18. Water & Irrigation - Soil Moisture Profile

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

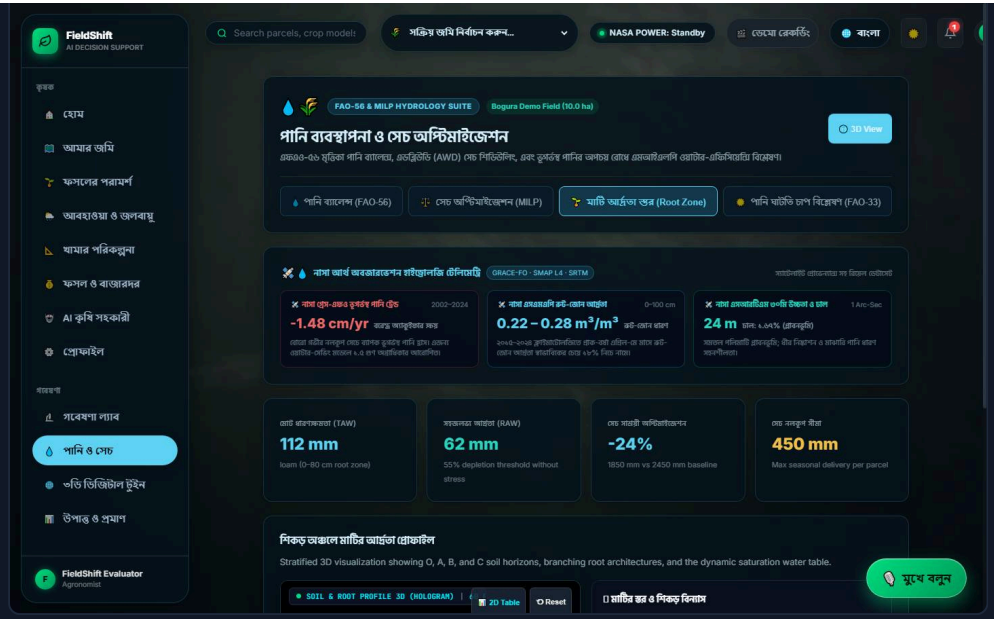


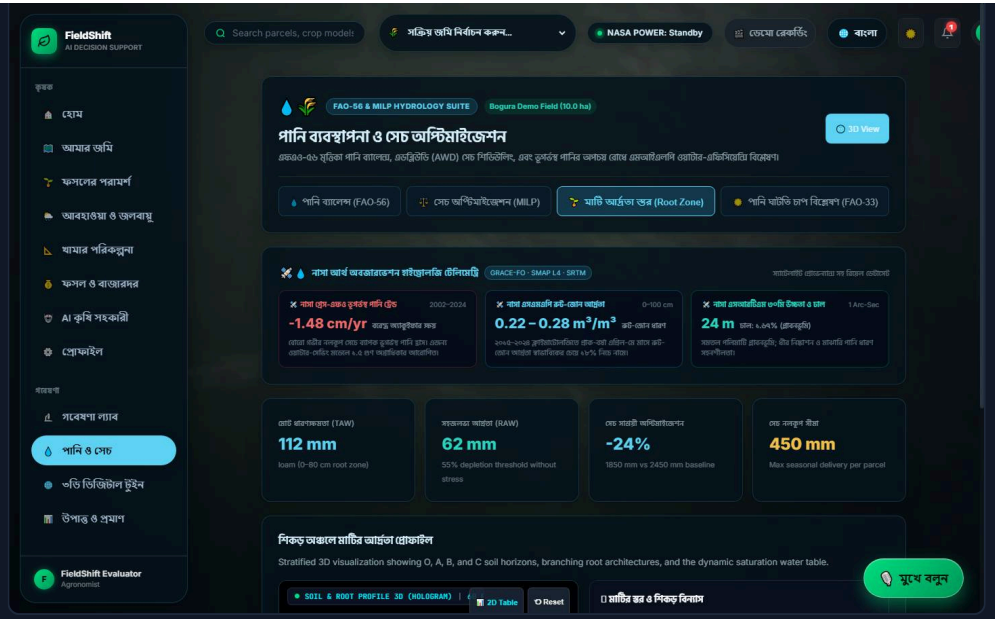

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



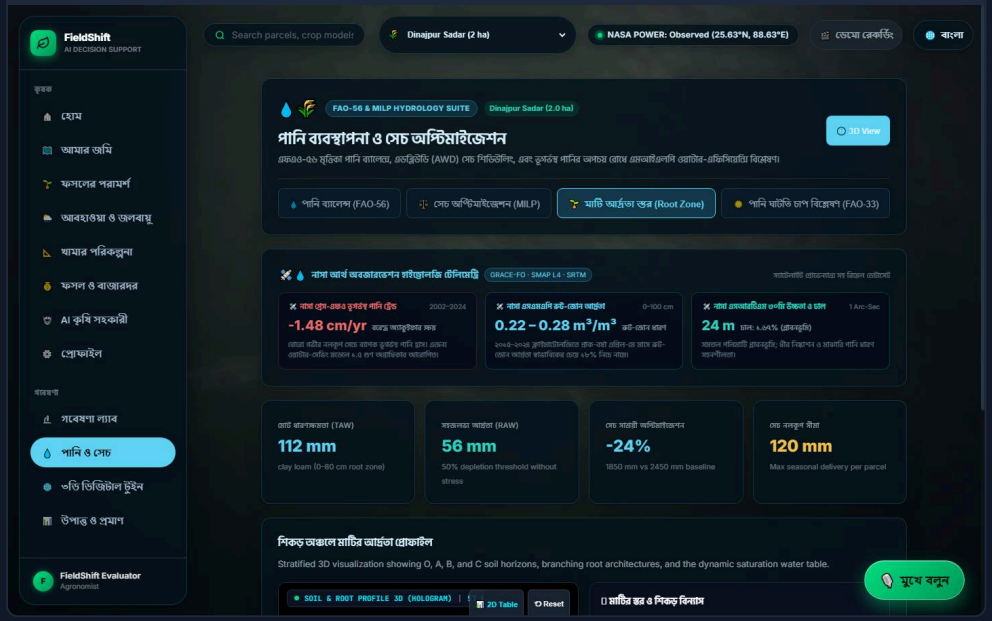

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 32% \| 18% \| 112 mm \| 228 mm (100%) \| 0.8 m | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 28.0000000000000004% \| 14.0000000000000002% \| 112 mm \| 228 mm (100%) \| 0.8 m | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 28.0000000000000004% \| 14.0000000000000002% \| 112 mm \| 228 mm (100%) \| 0.8 m | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 28.0000000000000004% \| 14.0000000000000002% \| 112 mm \| 228 mm (100%) \| 0.8 m |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

# 19. Water & Irrigation - Drought Stress Index

DIFFERS

## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)




## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



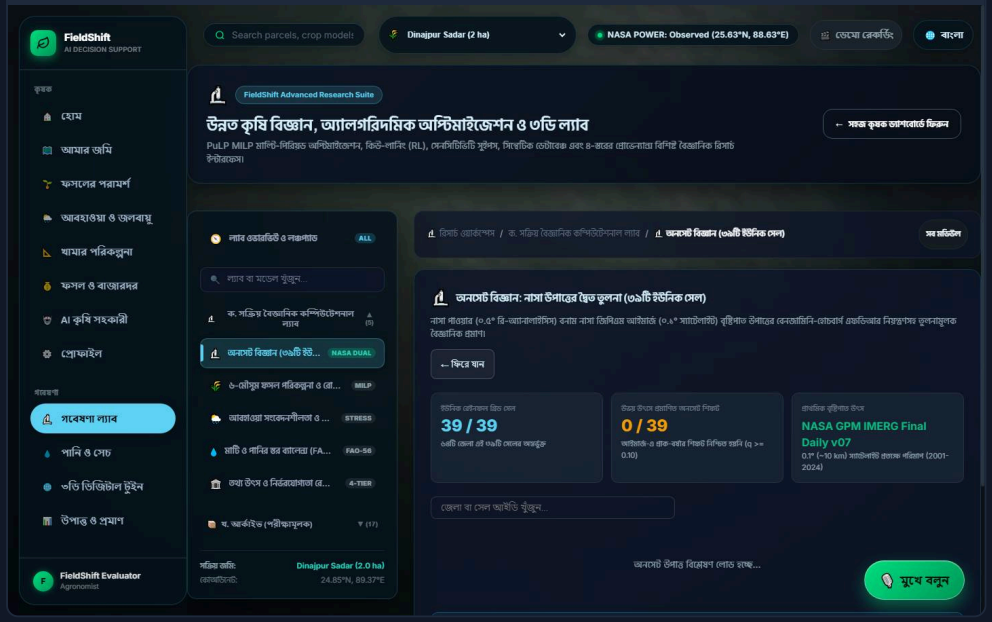
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 32% \| 18% \| 112 mm \| 228 mm (100%) \| 0.8 m | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** +18 mm \| +35 mm \| -4.5 mm \| -6.2 mm \| 102.3 mm / 112 mm |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

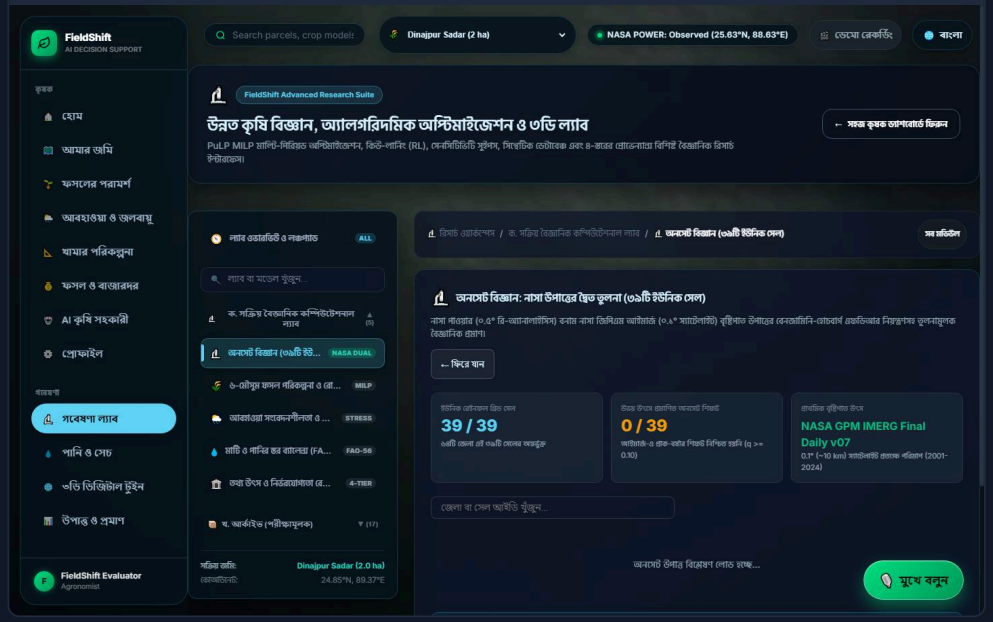
## 20. Research Lab - Onset Science (39 Unique Cells)

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

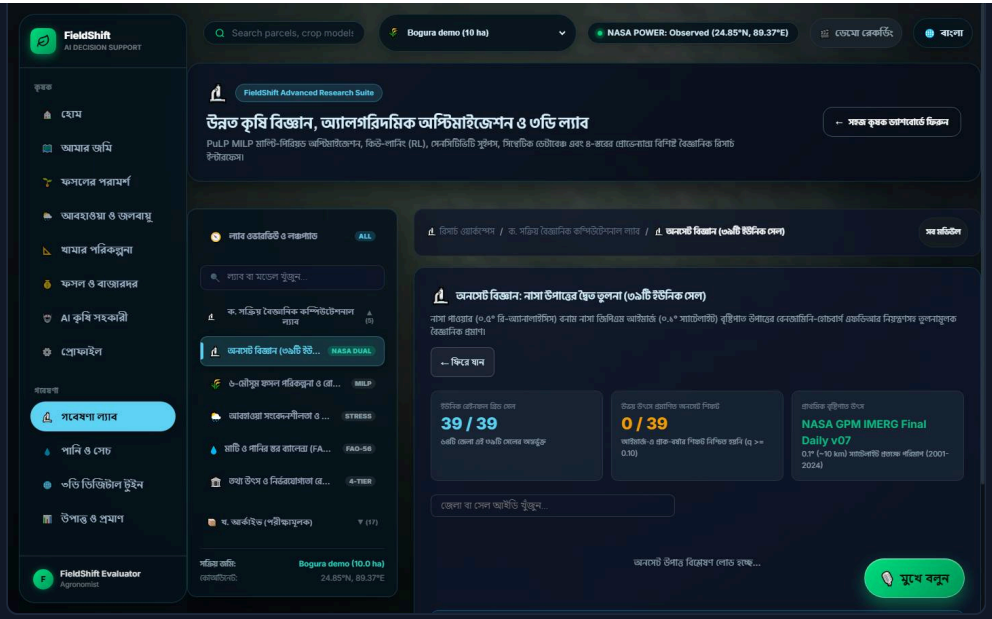


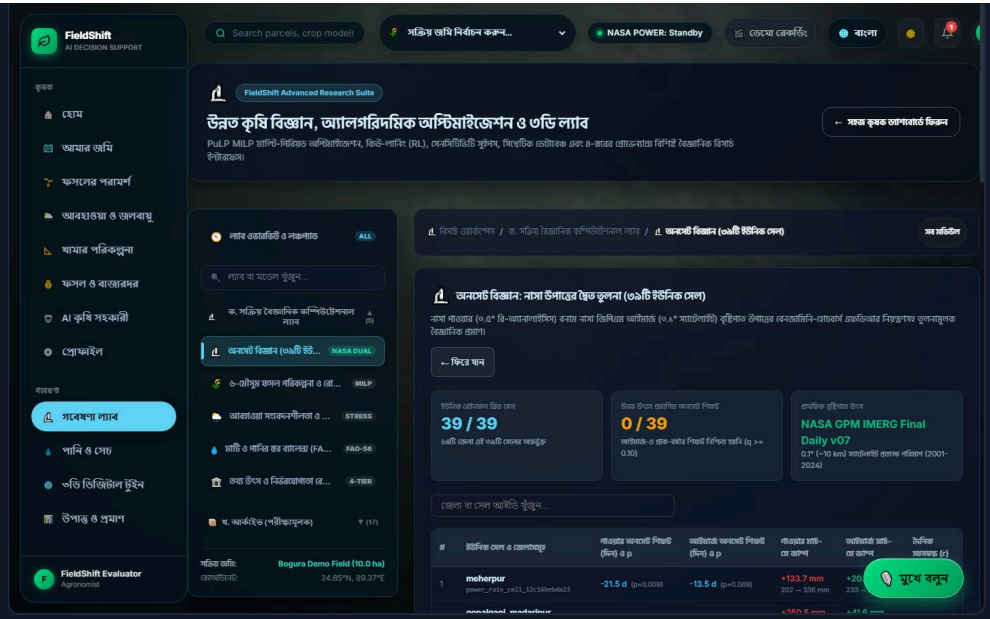

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



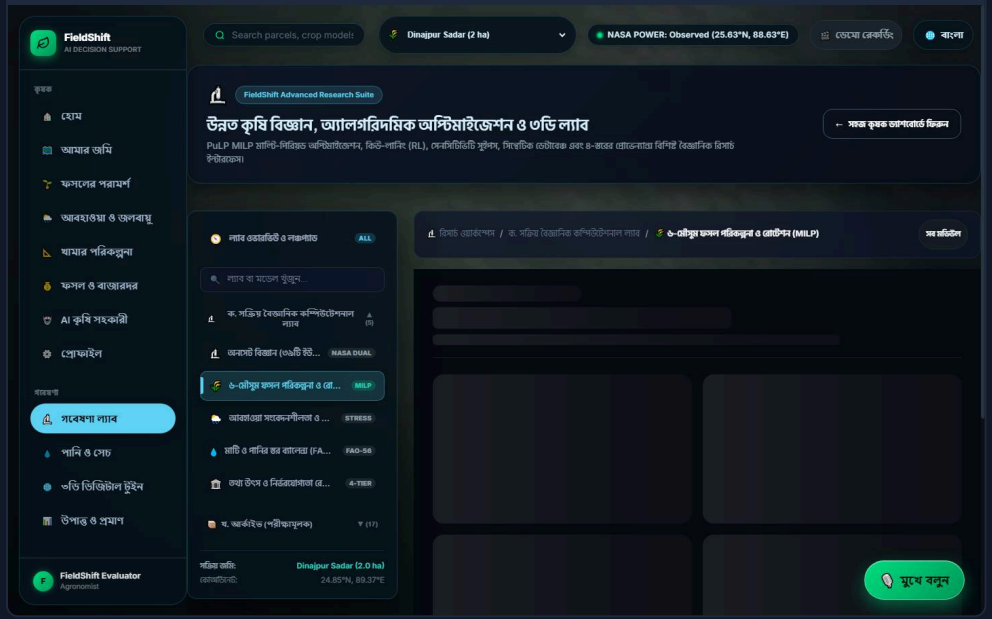

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 1 \| meherpur power_rain_cell_12c16bebda23 \| -21.5 d(p=0.009) \| -13.5 d(p=0.069) \| +133.7 mm 202 → 336 mm |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

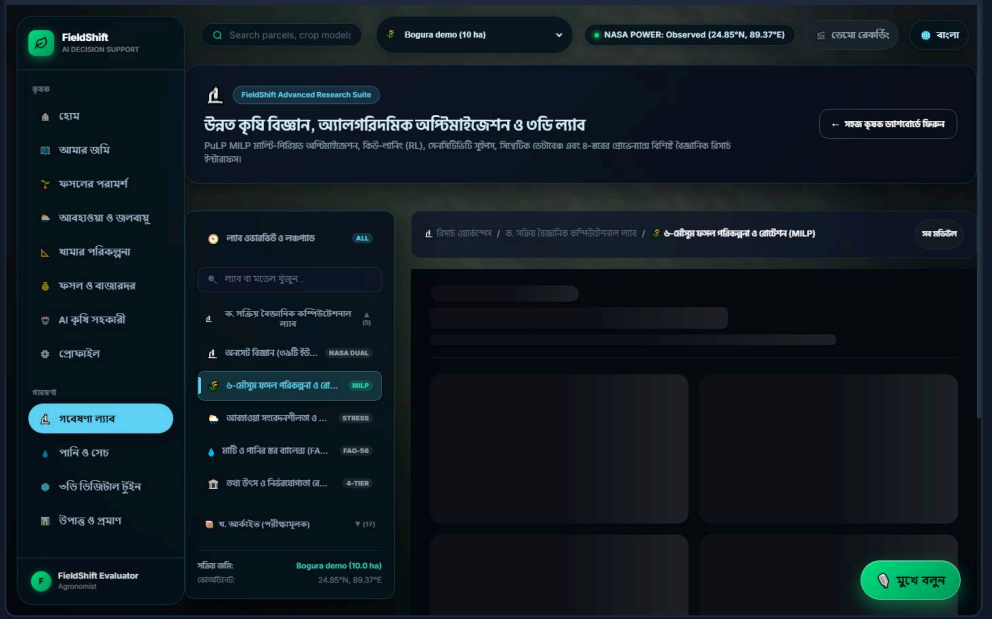
# 21. Research Lab - 6-Season MILP Lab

SAME

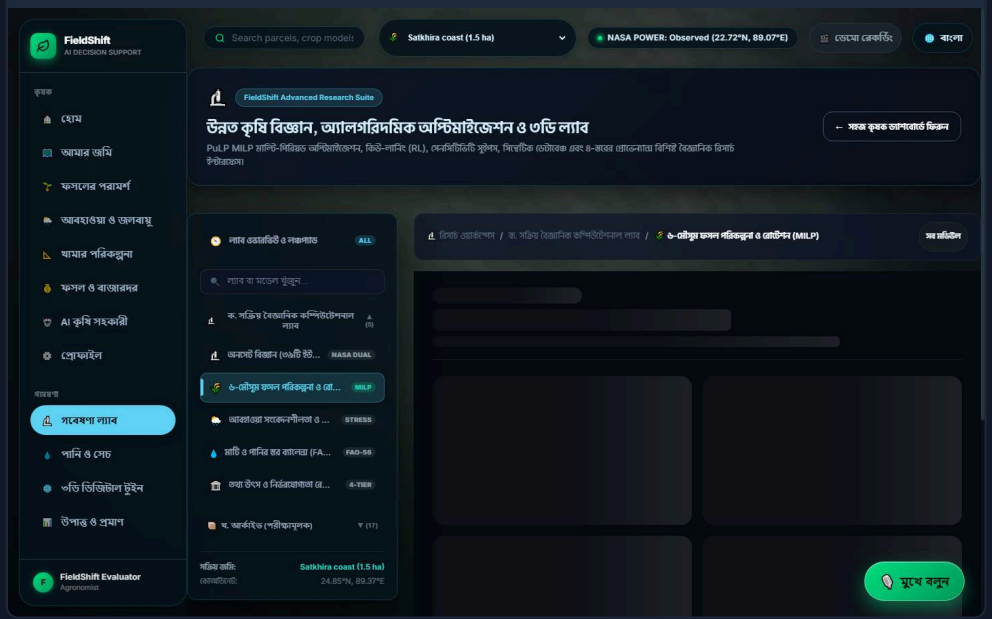
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



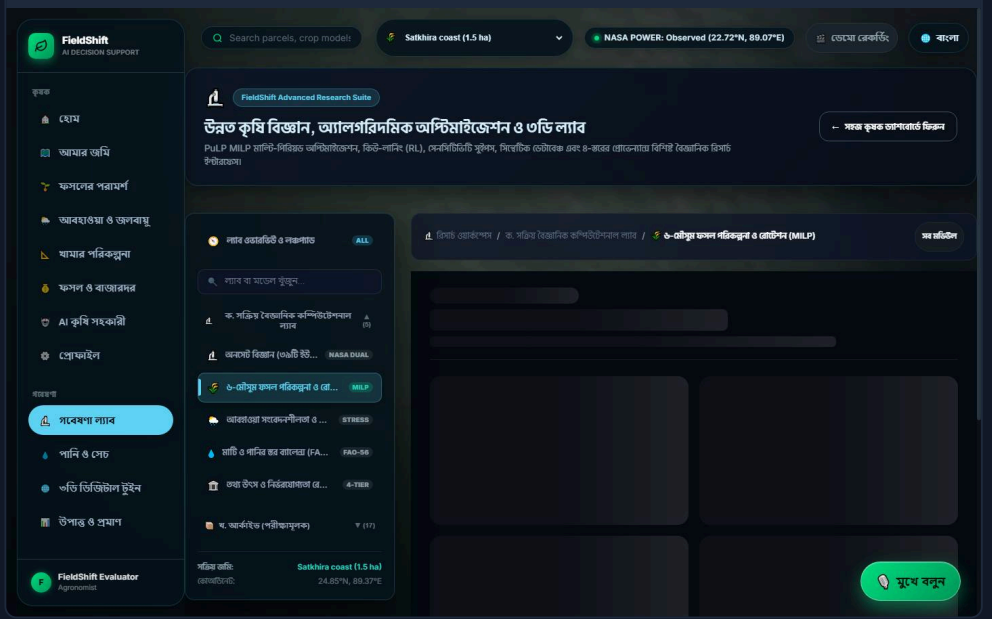

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



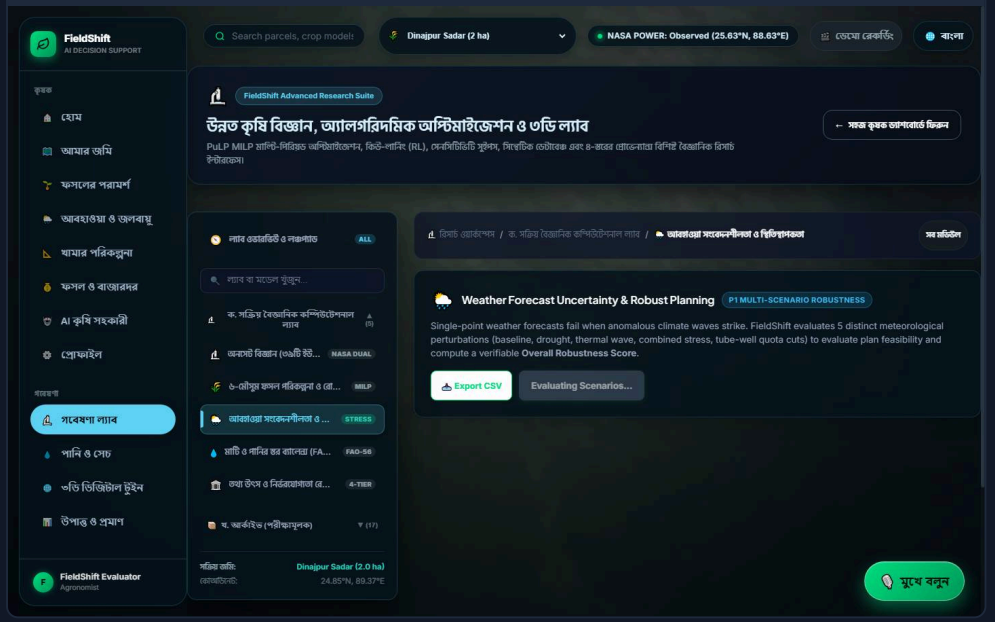
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

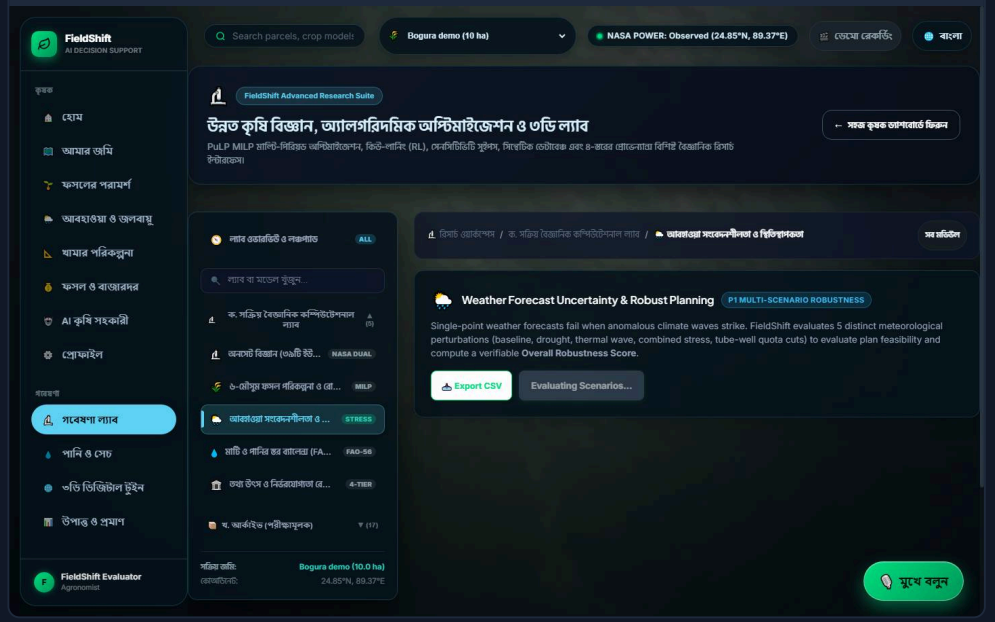
## 22. Research Lab - Weather Uncertainty

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

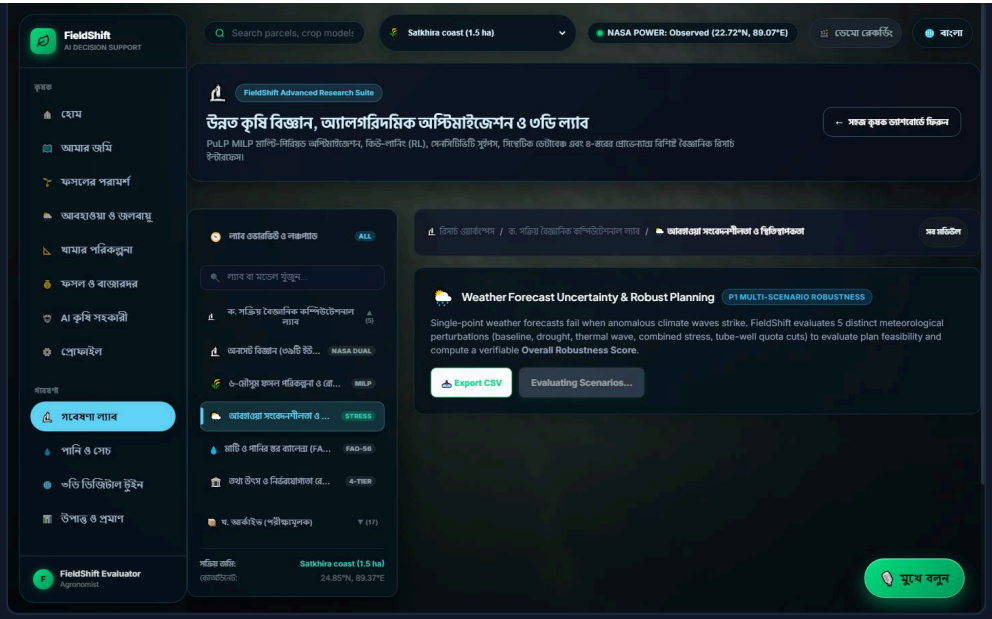


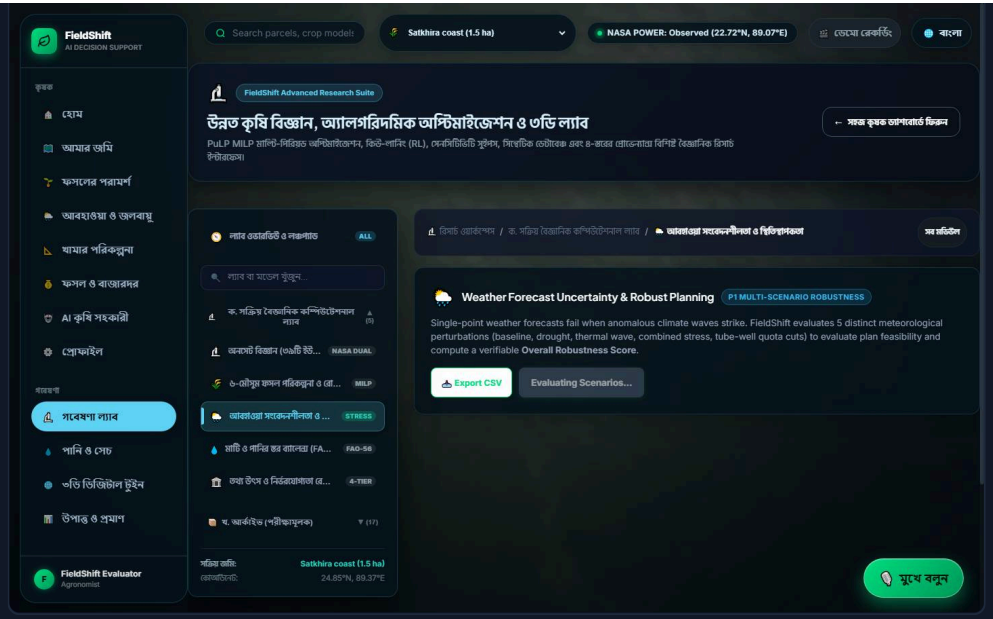

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



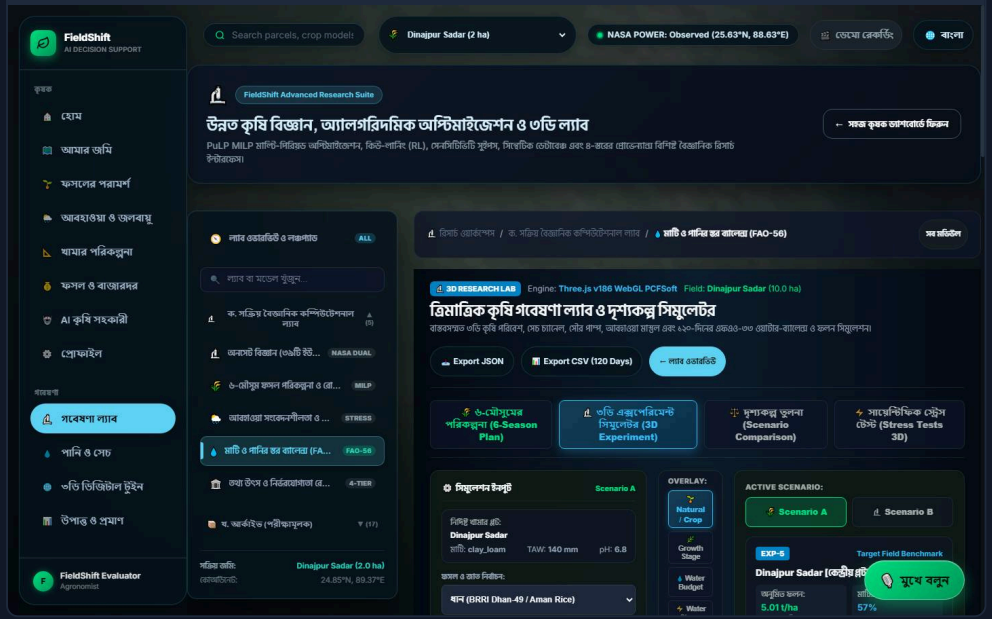

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

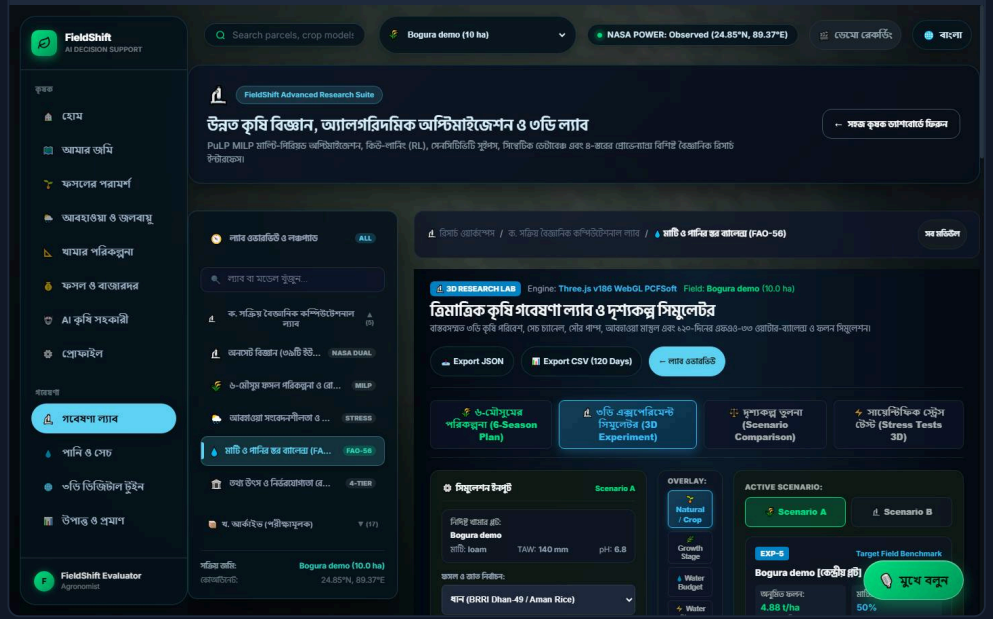
# 23. Research Lab - Soil & Water Hydrology FAO-56

DIFFERS

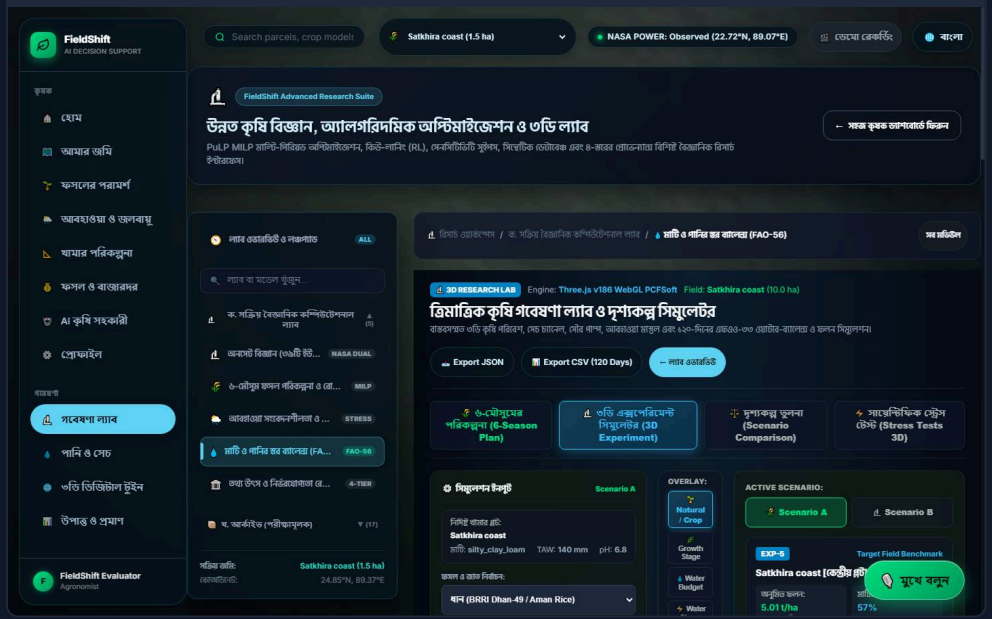
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



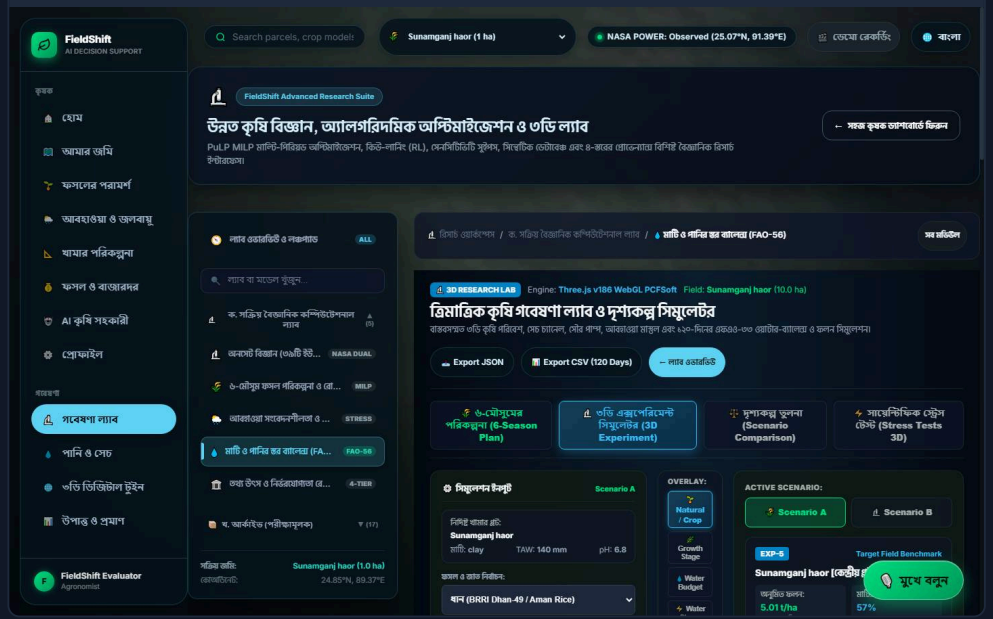

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



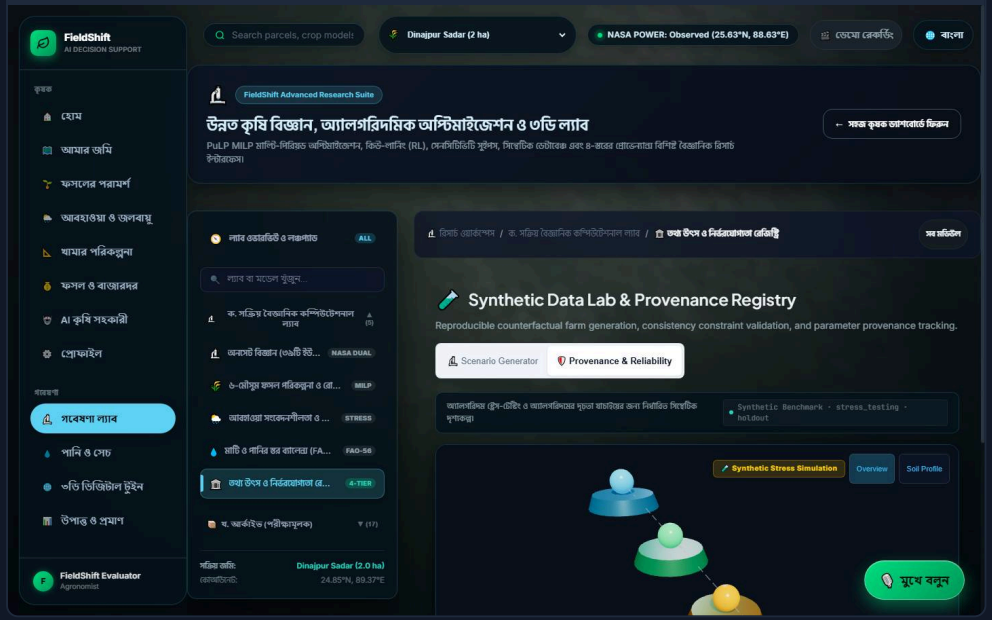
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 5.01 t/ha \| 561.3 mm | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 4.88 t/ha \| 561.3 mm | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 5.01 t/ha \| 561.3 mm | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** Three.js v186 WebGL PCFSoft \| 140 mm \| 6.8 \| 5.01 t/ha \| 561.3 mm |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

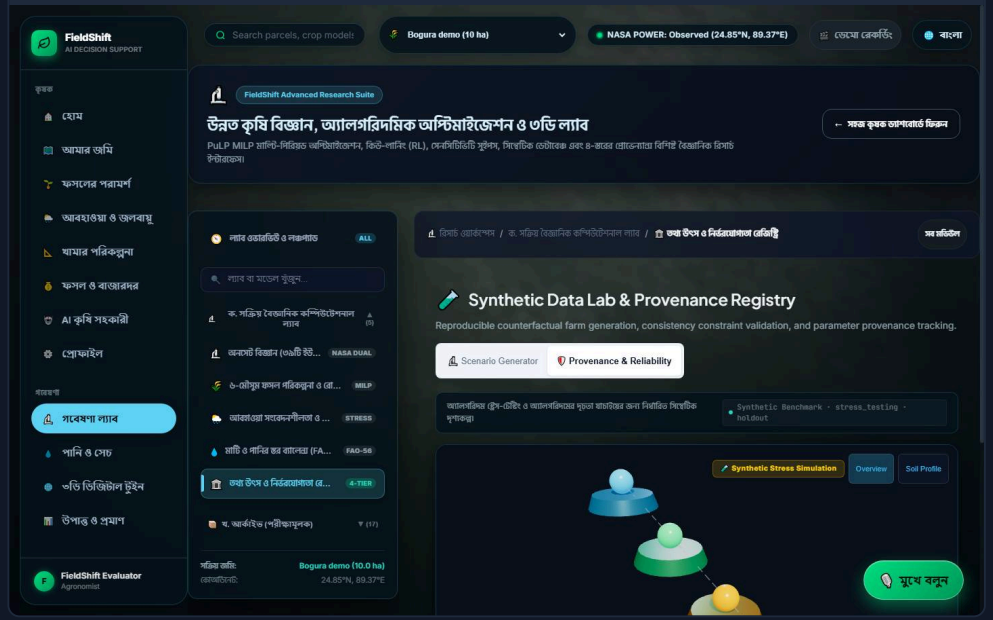
## 24. Research Lab - Data Provenance 4-Tier

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

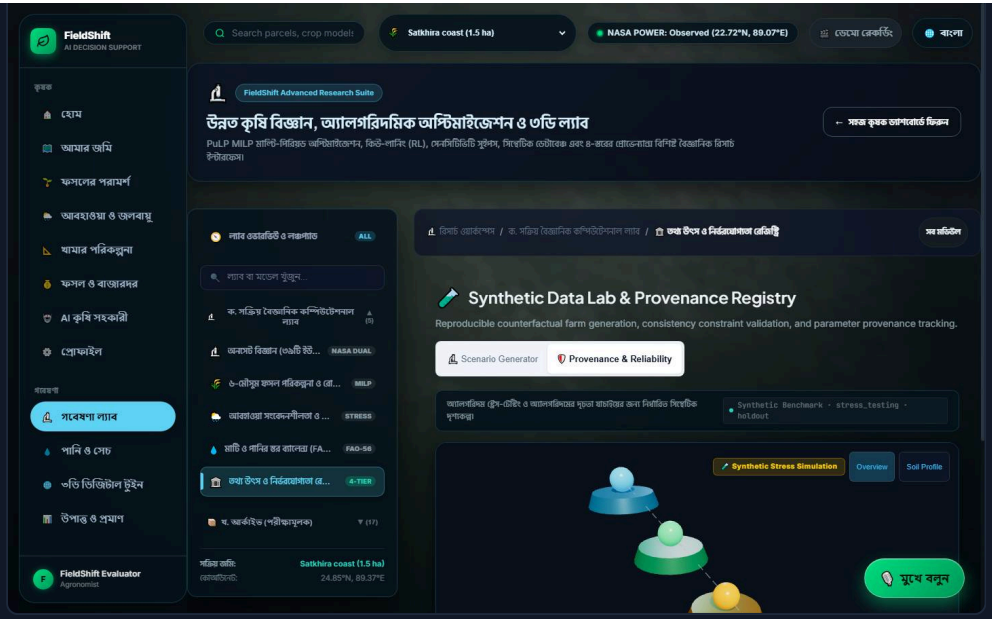


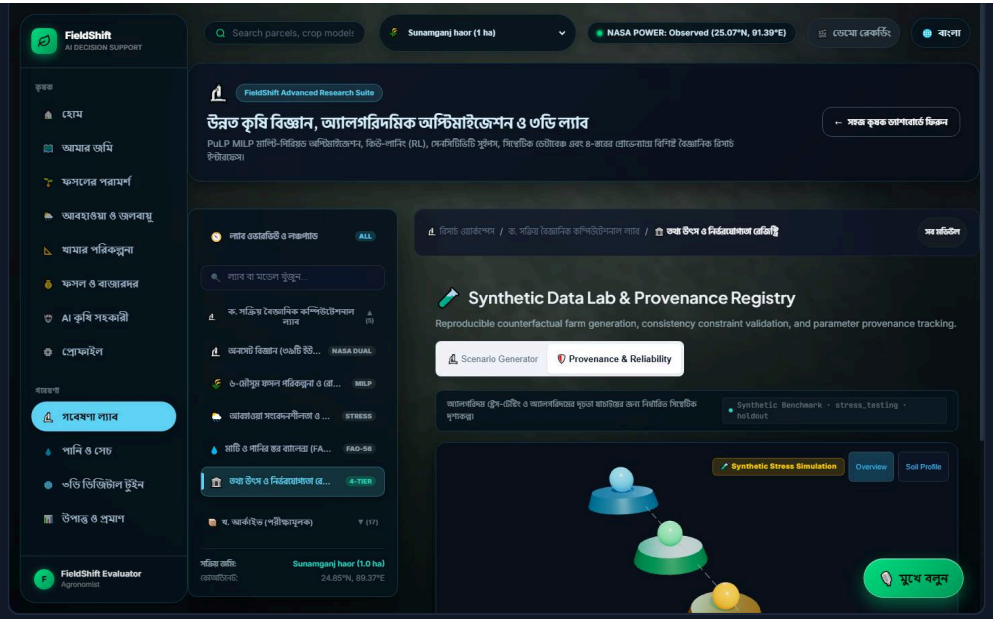

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



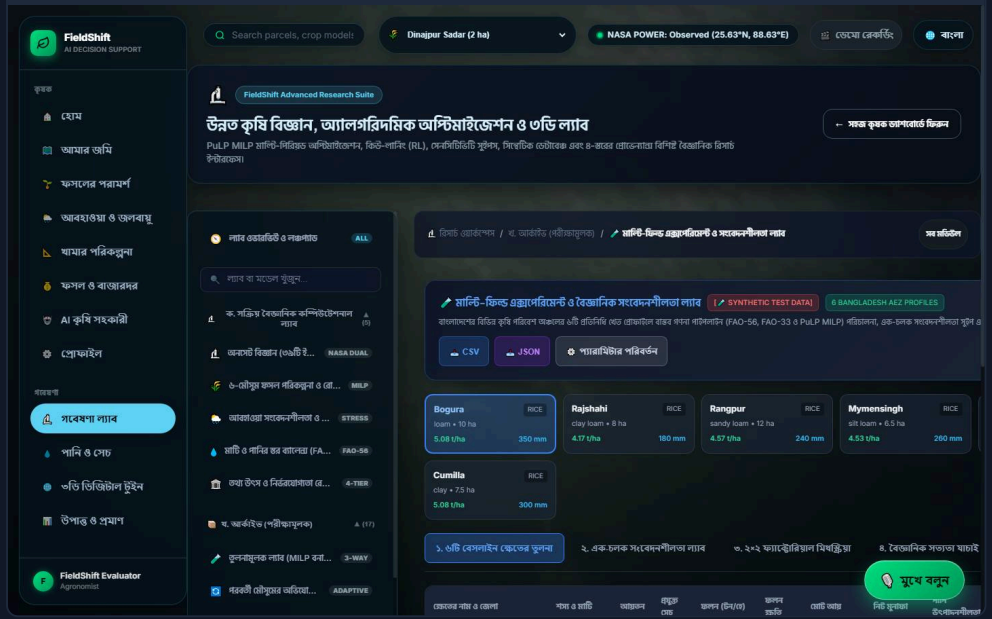

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

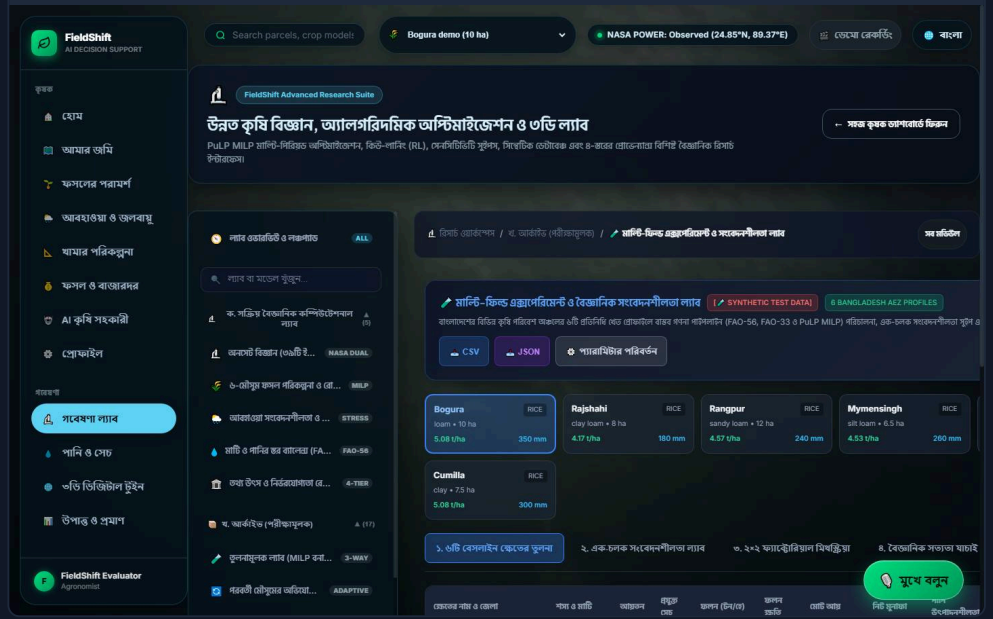
# 25. Research Lab Archive - Multi-Field Lab

SAME

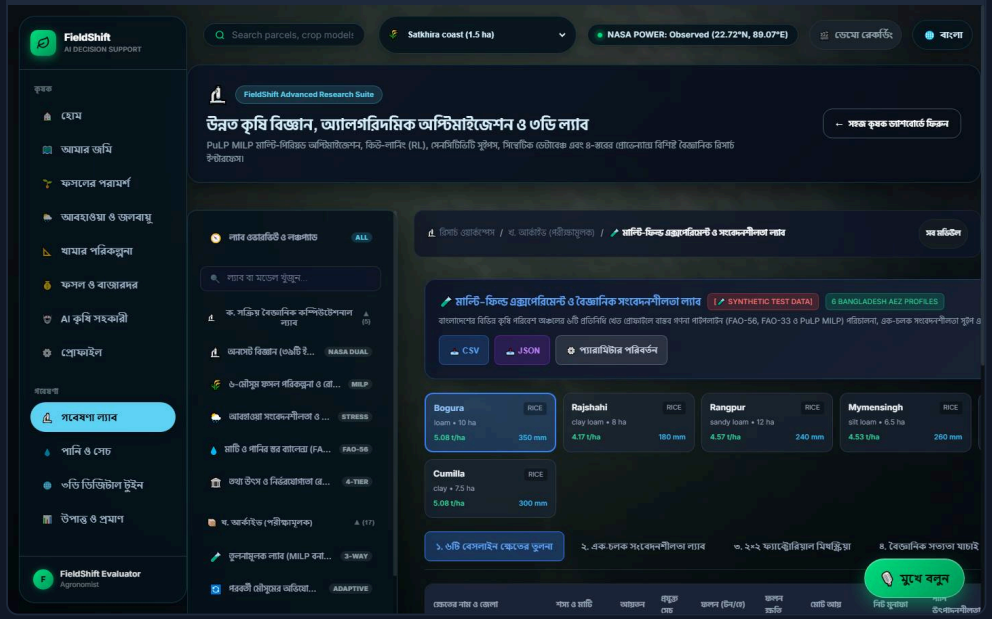
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



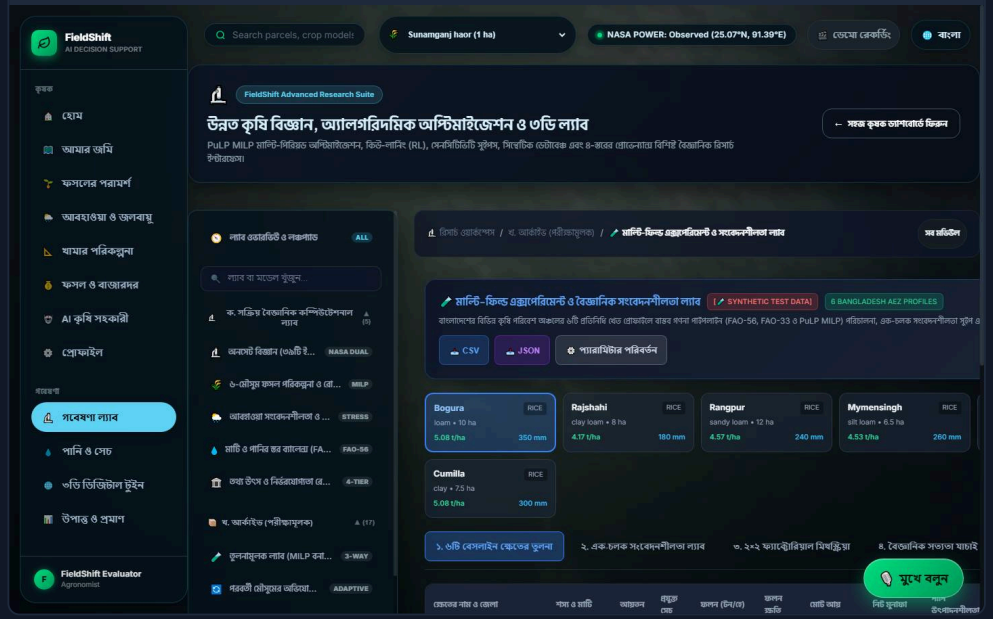

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



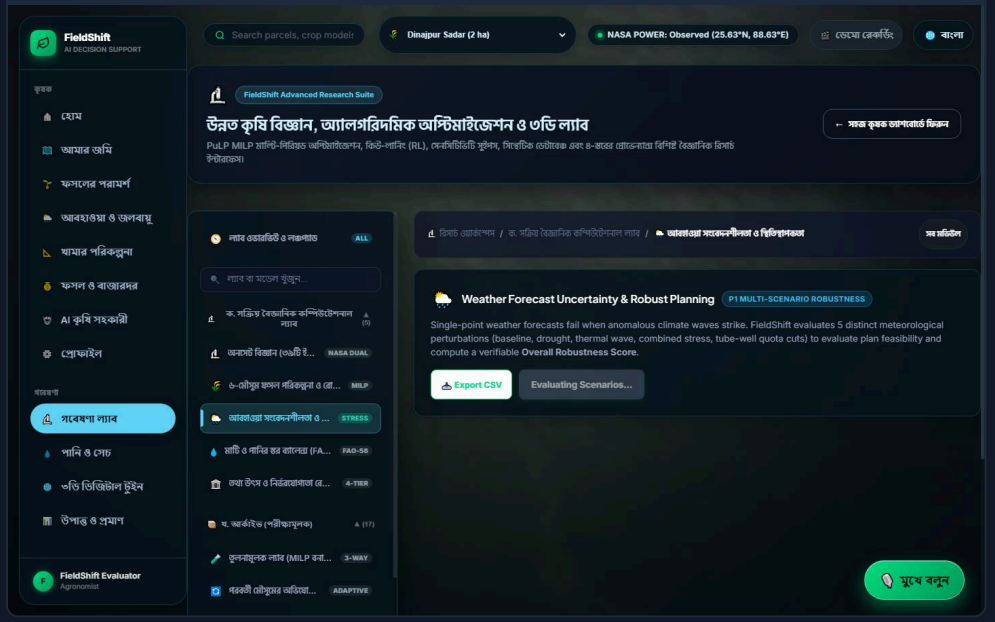
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 10 ha \| 350 mm \| 5.08 t/ha (50.8 t total) \| 7.7% \| 1,587,500 BDT | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 10 ha \| 350 mm \| 5.08 t/ha (50.8 t total) \| 7.7% \| 1,587,500 BDT | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 10 ha \| 350 mm \| 5.08 t/ha (50.8 t total) \| 7.7% \| 1,587,500 BDT | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 10 ha \| 350 mm \| 5.08 t/ha (50.8 t total) \| 7.7% \| 1,587,500 BDT |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

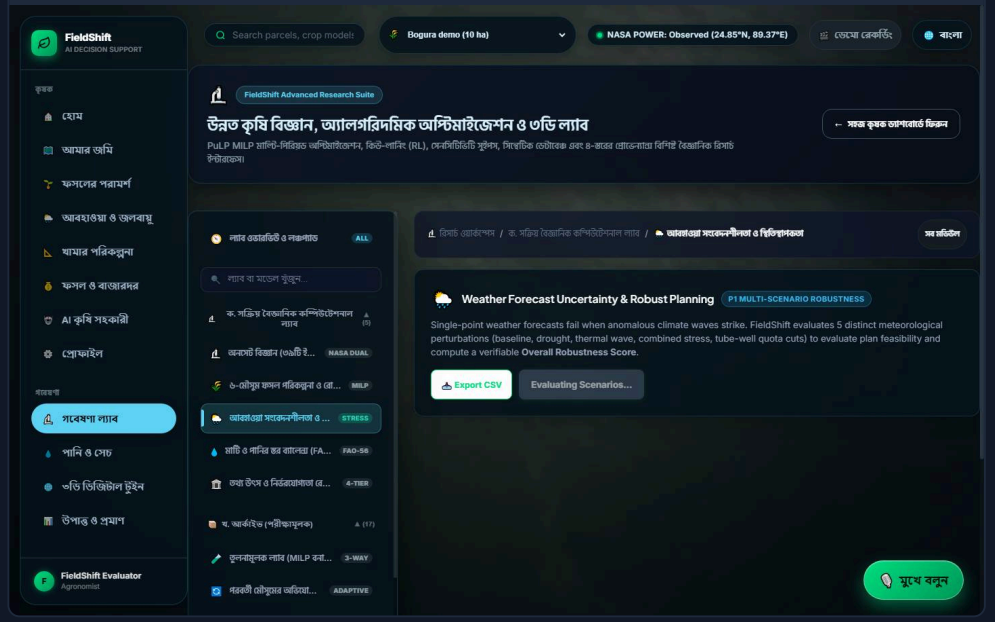
## 26. Research Lab Archive - Sensitivity Analysis

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

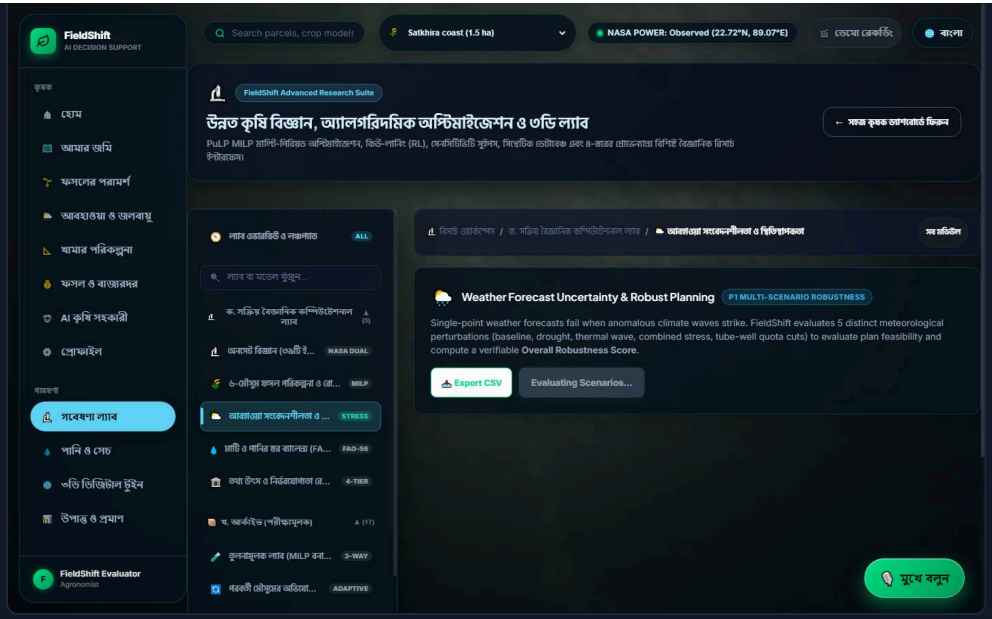


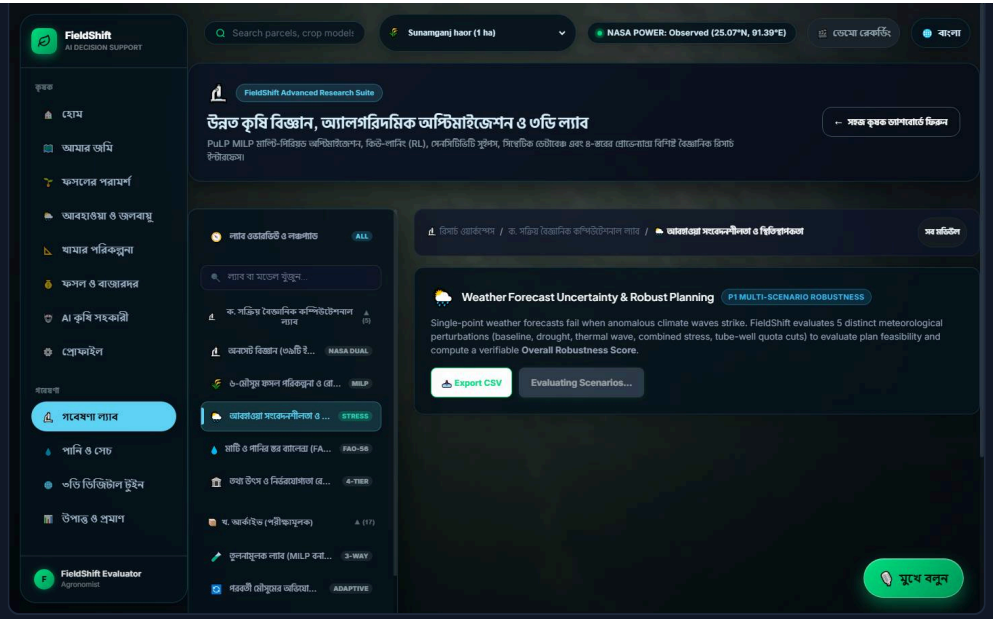

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



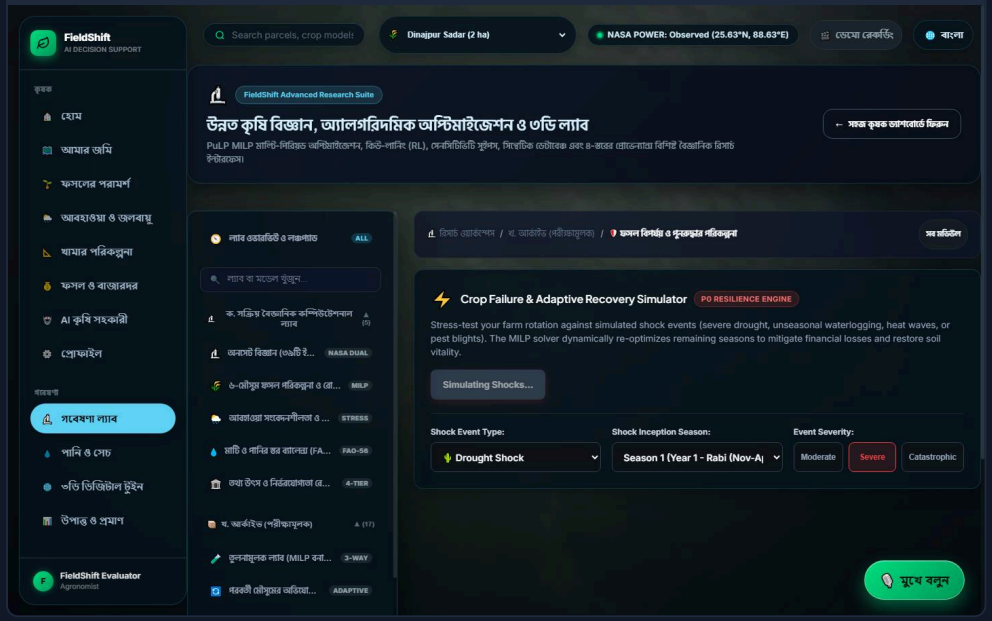

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

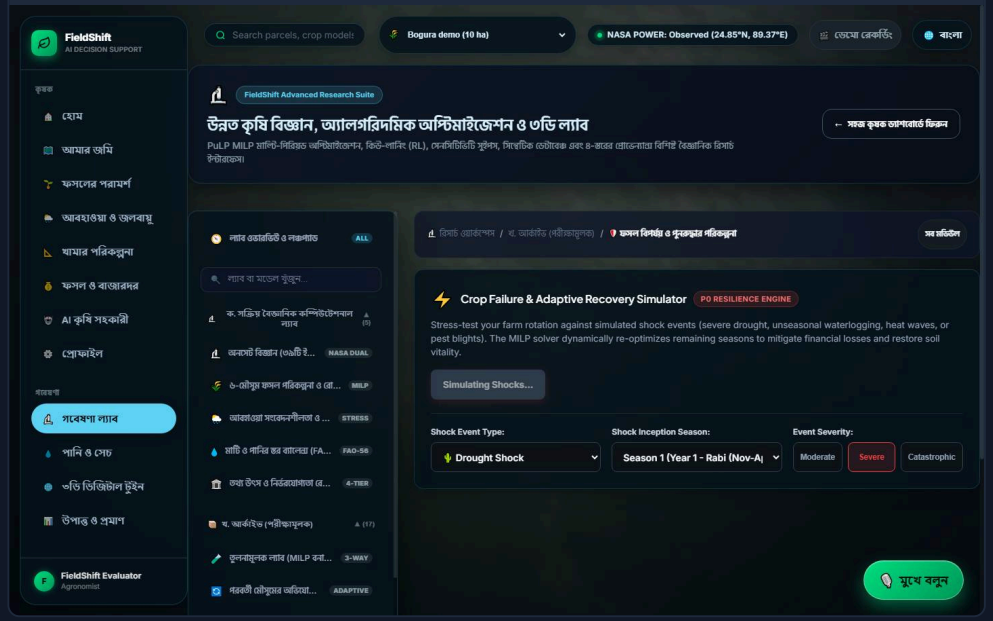
# 27. Research Lab Archive - Crop Failure Recovery

SAME

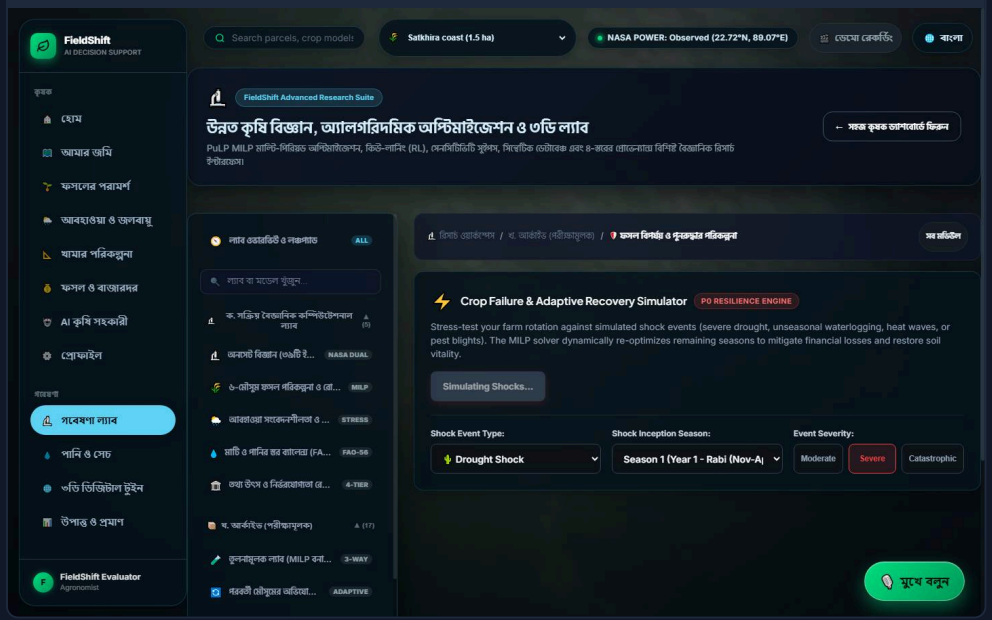
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



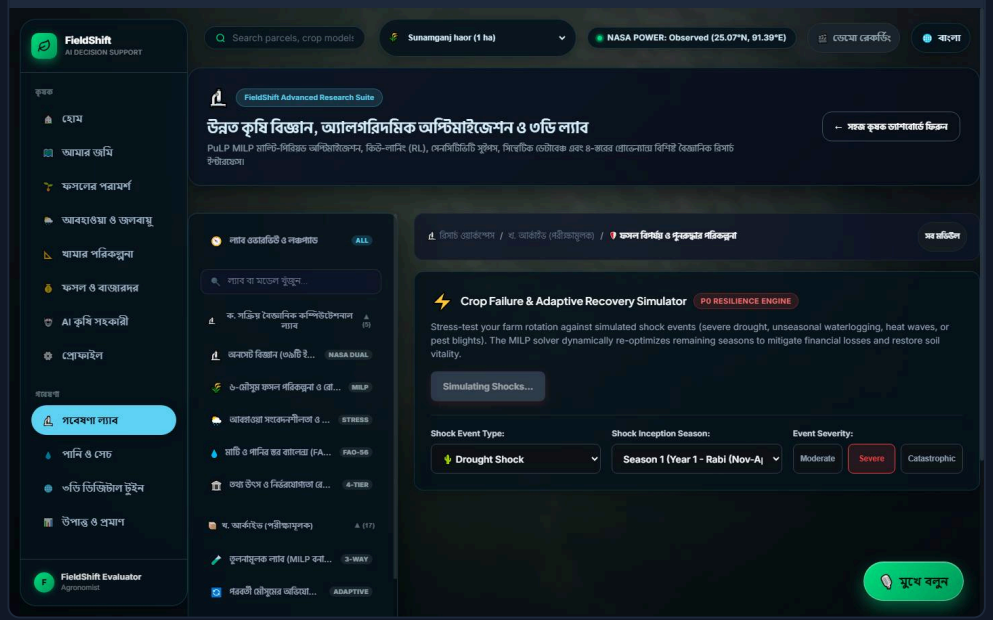

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



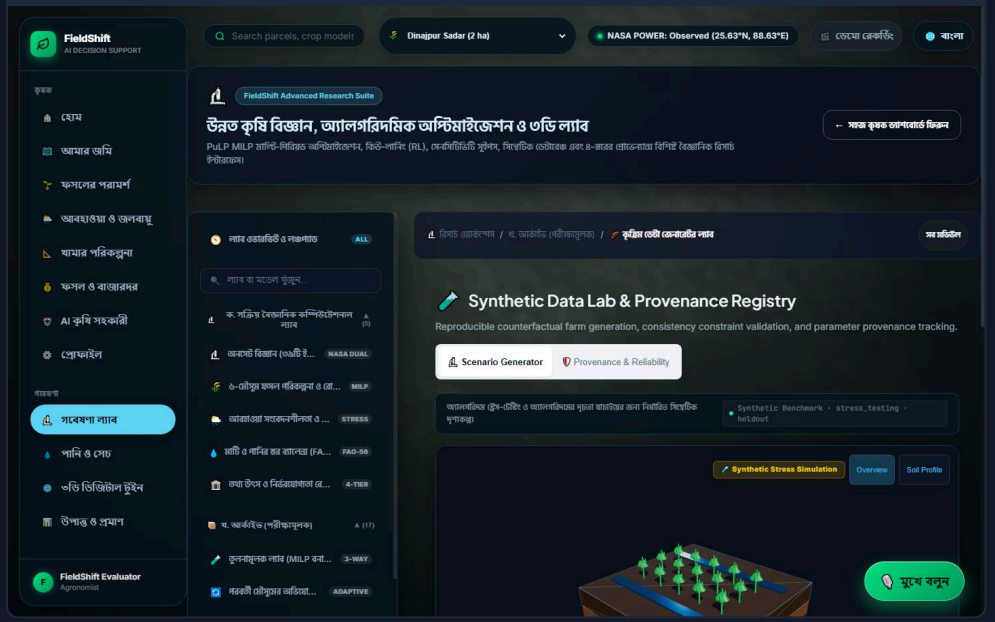
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

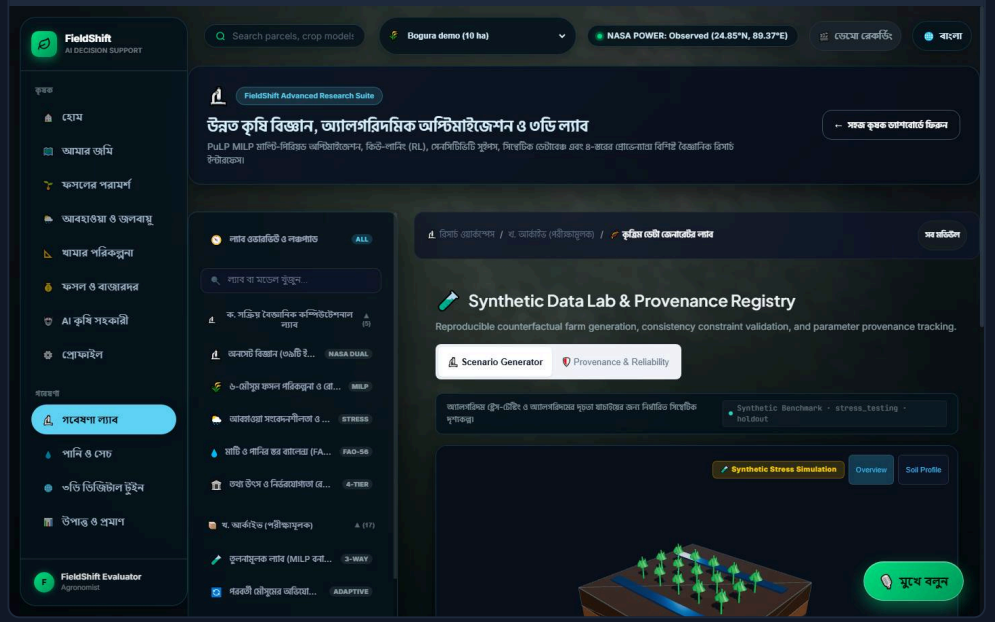
## 28. Research Lab Archive - Synthetic Data Lab

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

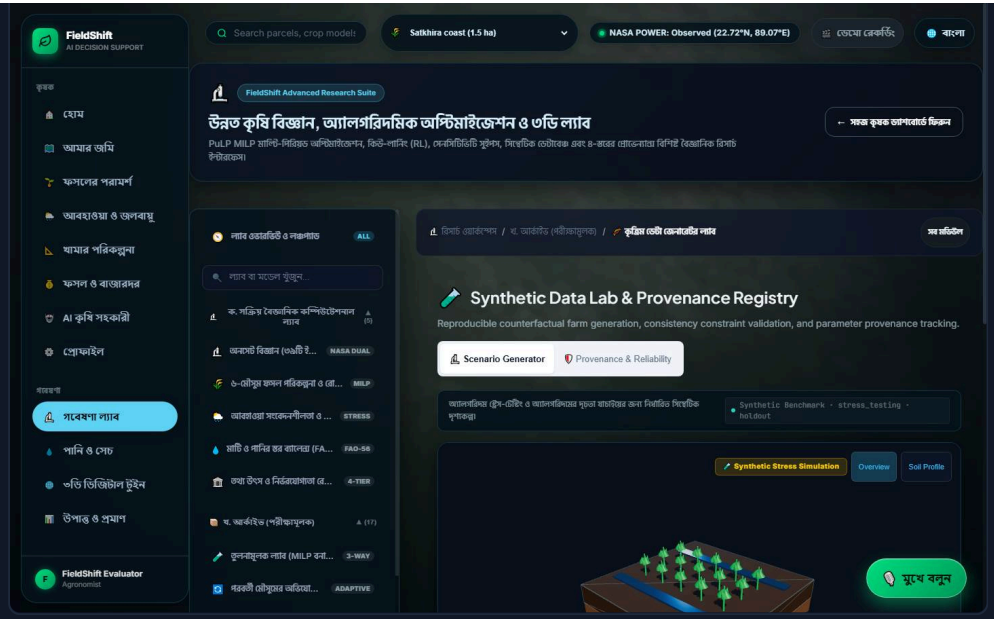


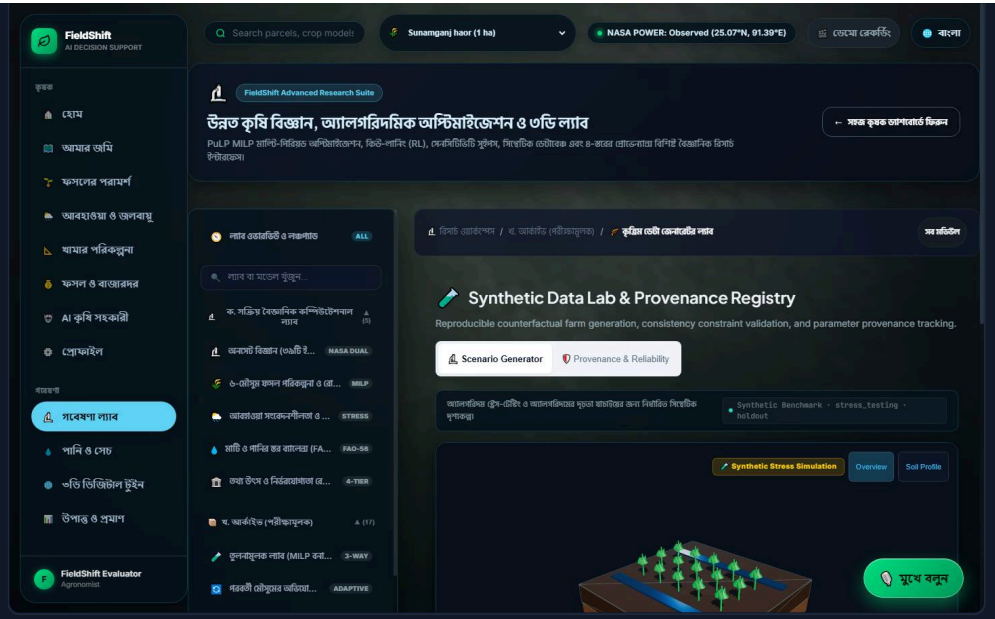

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



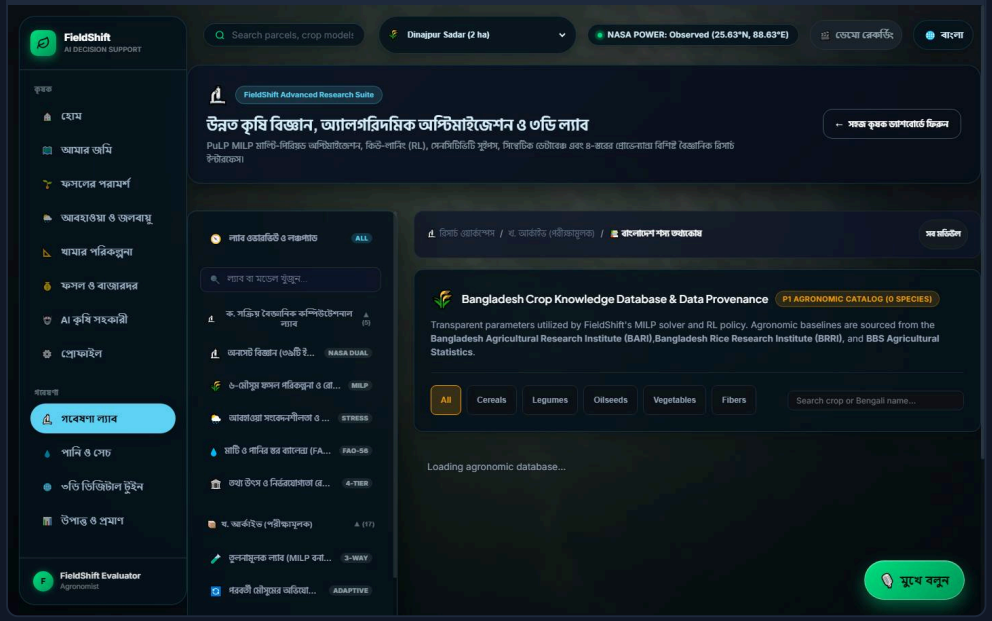

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

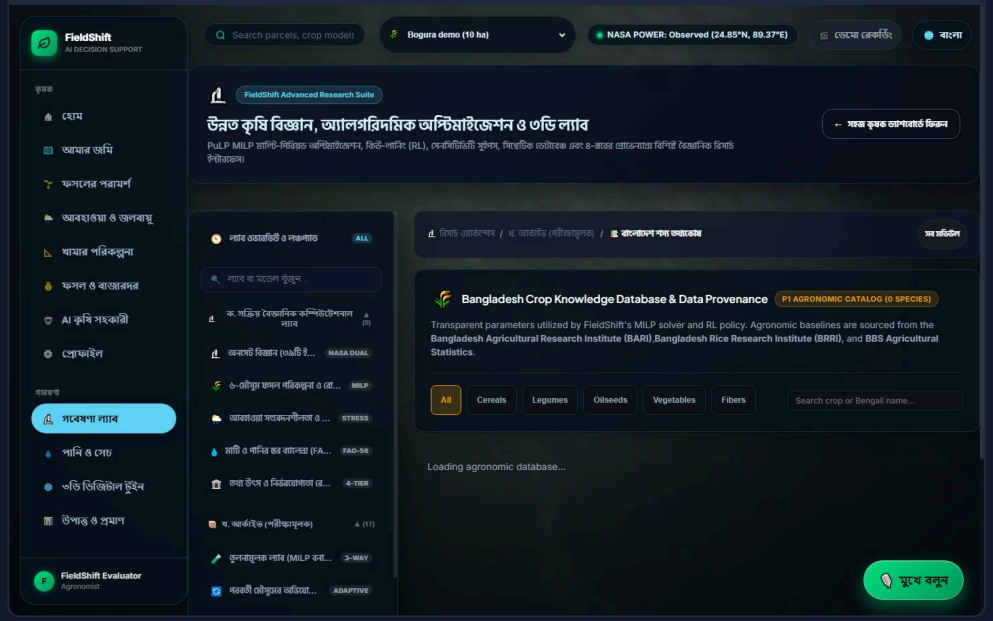
# 29. Research Lab Archive - Bangladesh Crop Database

SAME

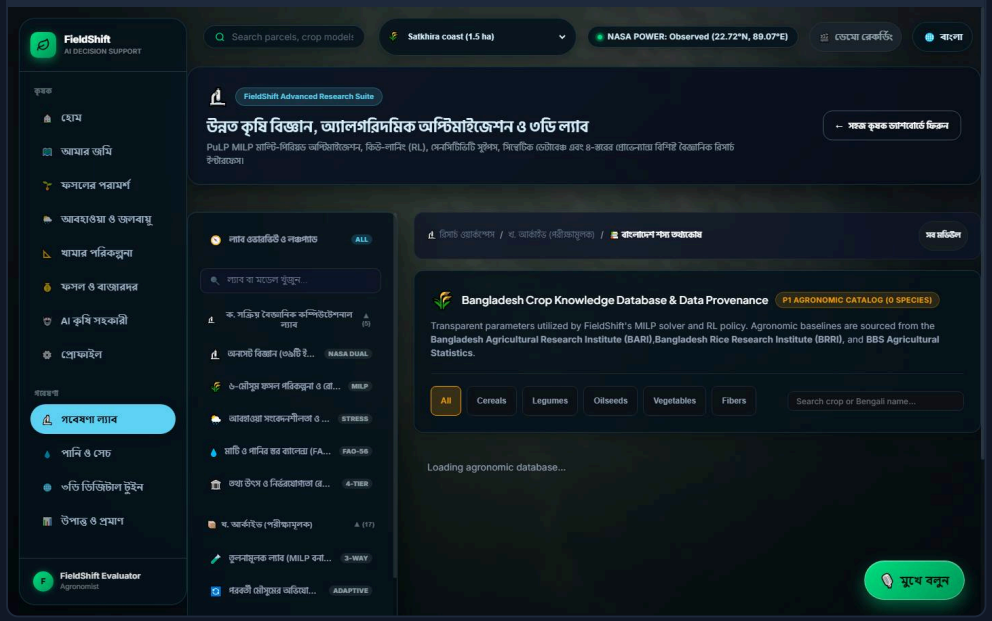
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



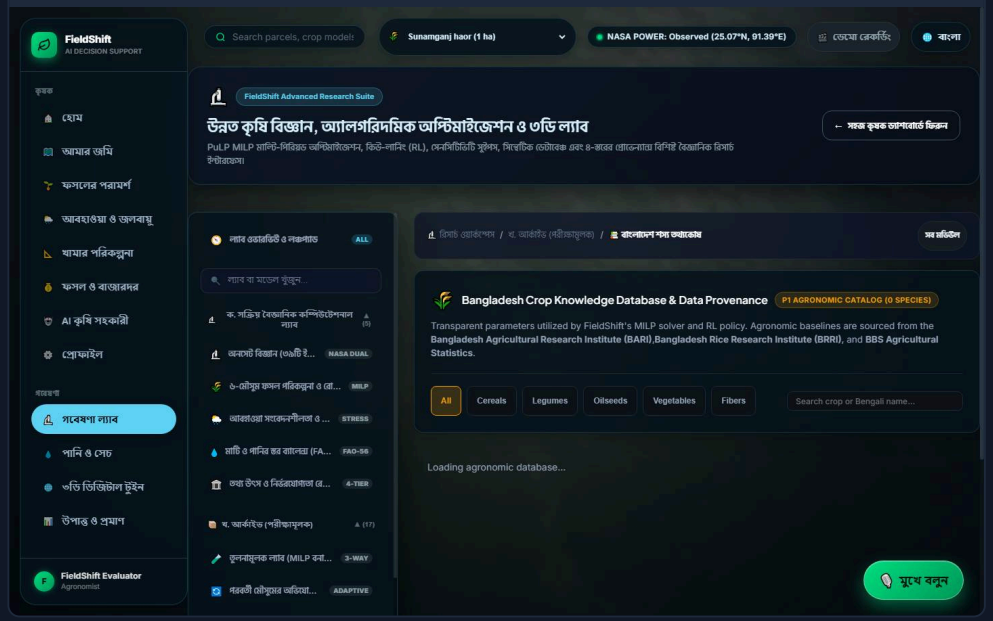

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



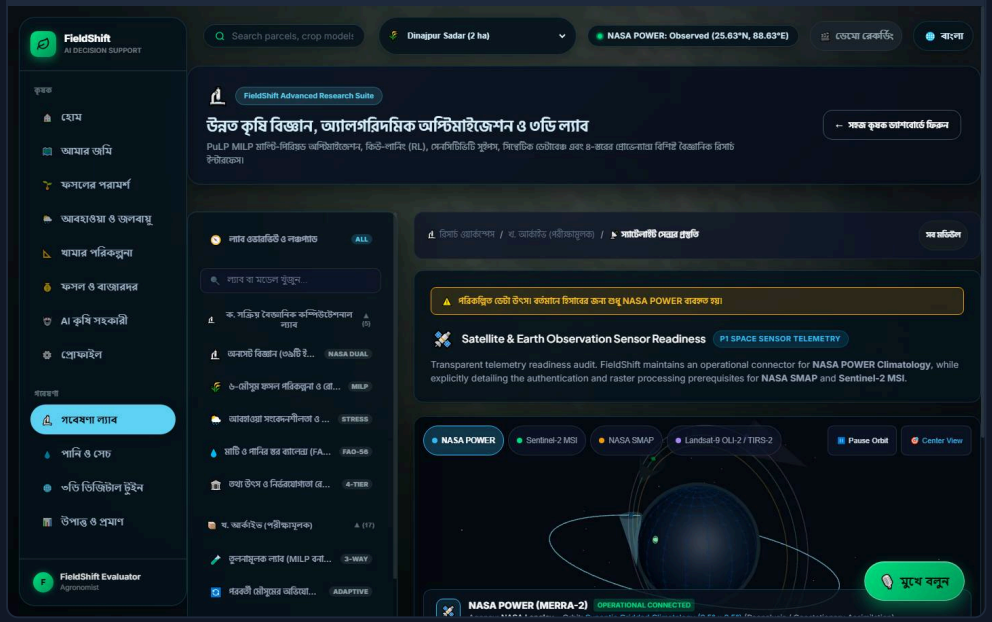
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

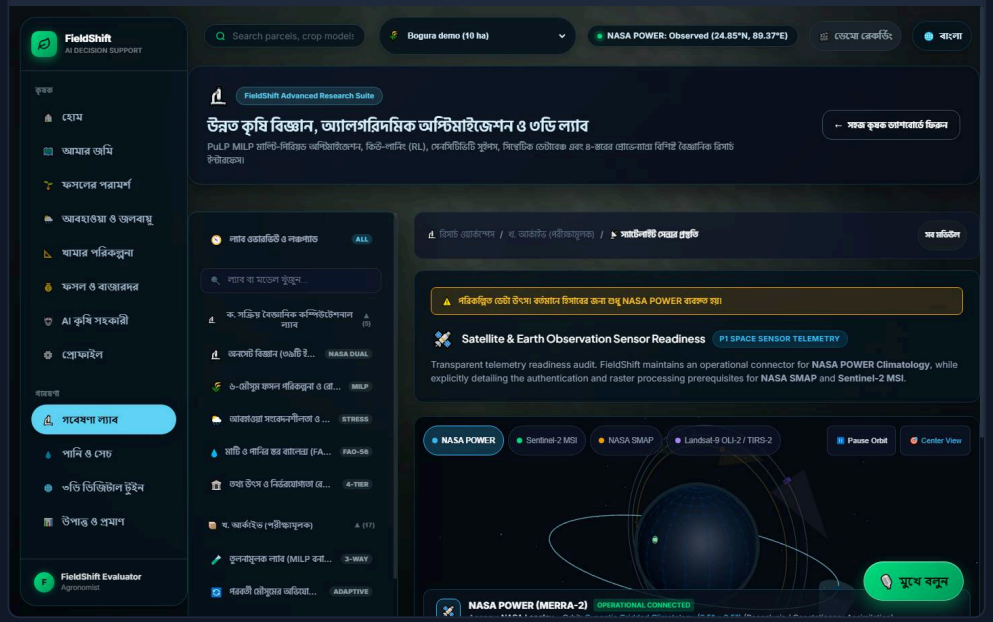
## 30. Research Lab Archive - Satellite Sensor Readiness

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

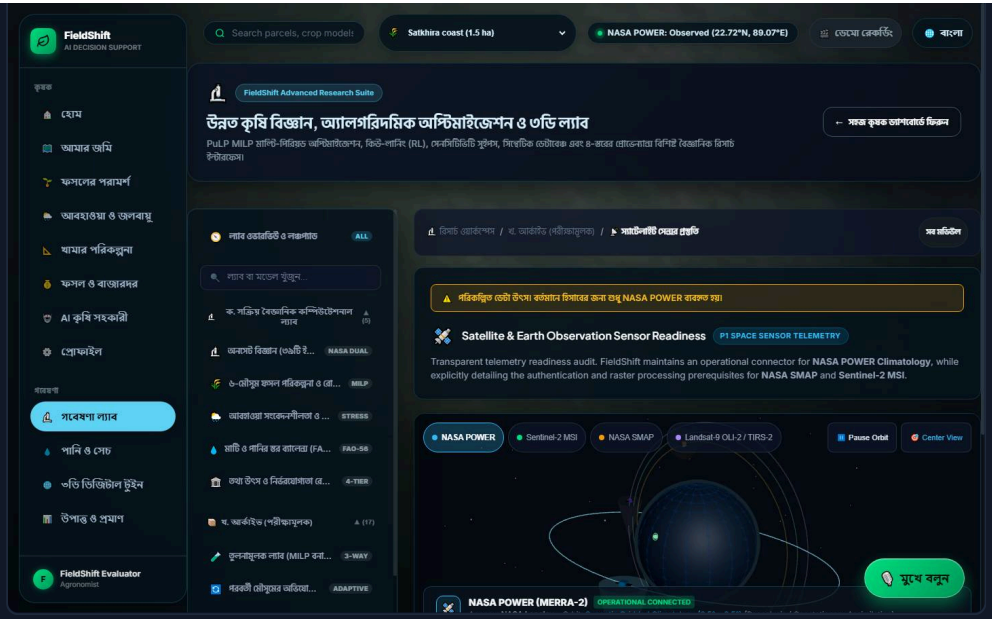


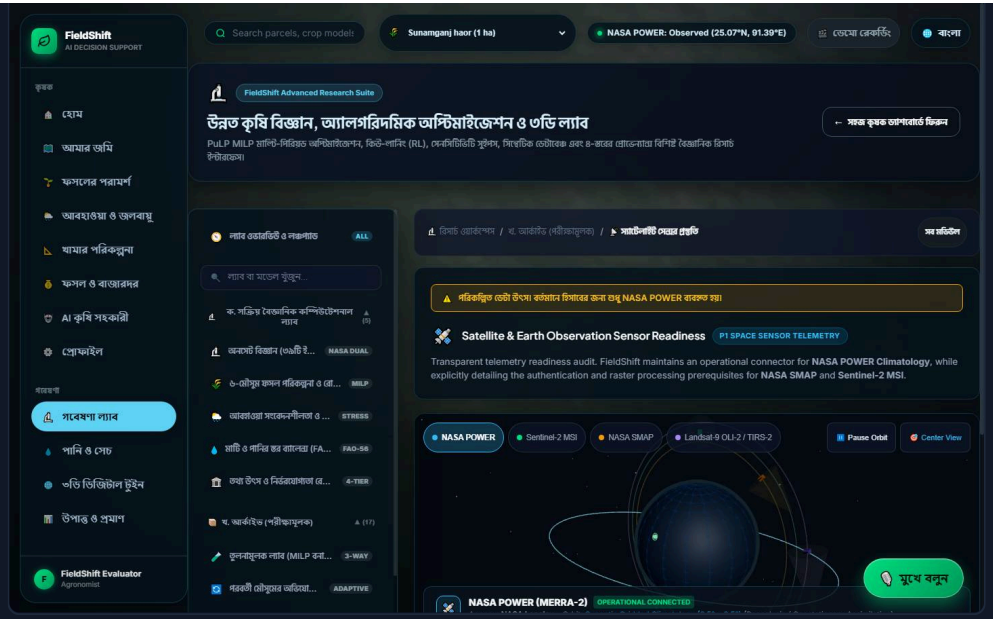

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



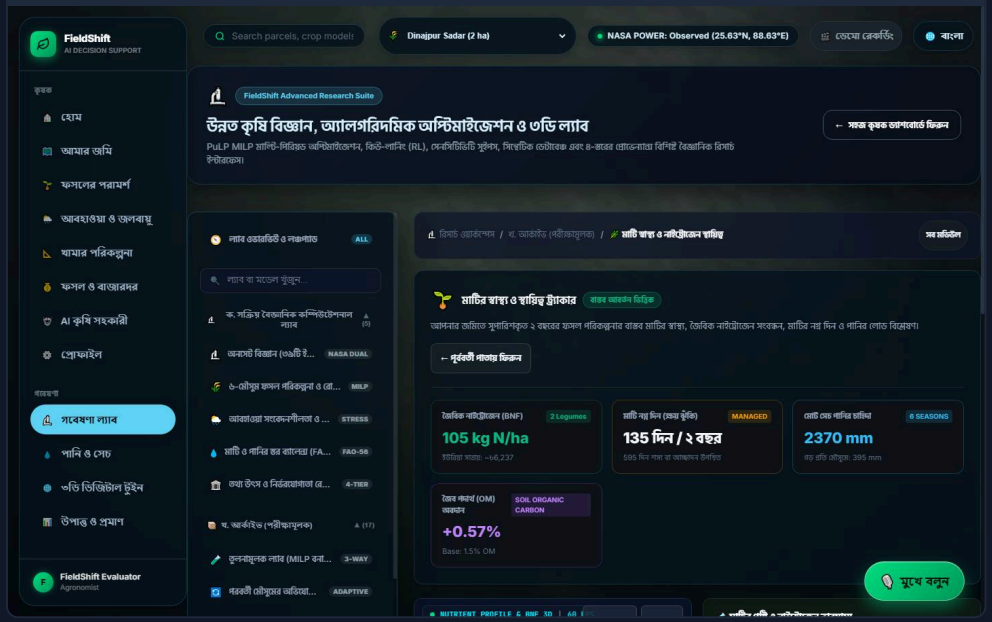

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** Sentinel-2 MSI | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** Sentinel-2 MSI | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** Sentinel-2 MSI | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** Sentinel-2 MSI |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

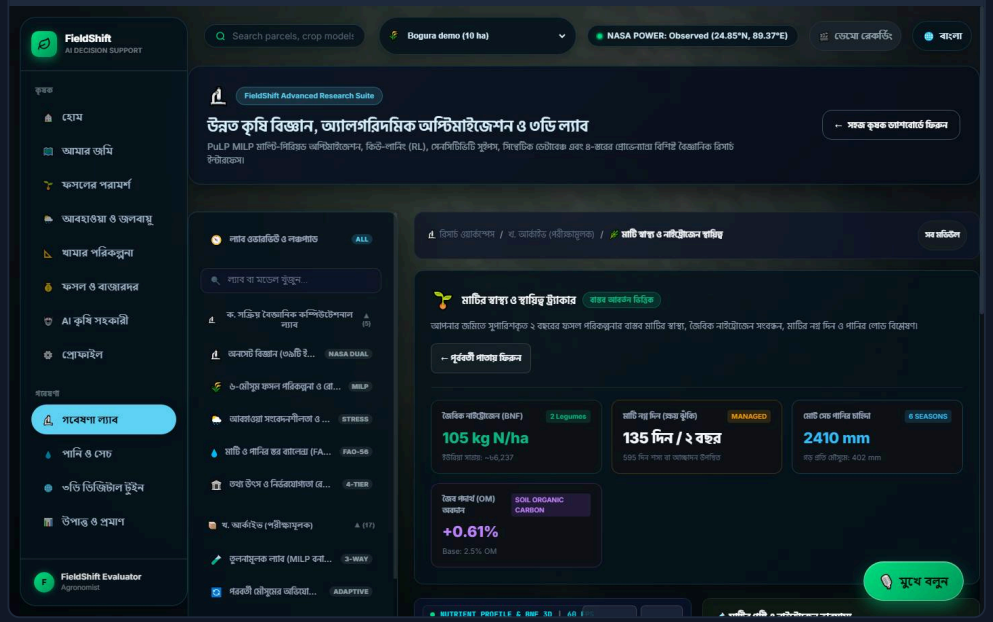
# 31. Research Lab Archive - Soil Health & Carbon

DIFFERS

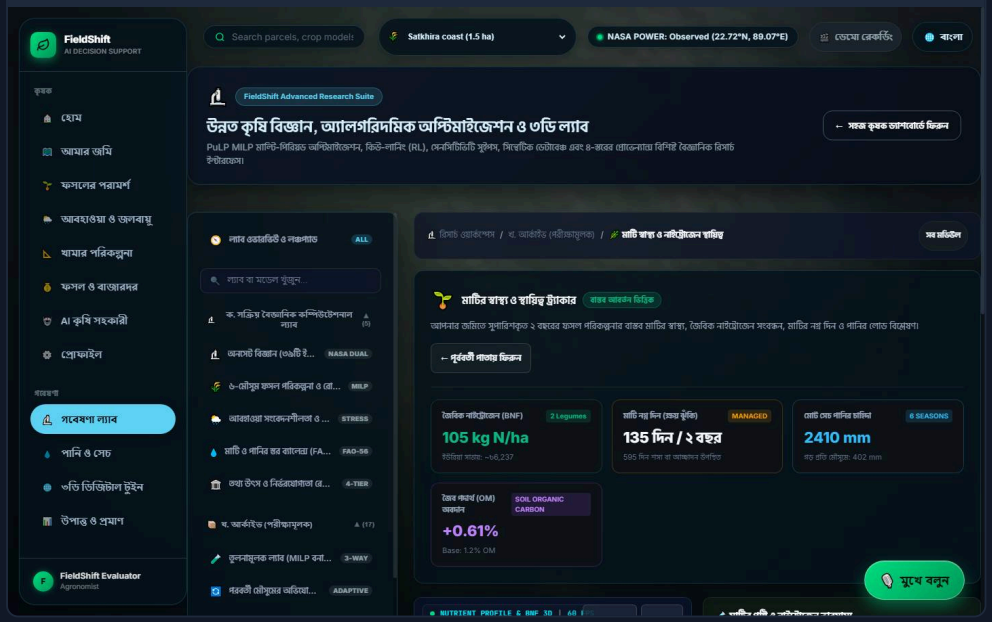
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



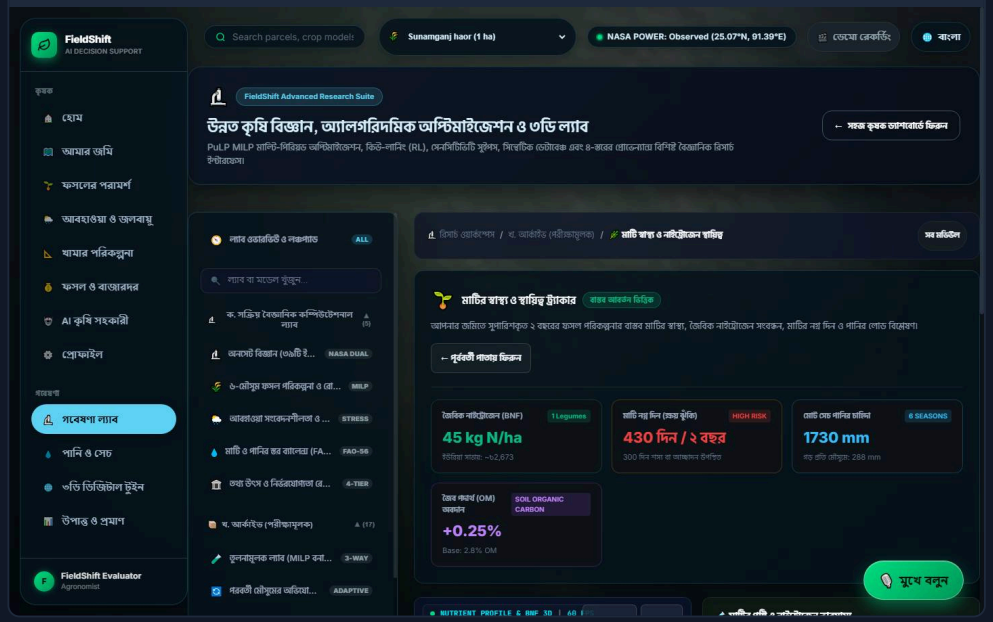

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



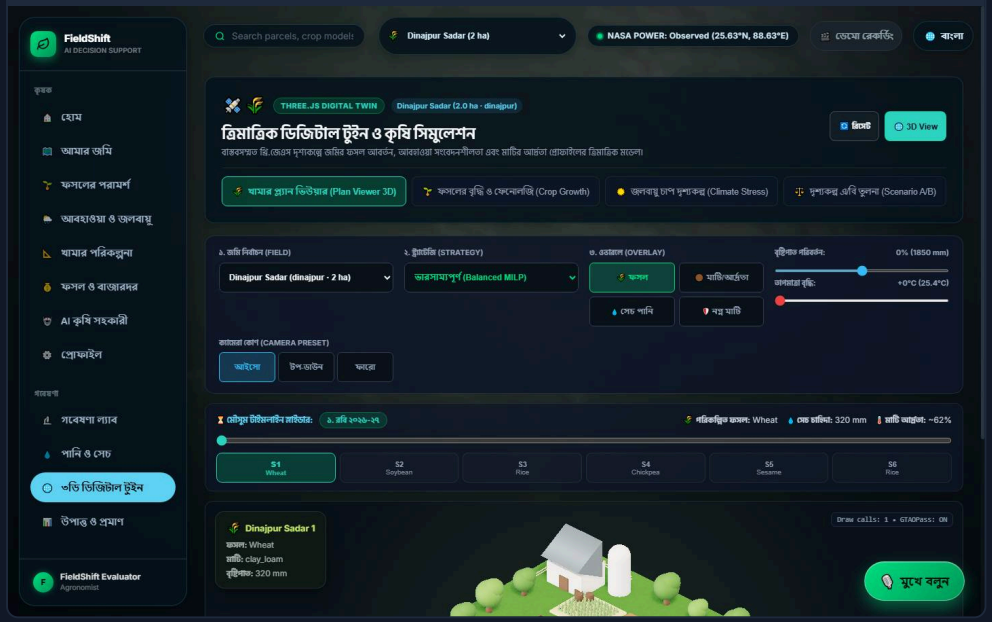
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 195 kg/ha \| 60 kg/ha \| 60 kg/ha \| 1.5% \| 5.8 | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 195 kg/ha \| 60 kg/ha \| 60 kg/ha \| 2.5% \| 6.5 | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 195 kg/ha \| 60 kg/ha \| 60 kg/ha \| 1.2% \| 7.4 | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 135 kg/ha \| 60 kg/ha \| 60 kg/ha \| 2.8% \| 6.2 |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

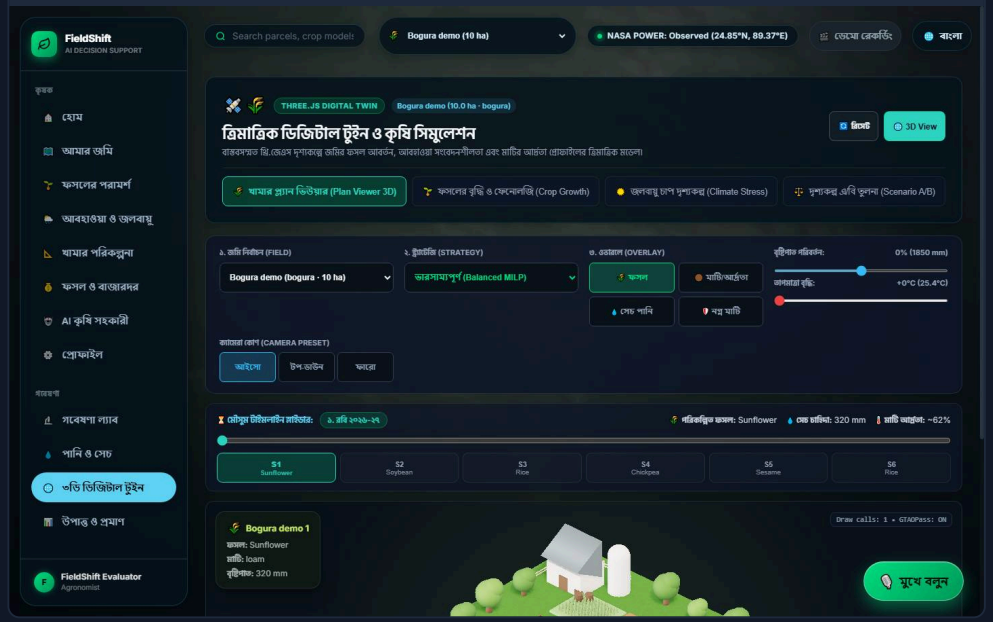
## 32. Digital Twin 3D - Farm Plan Diorama

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

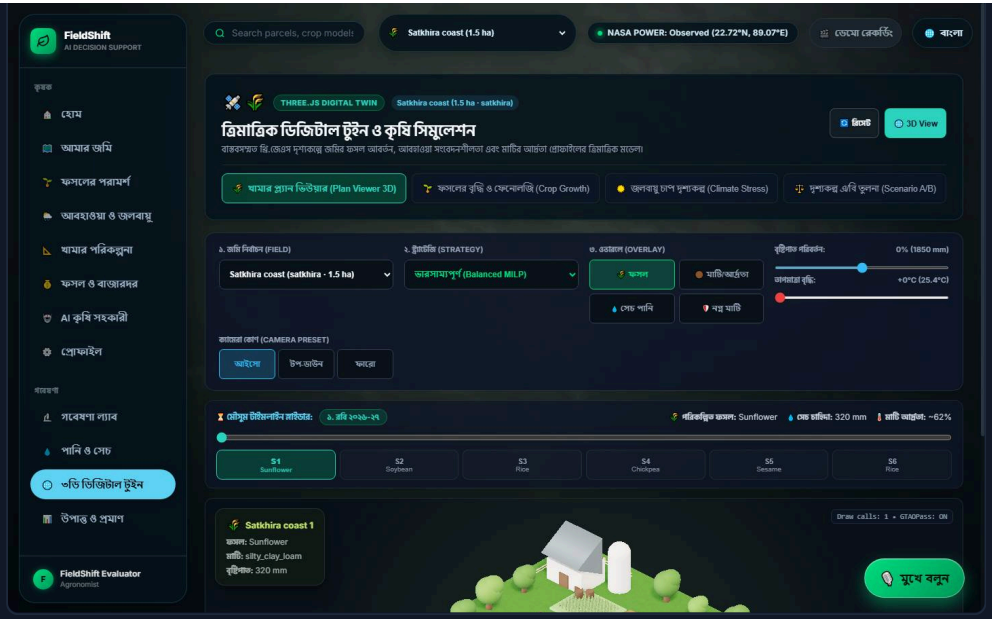


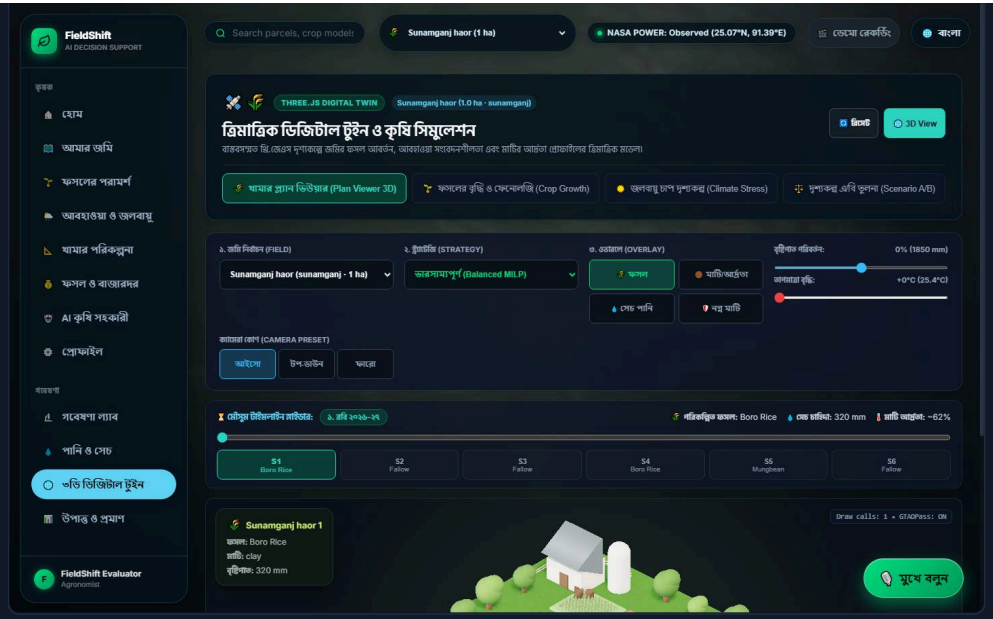

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



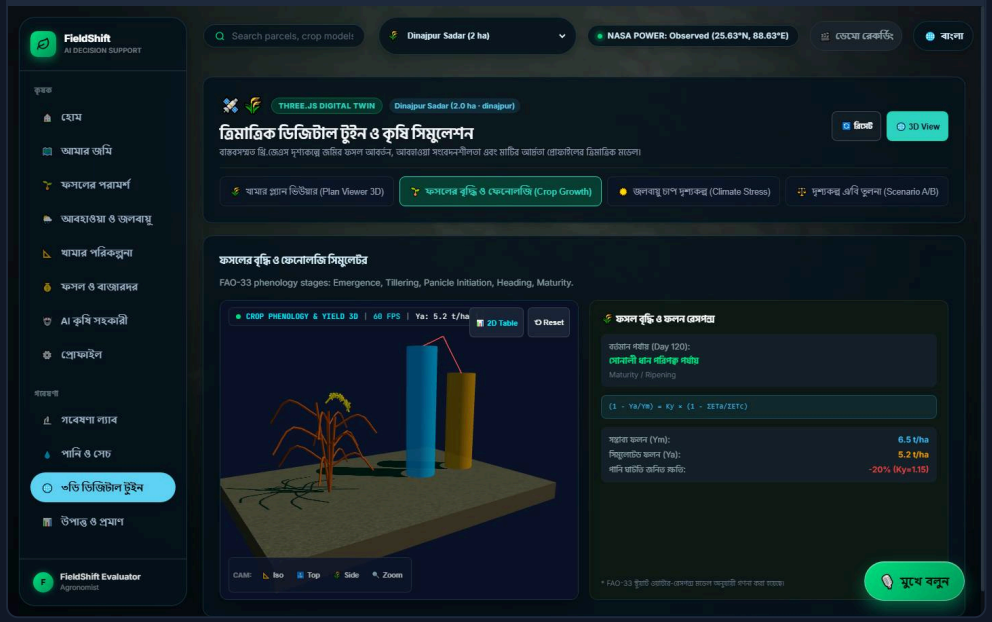

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** S1 \| S2 \| S3 \| S4 \| S5 | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** S1 \| S2 \| S3 \| S4 \| S5 | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** S1 \| S2 \| S3 \| S4 \| S5 | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** S1 \| S2 \| S3 \| S4 \| S5 |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

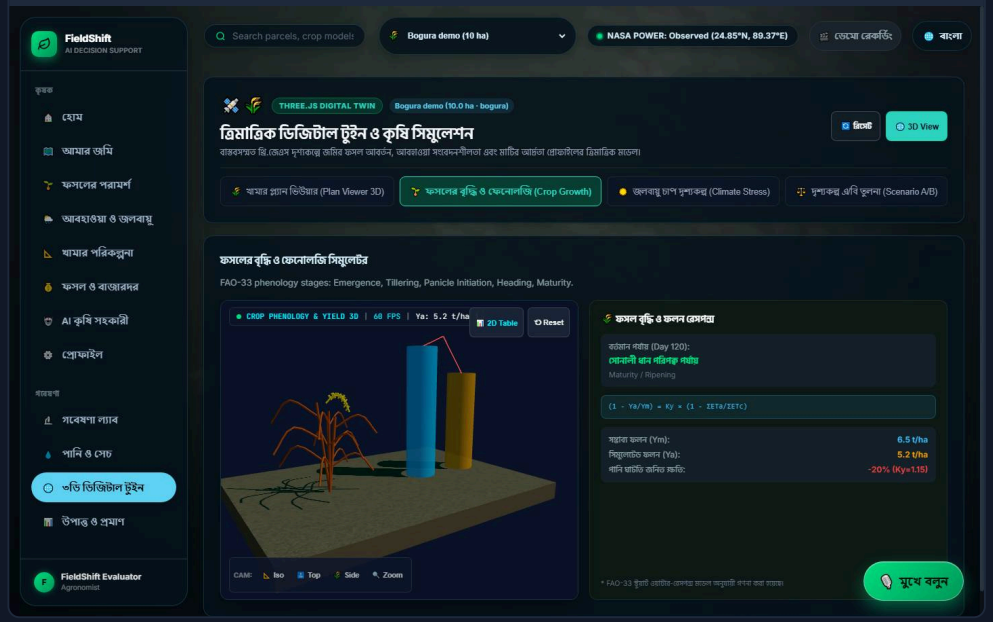
# 33. Digital Twin 3D - Crop Growth Phenology

SAME

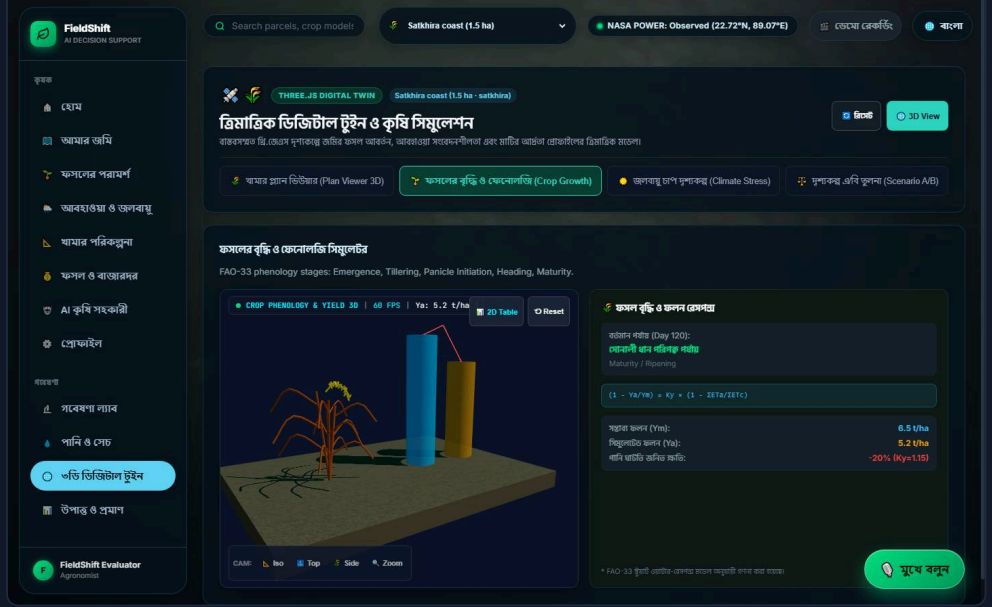
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



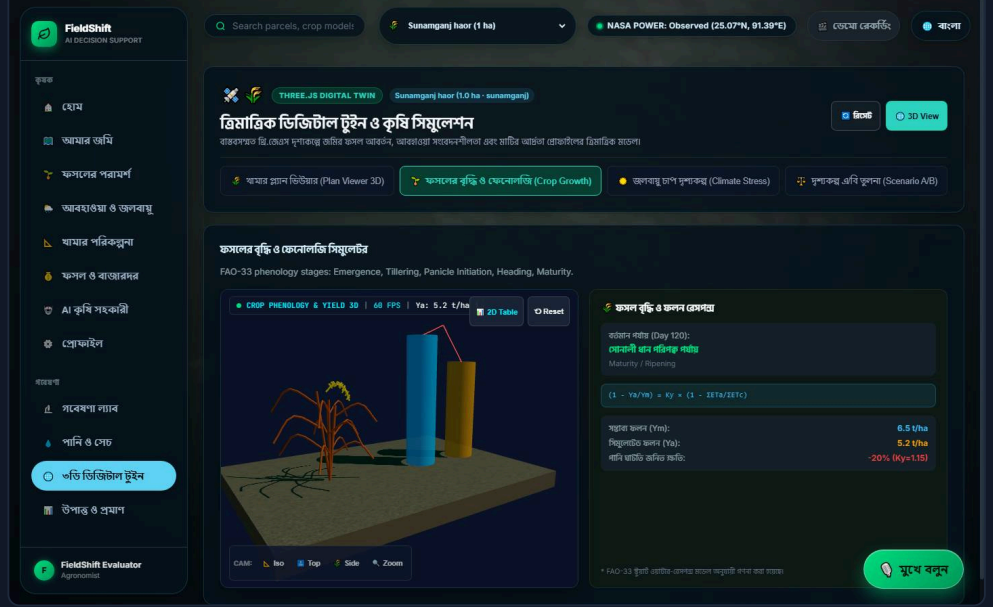

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



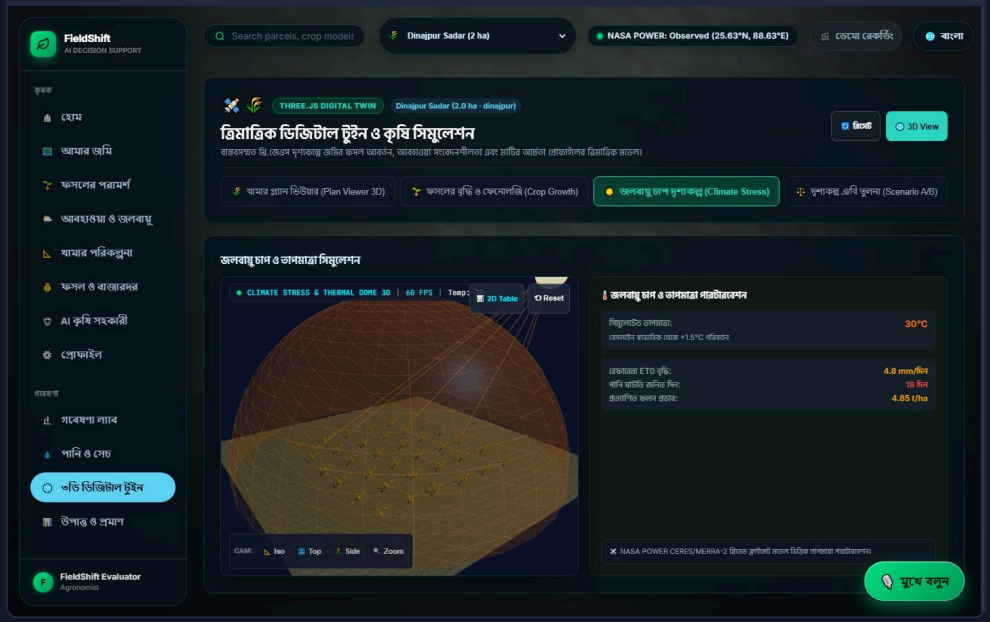
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 6.5 t/ha \| 5.2 t/ha \| -20% (Ky=1.15) | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 6.5 t/ha \| 5.2 t/ha \| -20% (Ky=1.15) | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 6.5 t/ha \| 5.2 t/ha \| -20% (Ky=1.15) | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 6.5 t/ha \| 5.2 t/ha \| -20% (Ky=1.15) |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

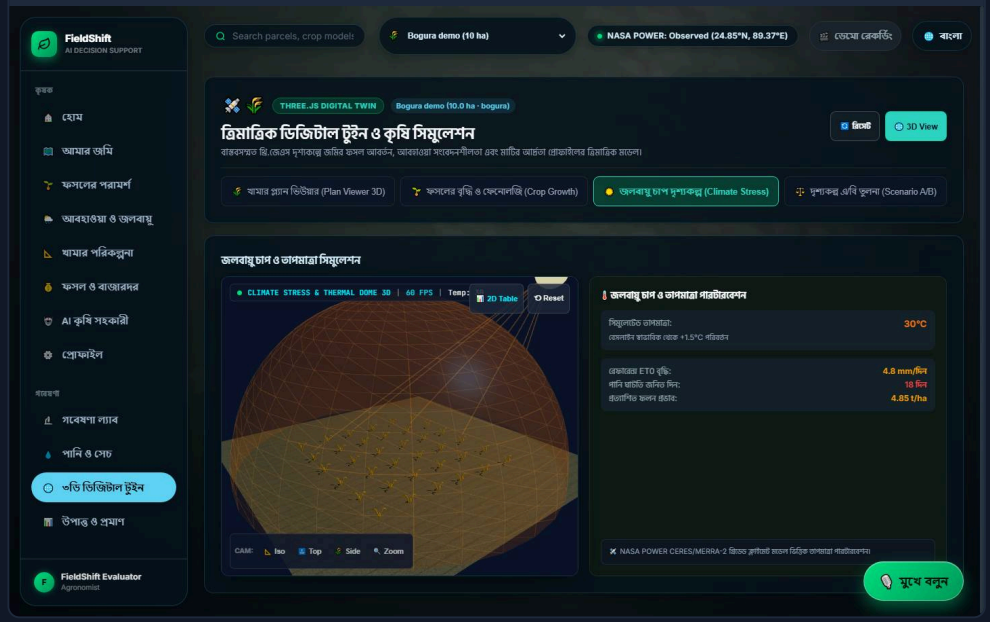
## 34. Digital Twin 3D - Climate Stress Visualization

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

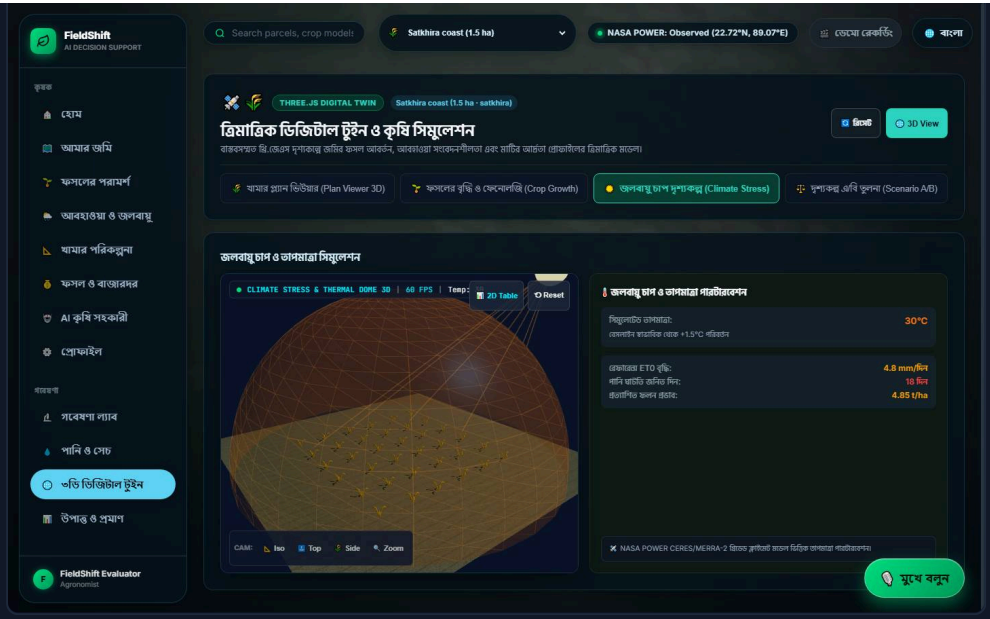


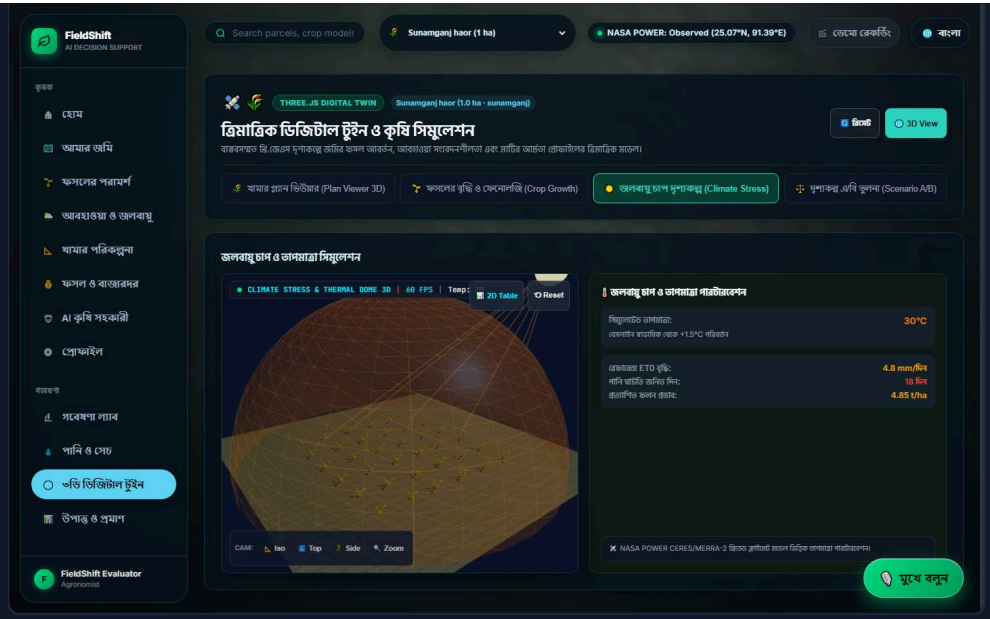

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



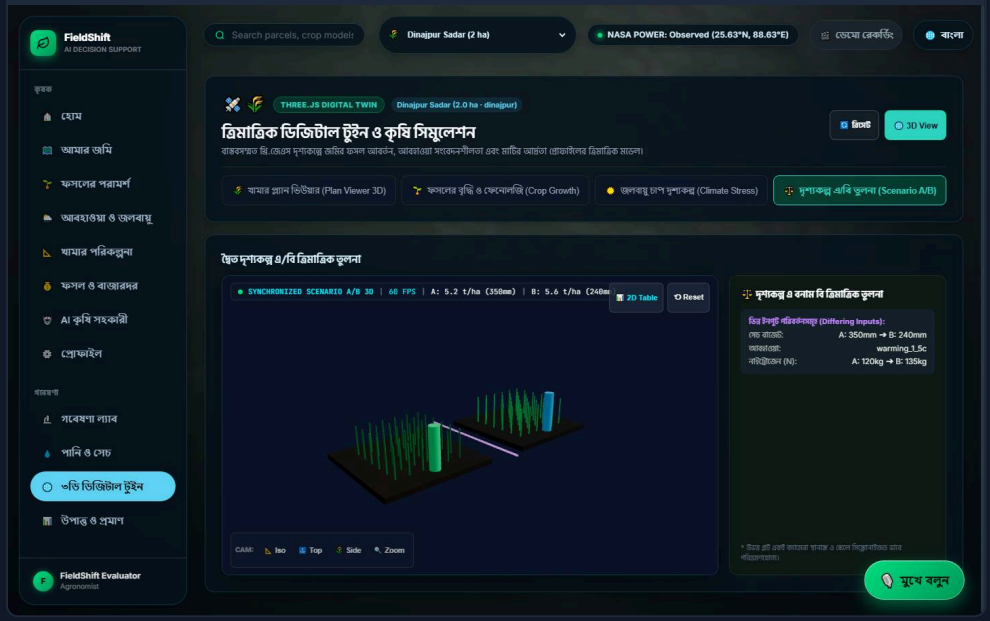

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 30°C \| 4.8 mm/দিন \| 18 দিন \| 4.85 t/ha | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 30°C \| 4.8 mm/দিন \| 18 দিন \| 4.85 t/ha | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 30°C \| 4.8 mm/দিন \| 18 দিন \| 4.85 t/ha | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 30°C \| 4.8 mm/দিন \| 18 দিন \| 4.85 t/ha |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

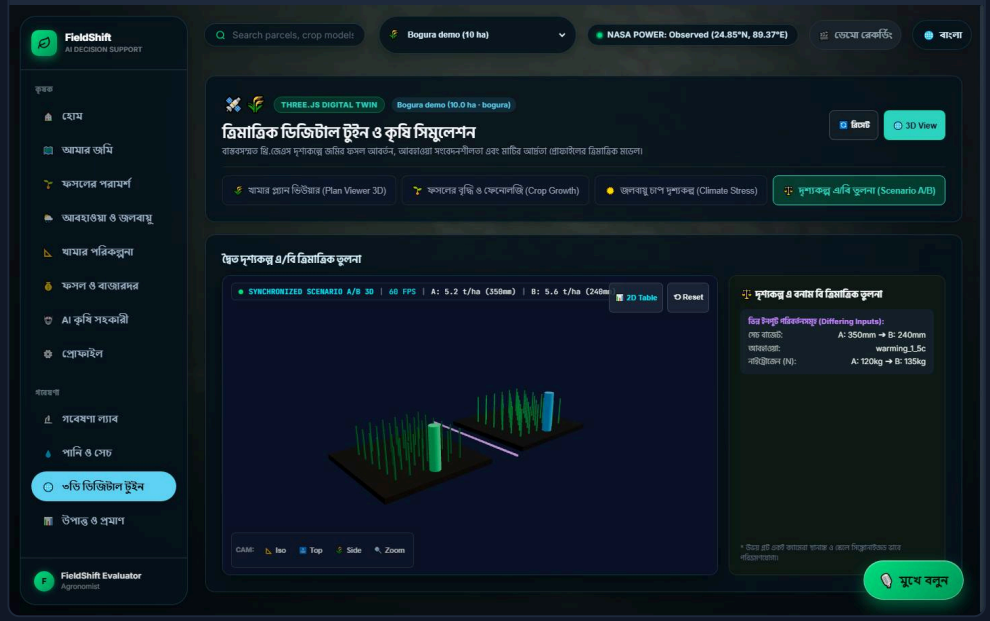
# 35. Digital Twin 3D - Scenario A/B Comparison

SAME

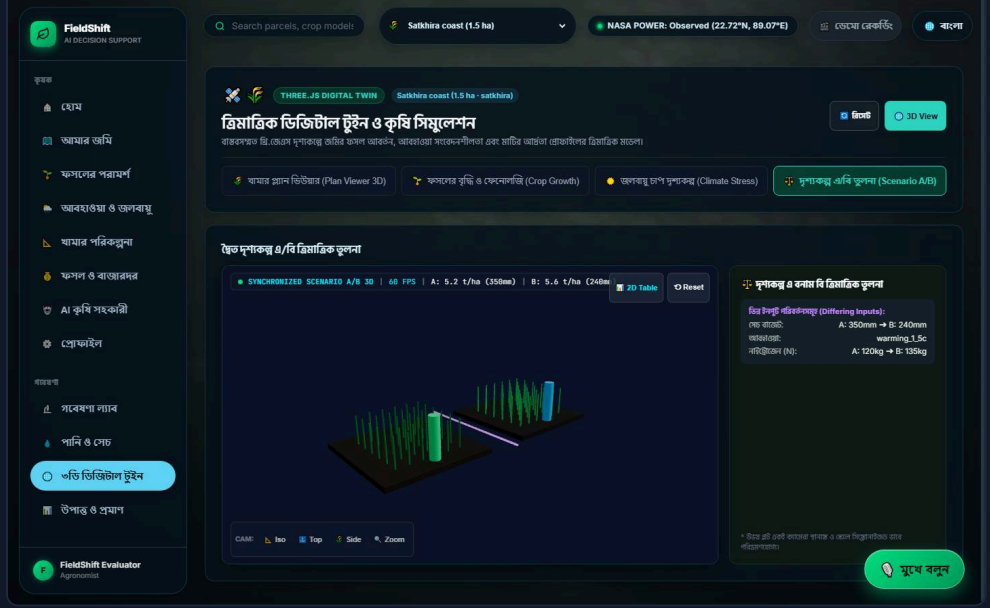
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



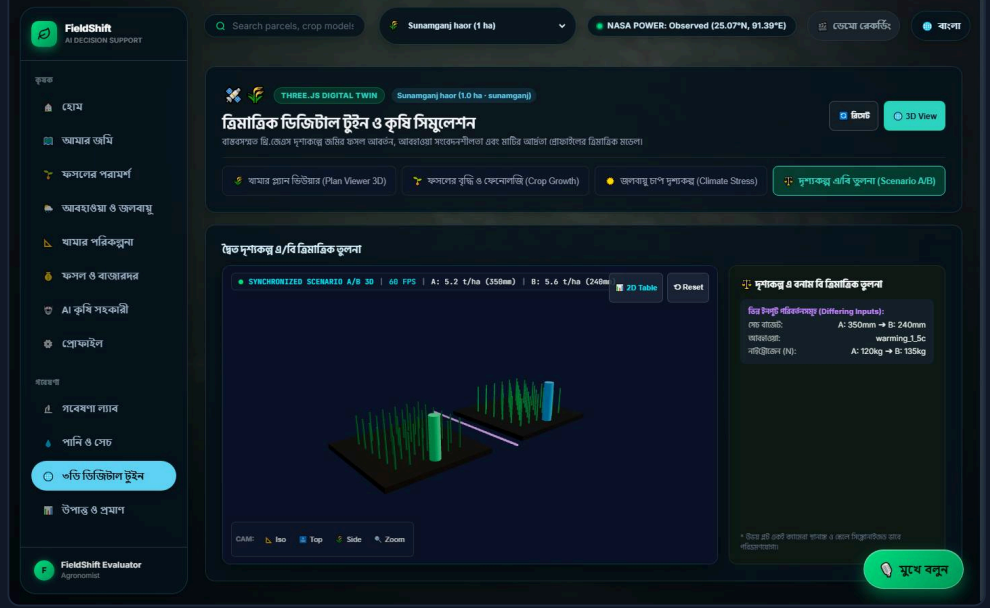

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



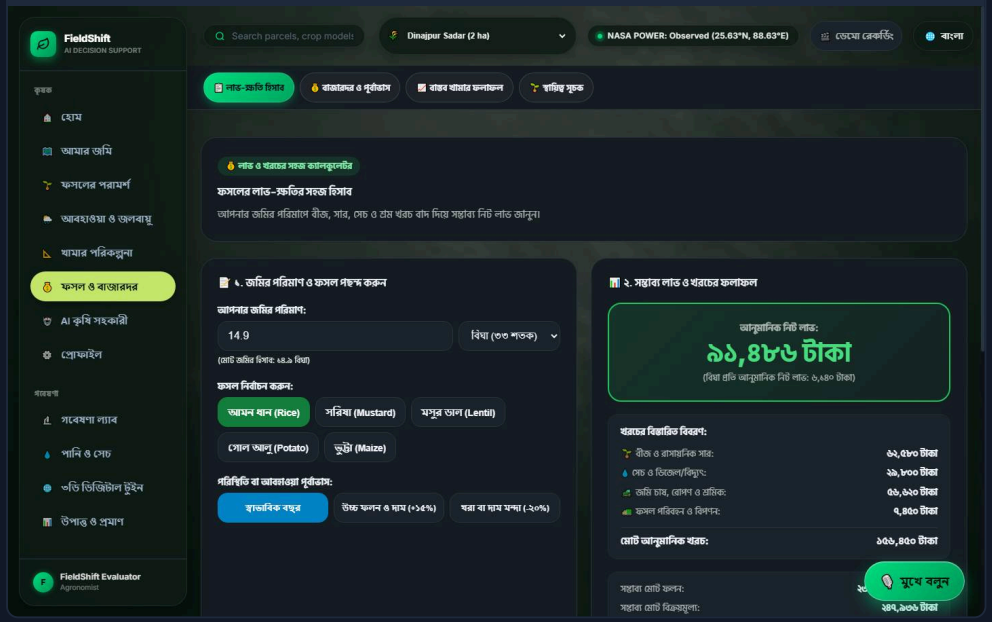
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

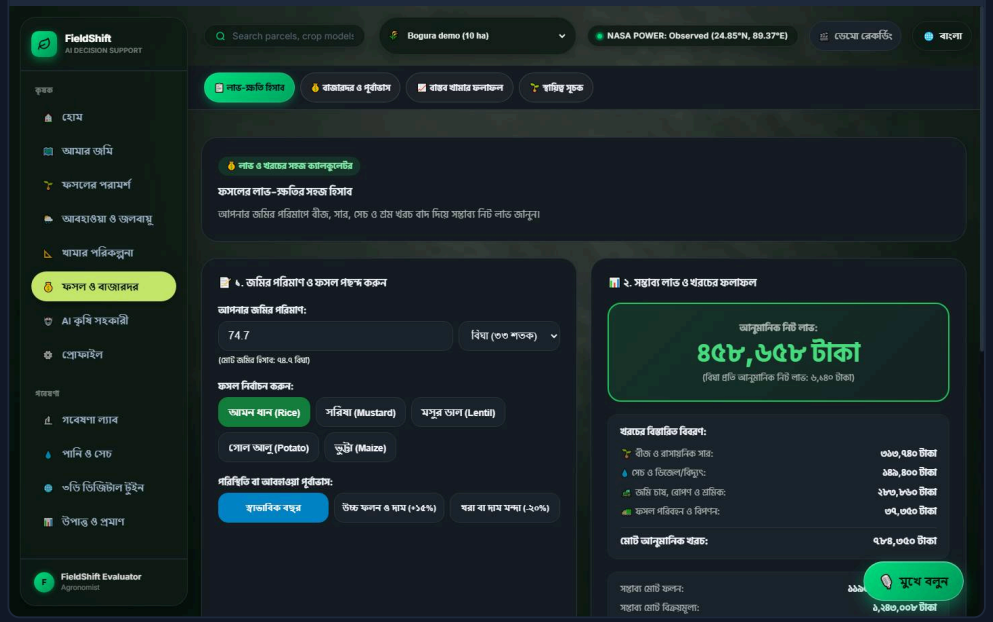
## 36. Market Intelligence & Profitability

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

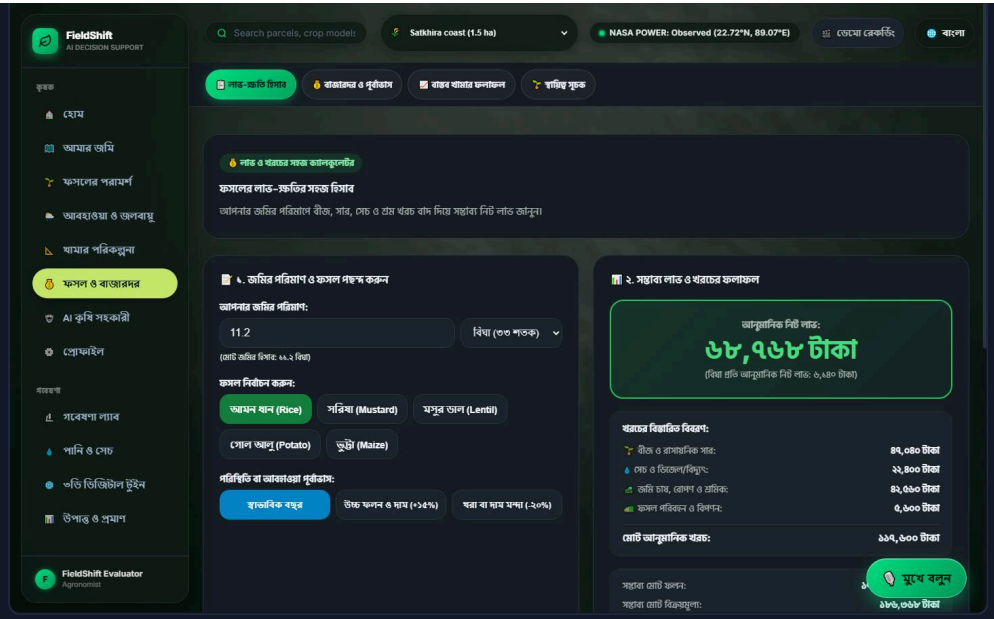


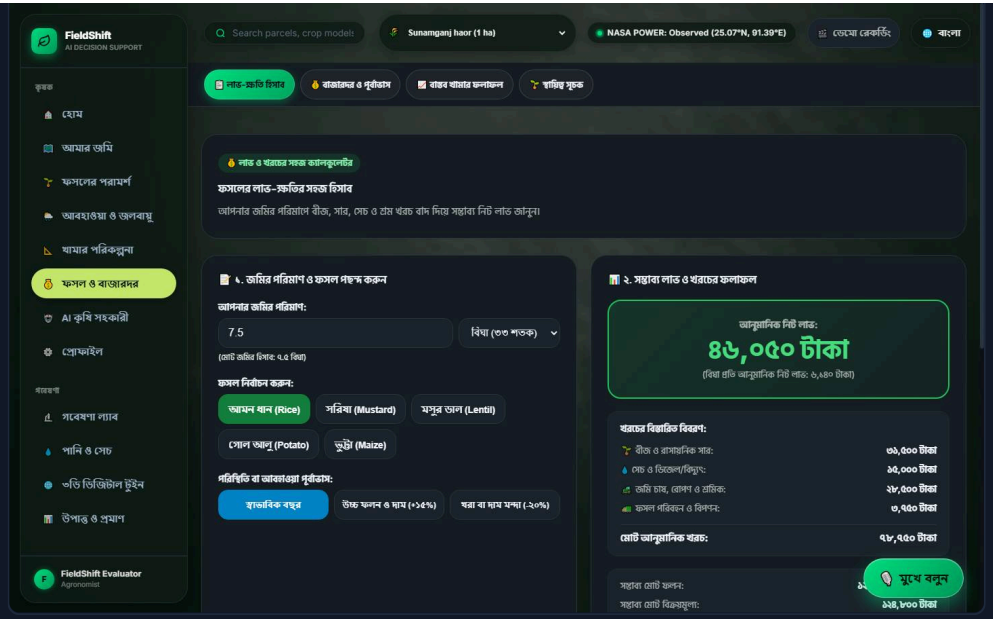

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



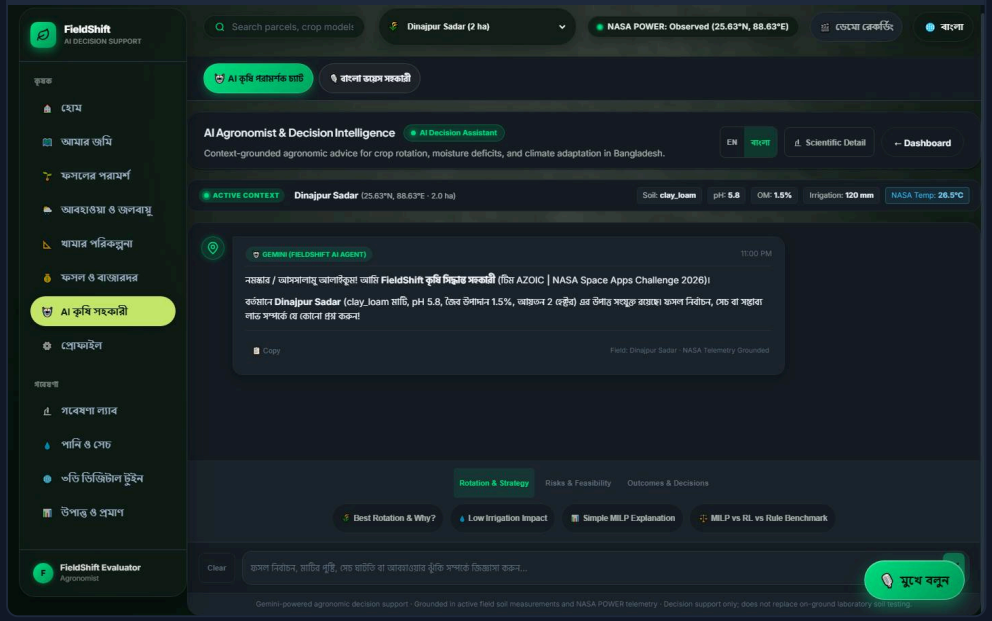

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Uniform values detected across distinct agro-climatic zones; parameter sweep sensitivity may require local ground truth calibration.

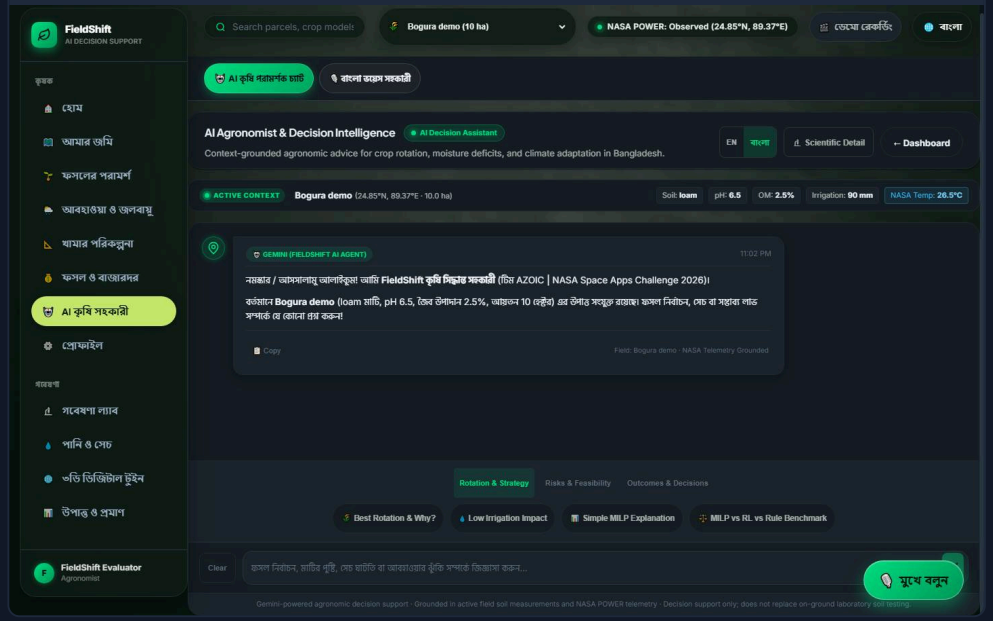
# 37. Agronomic AI Assistant

DIFFERS

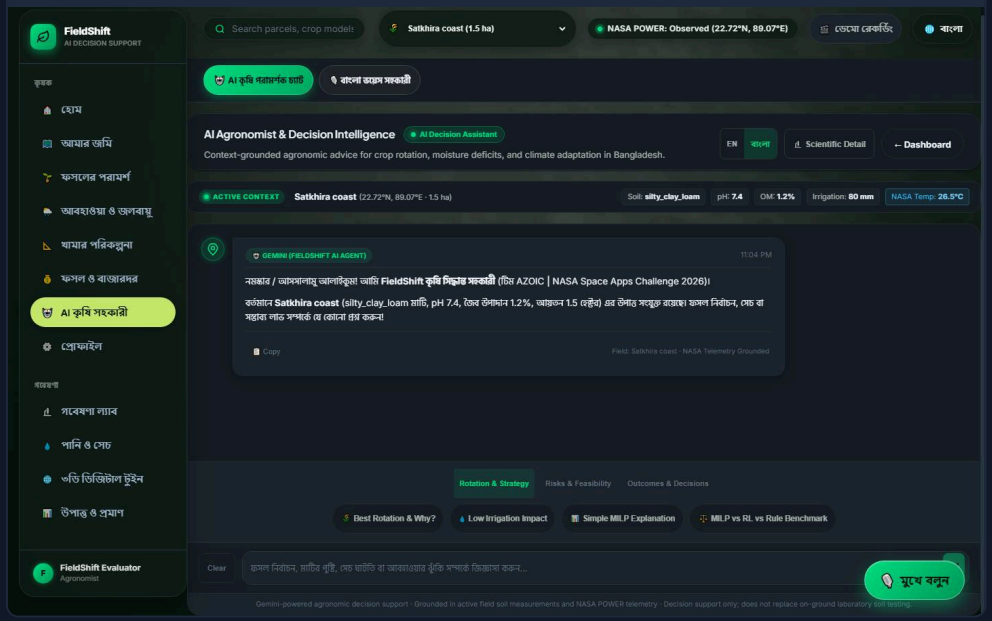
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



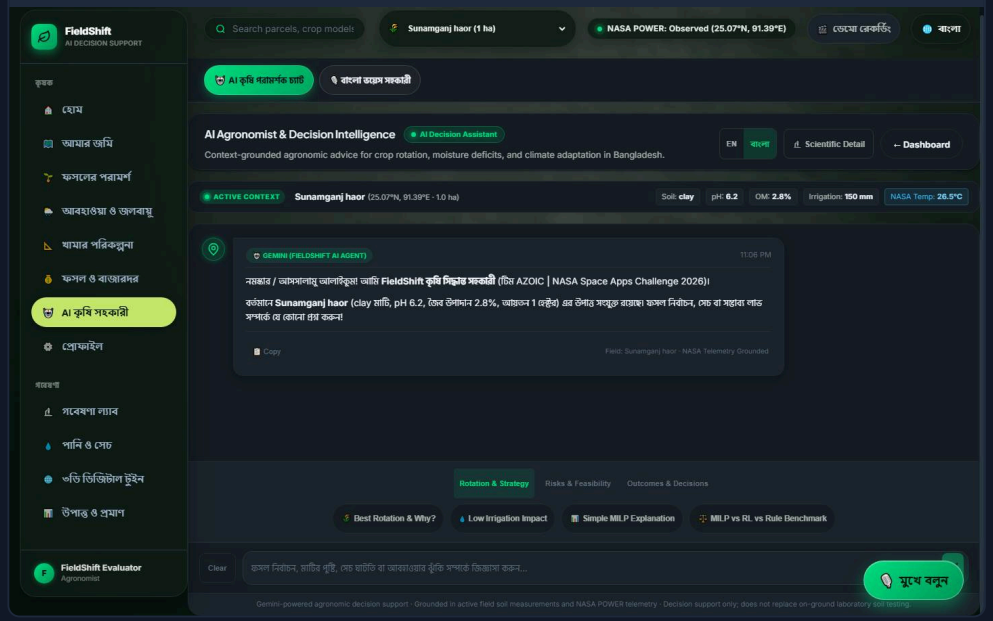

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



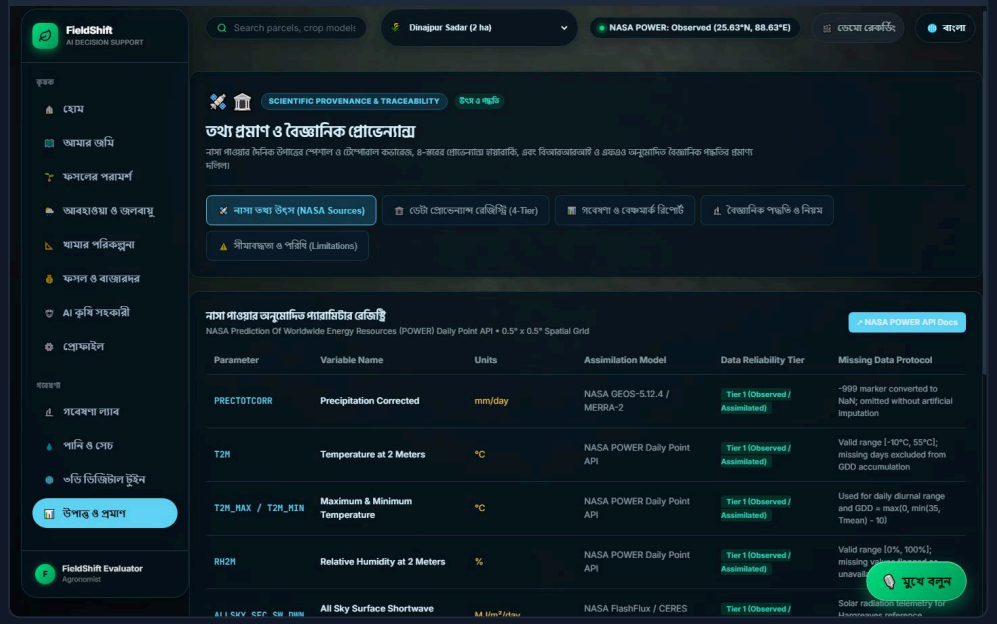
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 5.8 \| 1.5% \| 120 mm \| 26.5°C | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 6.5 \| 2.5% \| 90 mm \| 26.5°C | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 7.4 \| 1.2% \| 80 mm \| 26.5°C | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 6.2 \| 2.8% \| 150 mm \| 26.5°C |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

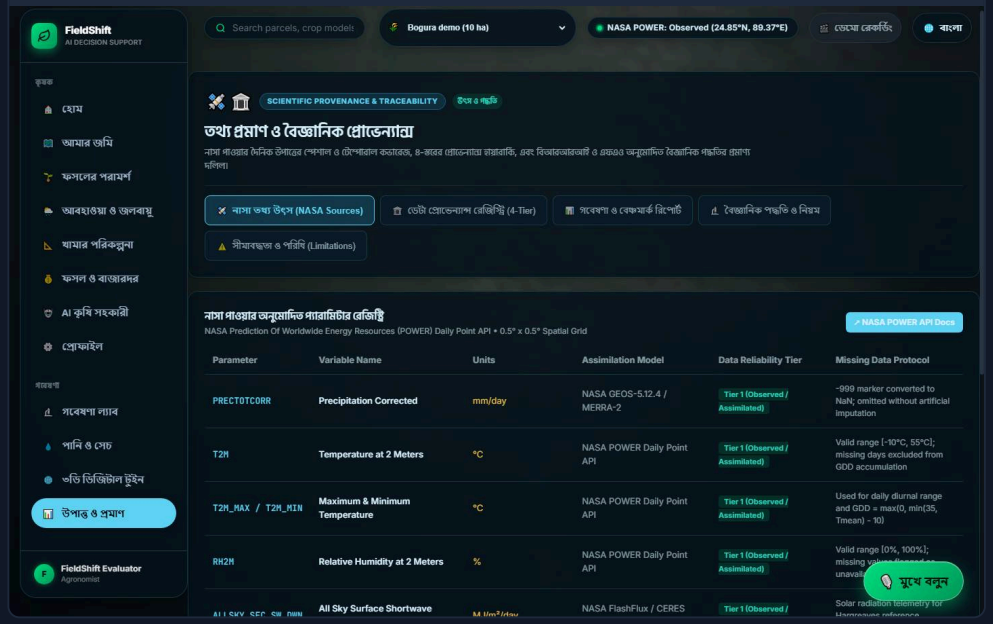
## 38. Data Evidence - NASA Satellite Feeds

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

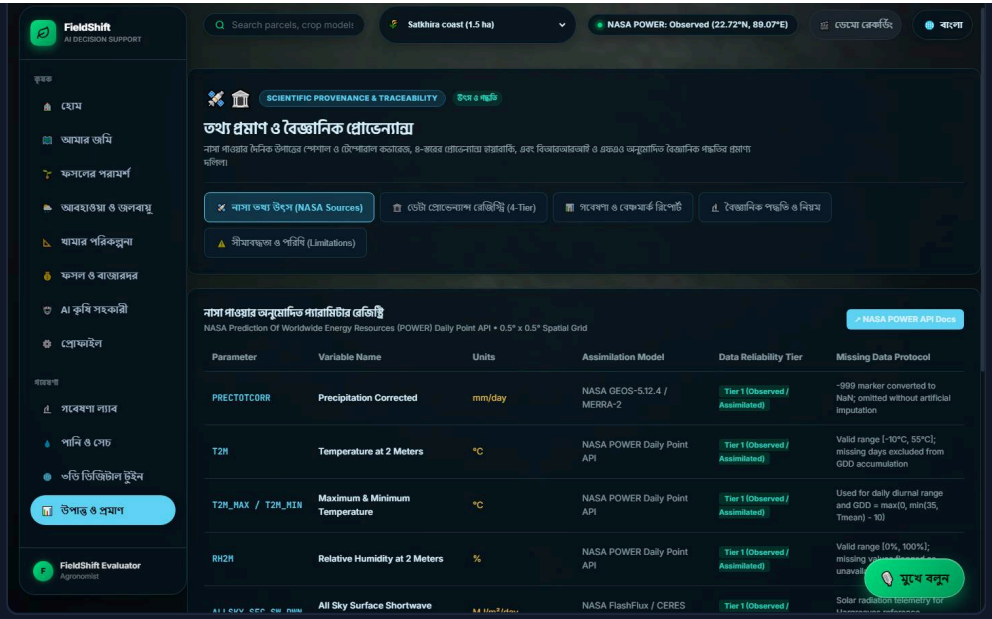


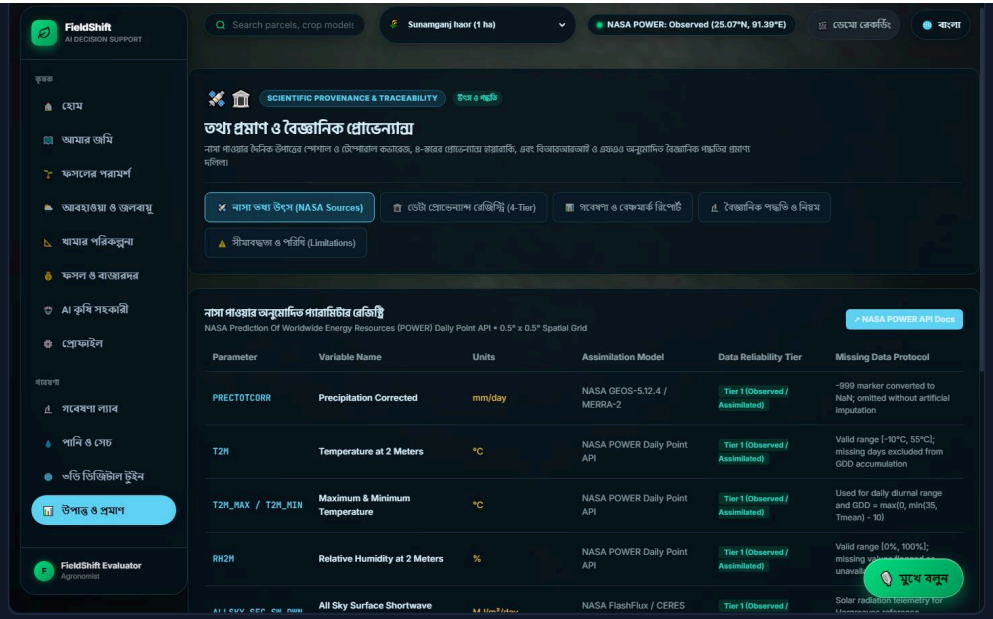

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



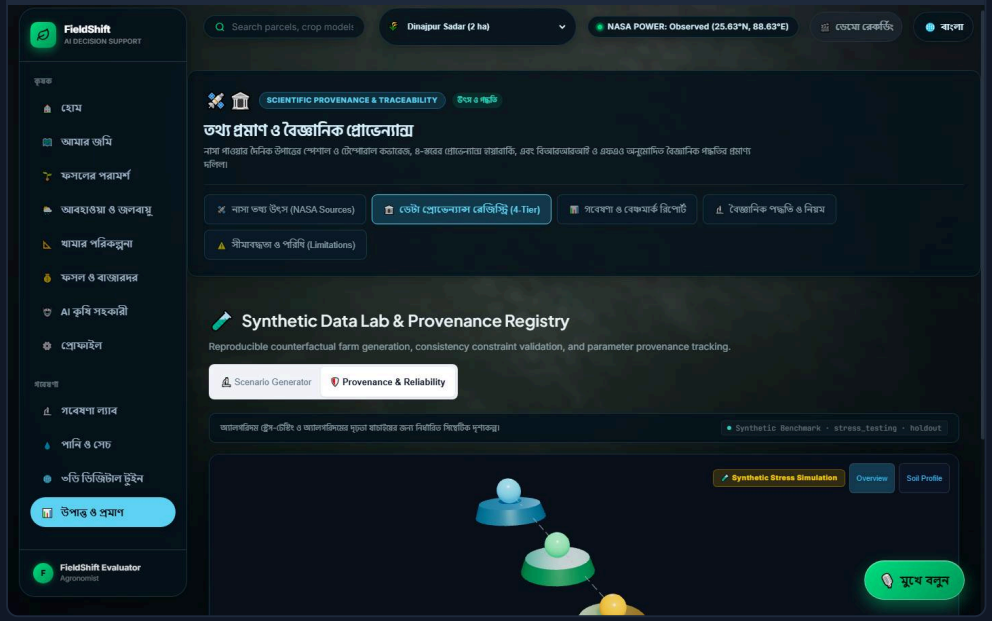

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** NASA GEOS-5.12.4 / MERRA-2 \| Tier 1 (Observed / Assimilated) \| -999 marker converted to NaN; omitted without artificial imputation \| T2M \| Temperature at 2 Meters | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** NASA GEOS-5.12.4 / MERRA-2 \| Tier 1 (Observed / Assimilated) \| -999 marker converted to NaN; omitted without artificial imputation \| T2M \| Temperature at 2 Meters | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** NASA GEOS-5.12.4 / MERRA-2 \| Tier 1 (Observed / Assimilated) \| -999 marker converted to NaN; omitted without artificial imputation \| T2M \| Temperature at 2 Meters | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** NASA GEOS-5.12.4 / MERRA-2 \| Tier 1 (Observed / Assimilated) \| -999 marker converted to NaN; omitted without artificial imputation \| T2M \| Temperature at 2 Meters |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

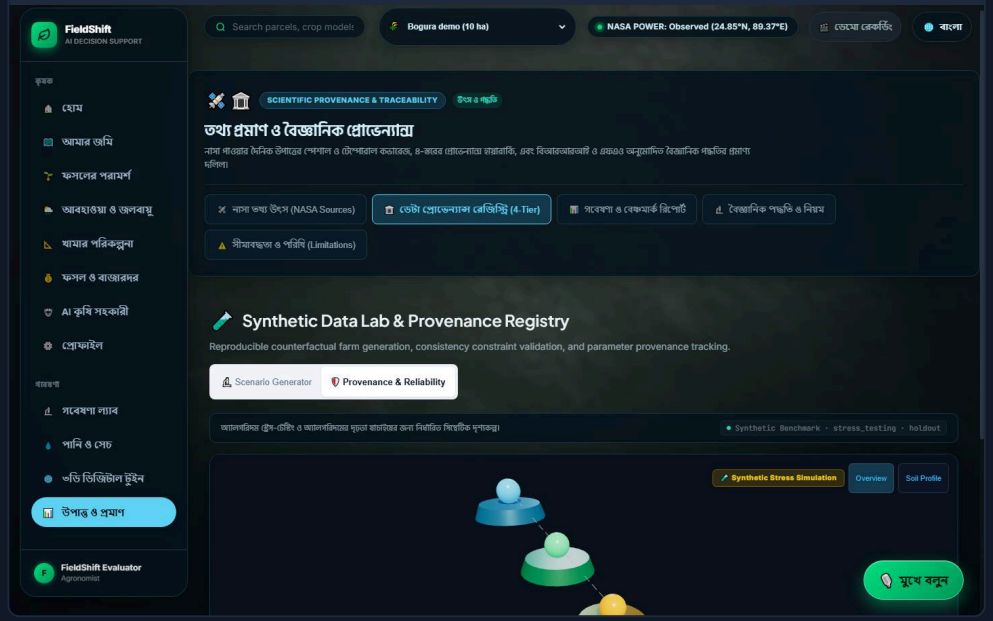
# 39. Data Evidence - 4-Tier Provenance Registry

DIFFERS

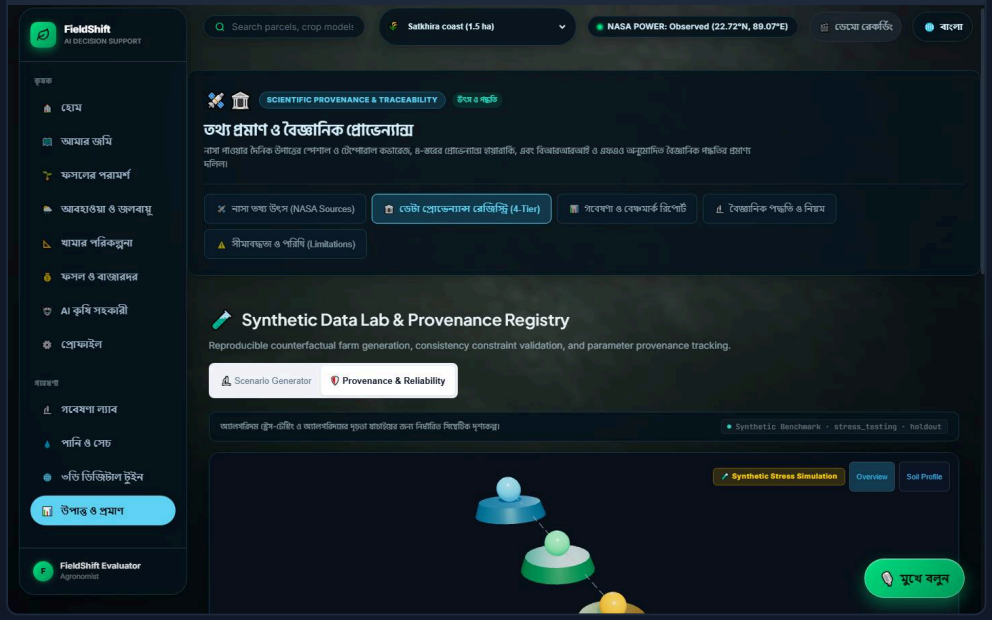
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



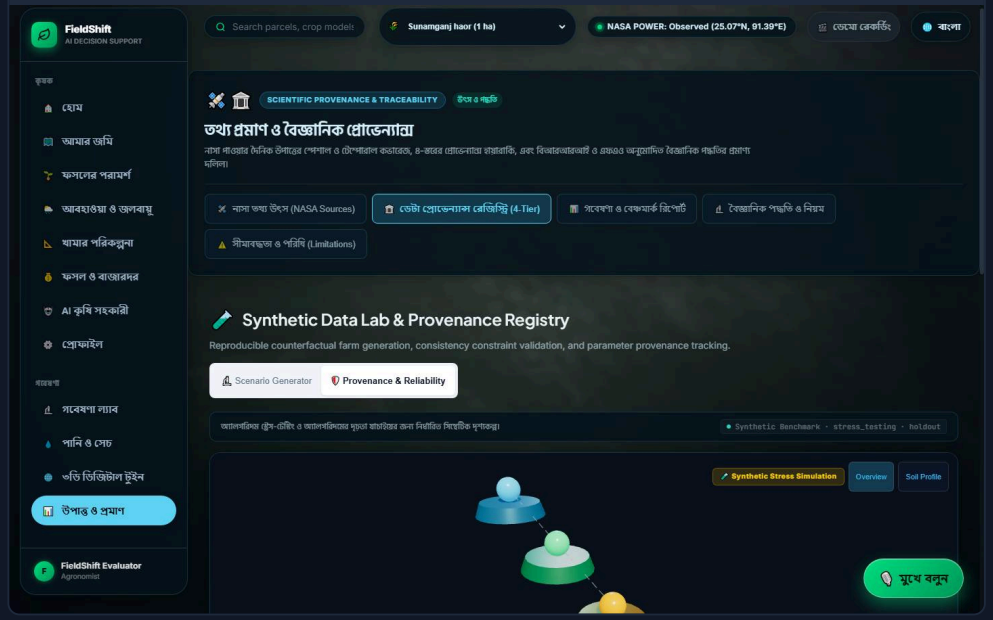

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



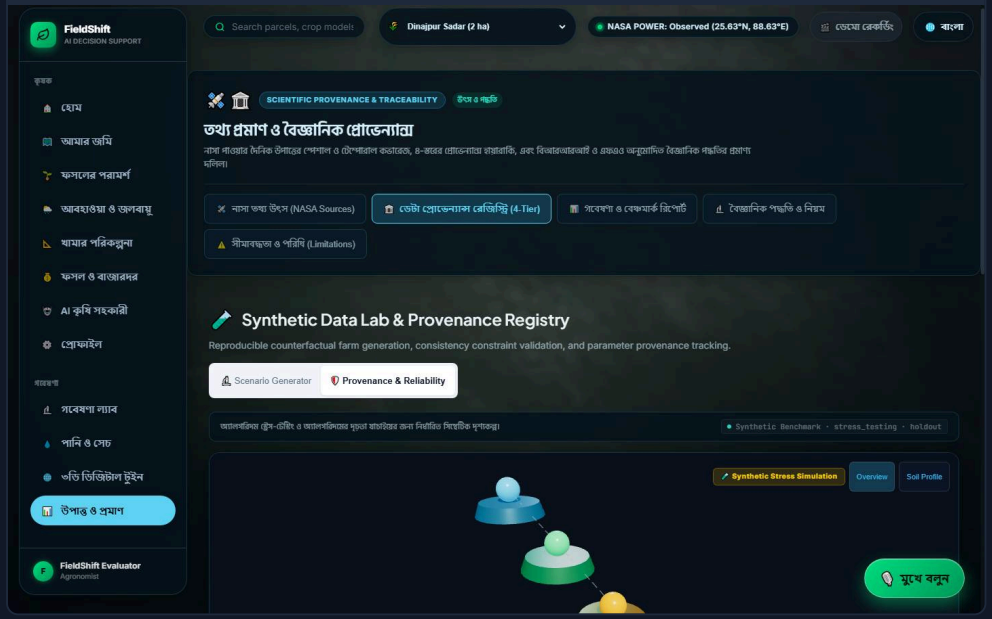
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** surface_temperature_t2m \| 25.4 °C \| NASA POWER Agroclimatology Daily Point API v2.0 ↗ \| ±1.2°C mean bias against Bangladesh Meteorological Department ground stations. \| 12.8 mm/day | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** surface_temperature_t2m \| 25.4 °C \| NASA POWER Agroclimatology Daily Point API v2.0 ↗ \| ±1.2°C mean bias against Bangladesh Meteorological Department ground stations. \| 12.8 mm/day |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

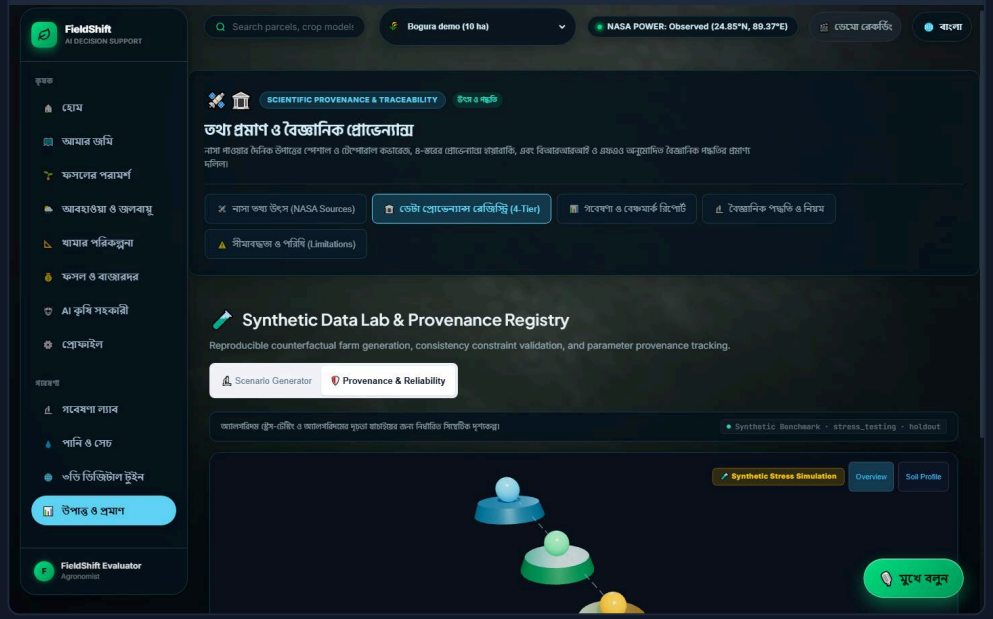
## 40. Data Evidence - IMERG & Climatology Reports

DIFFERS

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

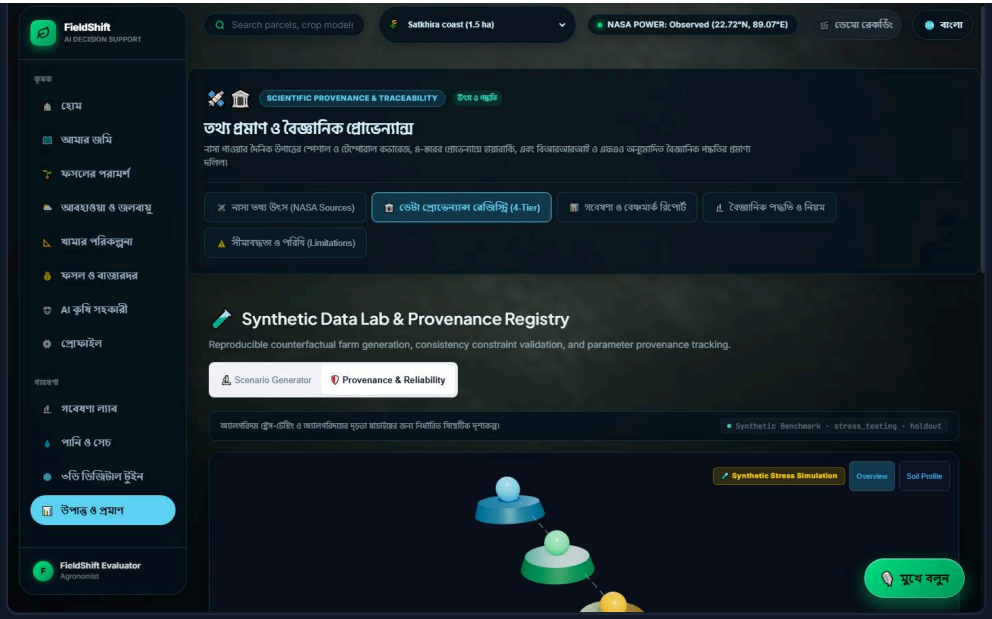


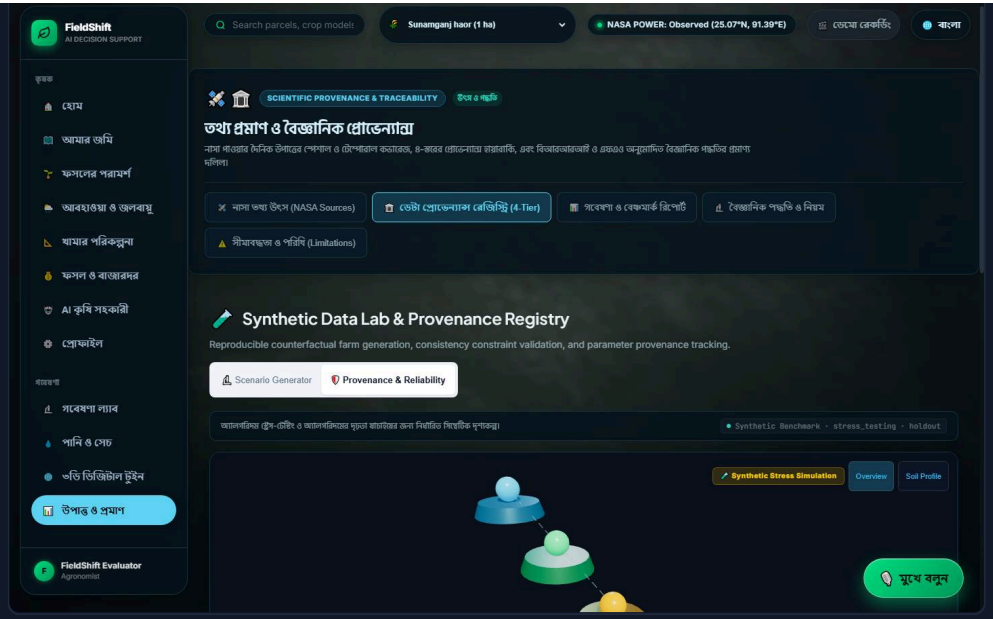

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



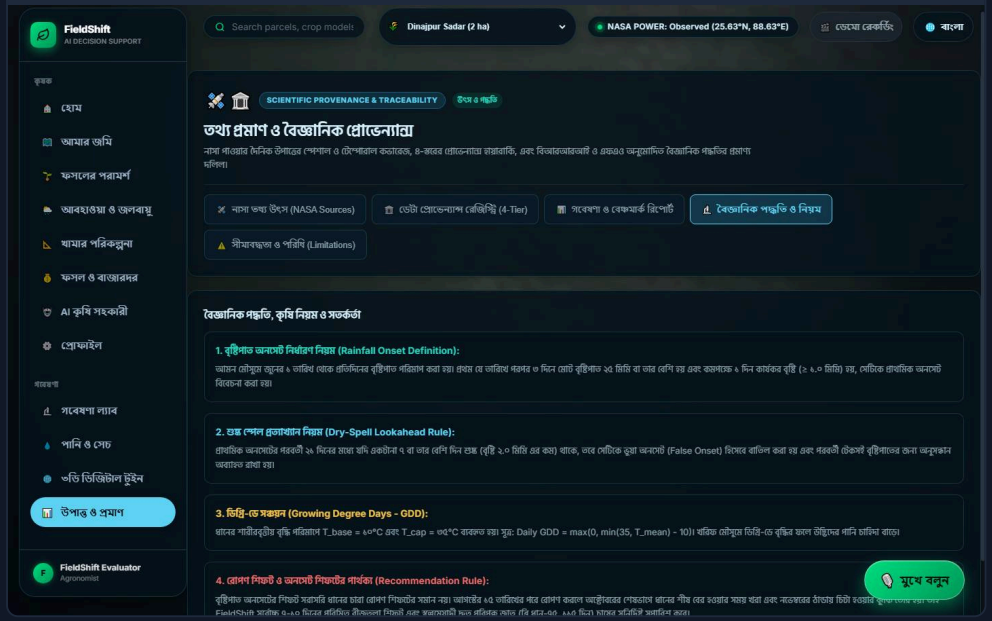

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** surface_temperature_t2m \| 25.4 °C \| NASA POWER Agroclimatology Daily Point API v2.0 ↗ \| ±1.2°C mean bias against Bangladesh Meteorological Department ground stations. \| 12.8 mm/day | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** surface_temperature_t2m \| 25.4 °C \| NASA POWER Agroclimatology Daily Point API v2.0 ↗ \| ±1.2°C mean bias against Bangladesh Meteorological Department ground stations. \| 12.8 mm/day | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** surface_temperature_t2m \| 25.4 °C \| NASA POWER Agroclimatology Daily Point API v2.0 ↗ \| ±1.2°C mean bias against Bangladesh Meteorological Department ground stations. \| 12.8 mm/day |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.

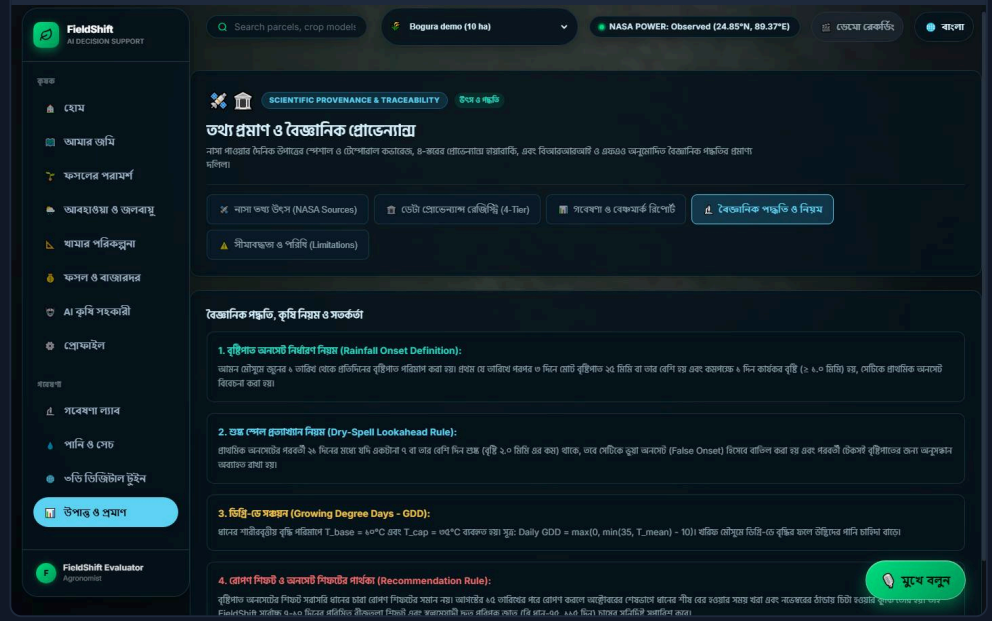
# 41. Data Evidence - Scientific Methodology

SAME

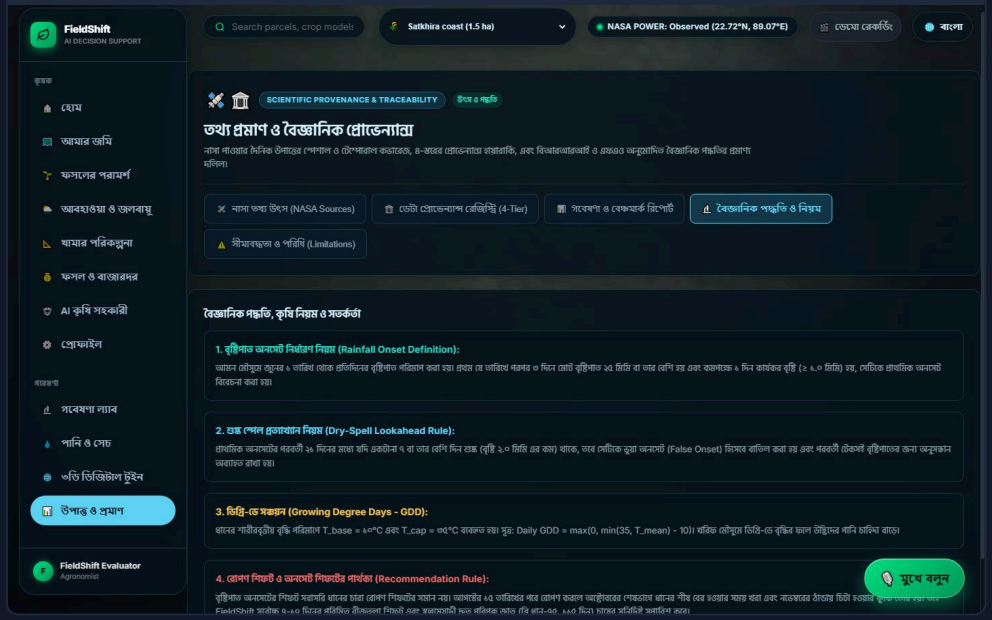
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



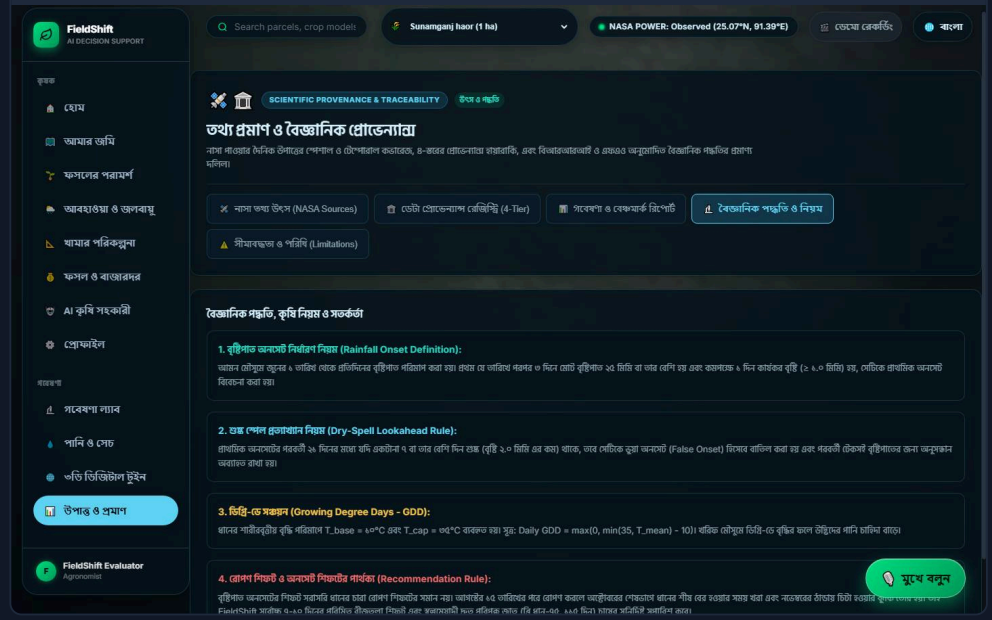

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



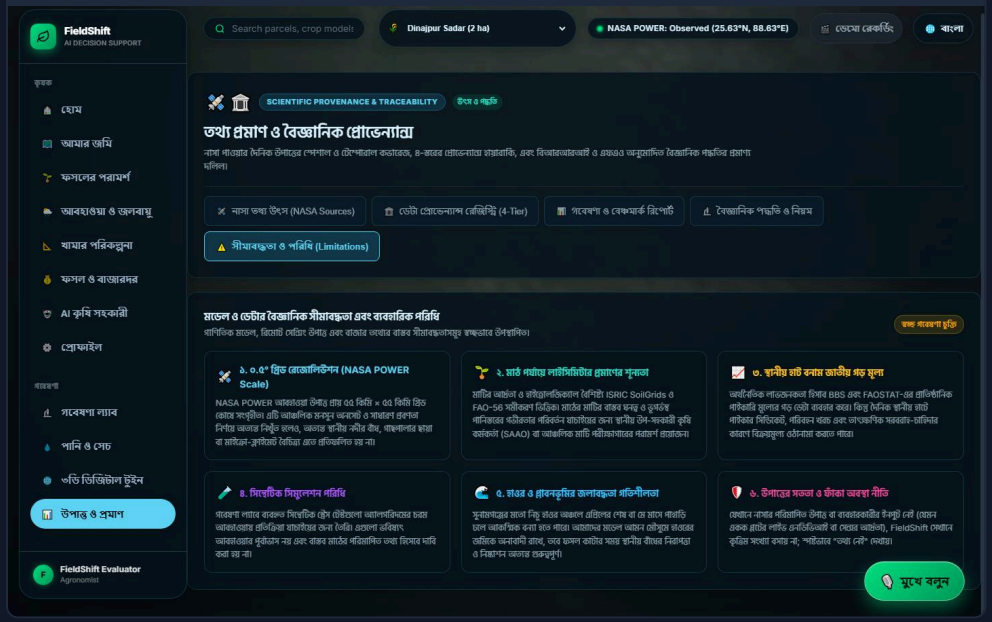
| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition): \| 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule): \| 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): \| 4. রোপণ শিফট ও অনসেট শিফটের পার্থক্য (Recommendation Rule): | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition): \| 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule): \| 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): \| 4. রোপণ শিফট ও অনসেট শিফটের পার্থক্য (Recommendation Rule): | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition): \| 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule): \| 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): \| 4. রোপণ শিফট ও অনসেট শিফটের পার্থক্য (Recommendation Rule): | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 1. বৃষ্টিপাত অনসেট নির্ধারণ নিয়ম (Rainfall Onset Definition): \| 2. শুষ্ক স্পেল প্রত্যাখ্যান নিয়ম (Dry-Spell Lookahead Rule): \| 3. ডিগ্রি-ডে সঞ্চয়ন (Growing Degree Days - GDD): \| 4. রোপণ শিফট ও অনসেট শিফটের পার্থক্য (Recommendation Rule): |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

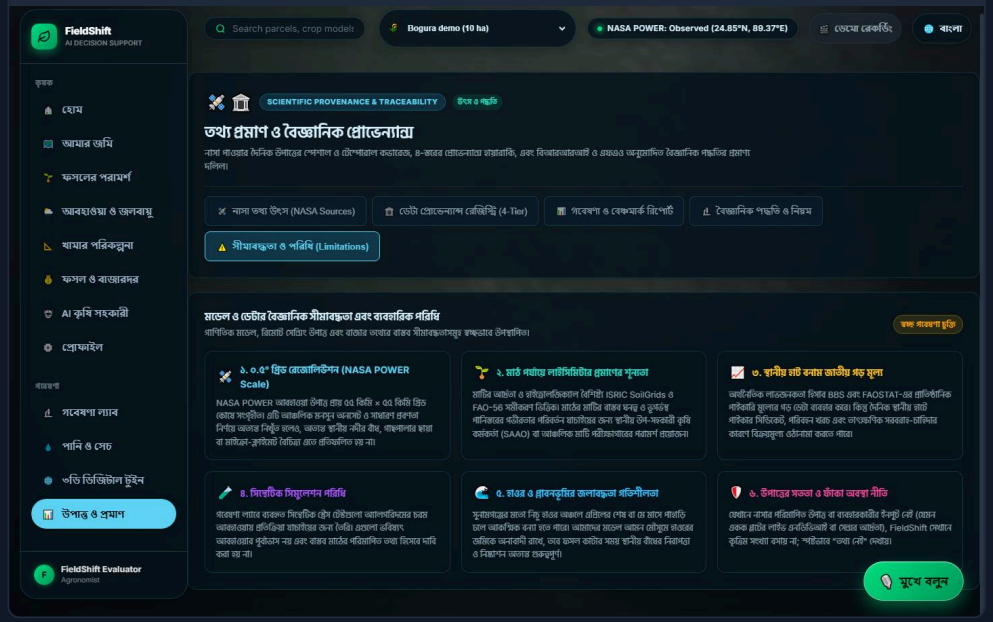
## 42. Data Evidence - Boundaries & Disclaimers

SAME

### 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)

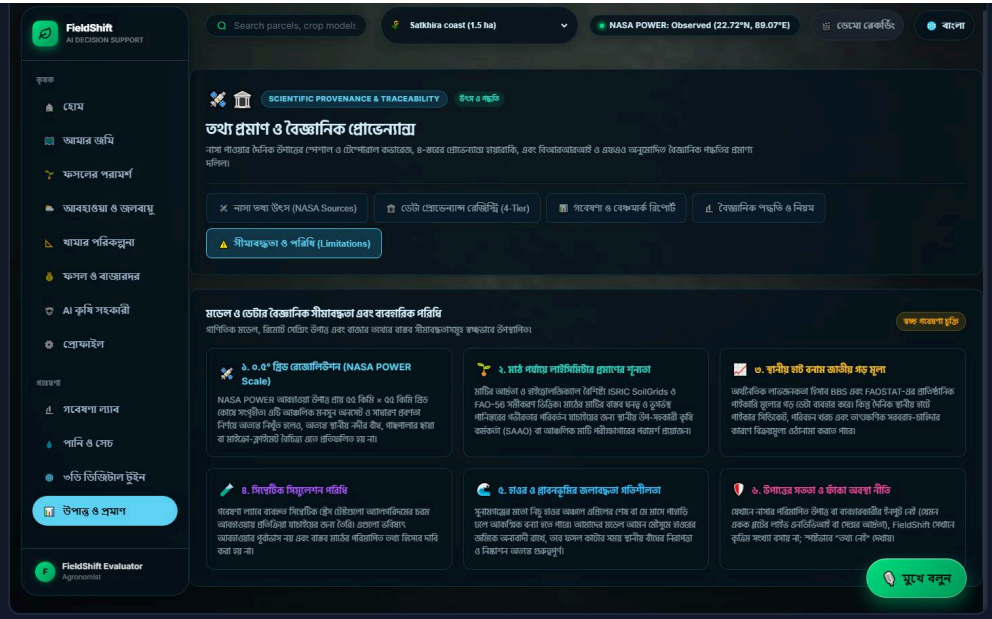


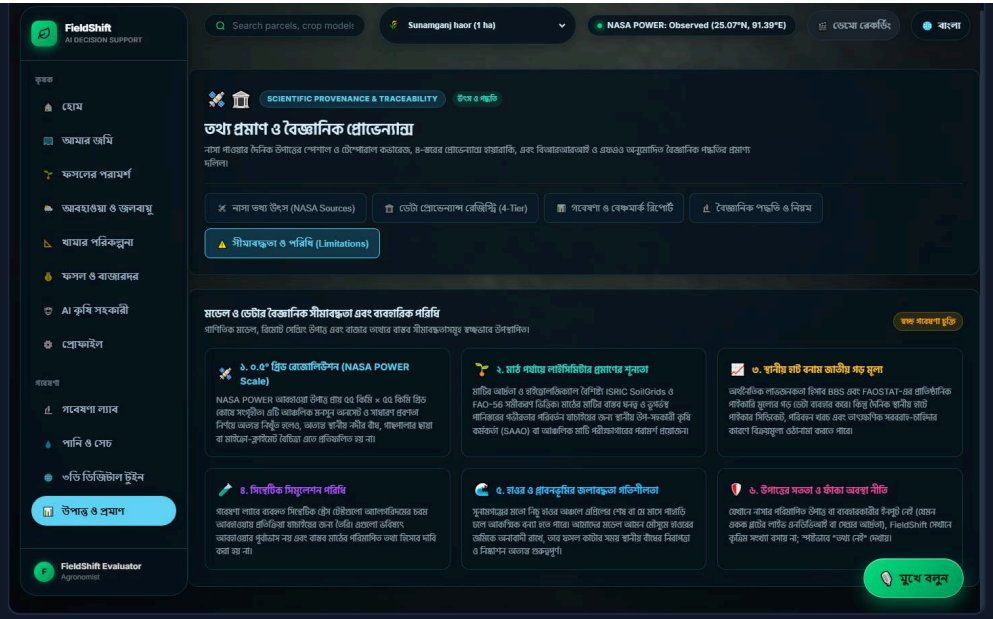

### 2. Bogura Demo (Floodplain | 10.0 ha)




### 3. Satkhira Coast (Coastal Saline | 1.5 ha)

### 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



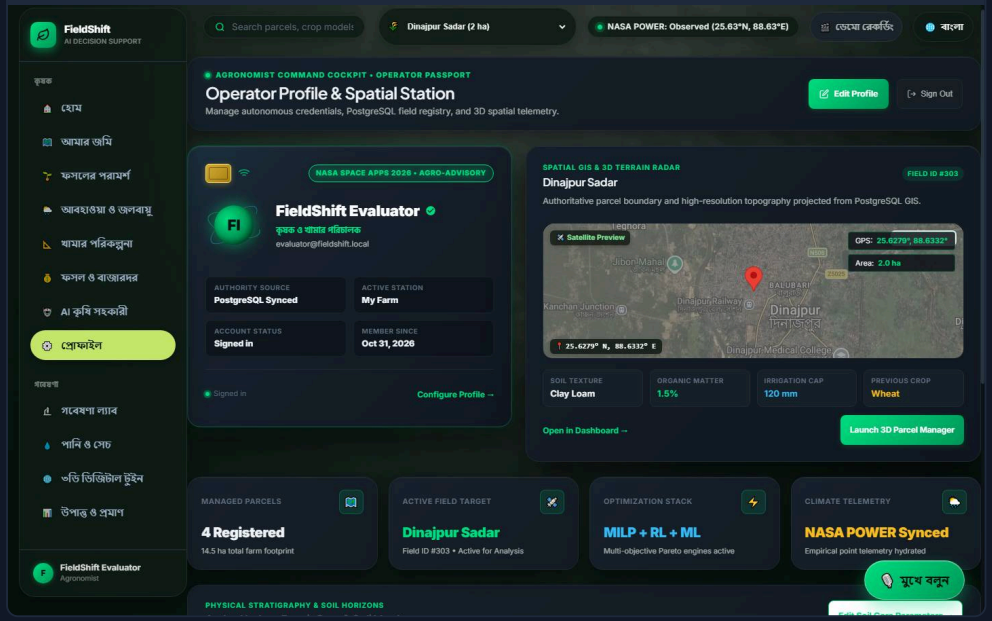

| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** N/A | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** N/A | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** N/A | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** N/A |

**Divergence Analysis:** Platform-wide shared dataset / scientific methodology reference. Uniformity is scientifically intended.

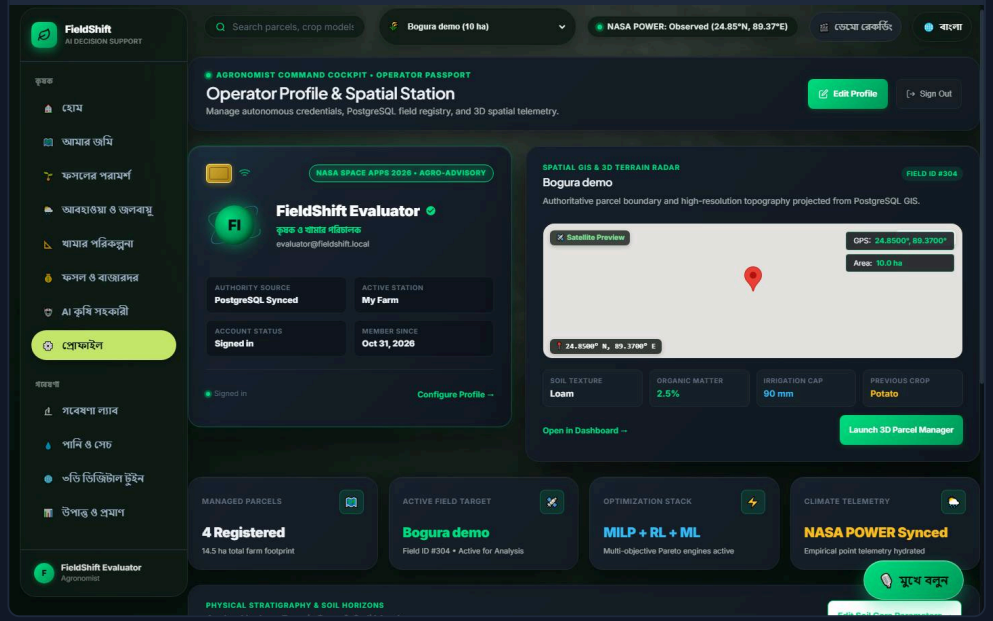
# 43. Operator Cockpit & Account Settings

DIFFERS

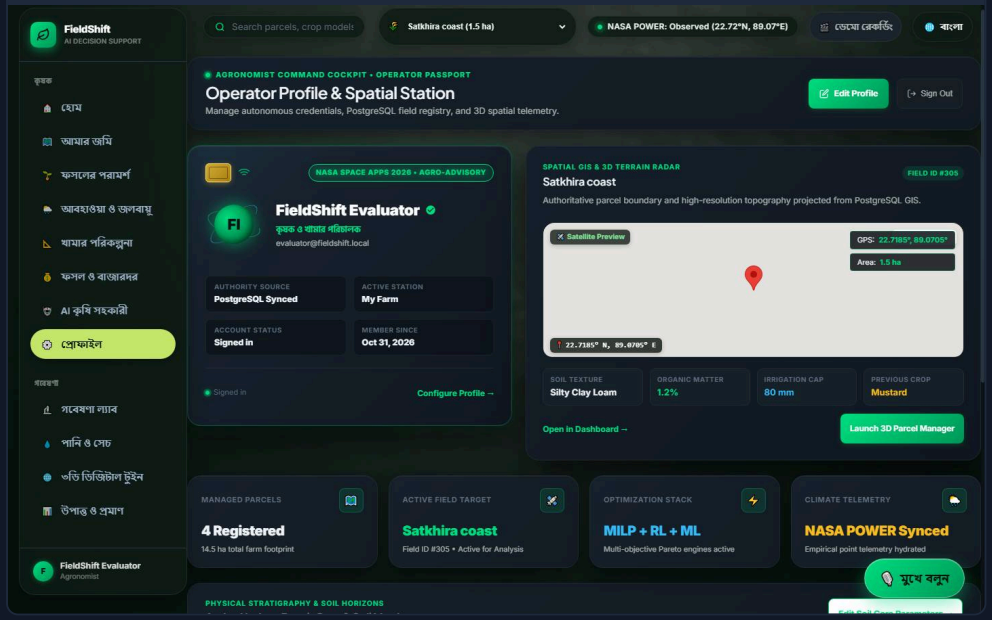
## 1. Dinajpur Sadar (Northwest Grain Belt | 2.0 ha)



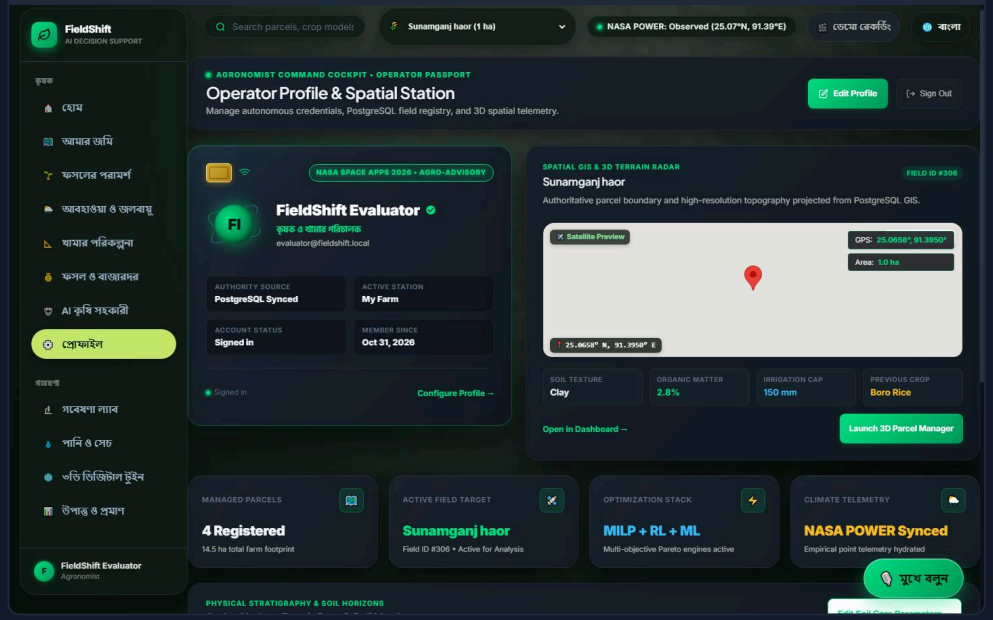

## 2. Bogura Demo (Floodplain | 10.0 ha)




## 3. Satkhira Coast (Coastal Saline | 1.5 ha)




## 4. Sunamganj Haor (Wetland Basin | 1.0 ha)



| Dinajpur Values | Bogura Values | Satkhira Values | Sunamganj Values |
|---|---|---|---|
| **Coord:** 25.63°N, 88.63°E<br>**Metrics:** 25.6279°, 88.6332° \| 2.0 ha \| 1.5% \| 120 mm \| Organic Matter: 2.4% (ISRIC SoilGrids) | **Coord:** 24.85°N, 89.37°E<br>**Metrics:** 24.8500°, 89.3700° \| 10.0 ha \| 2.5% \| 90 mm \| Organic Matter: 2.4% (ISRIC SoilGrids) | **Coord:** 22.72°N, 89.07°E<br>**Metrics:** 22.7185°, 89.0705° \| 1.5 ha \| 1.2% \| 80 mm \| Organic Matter: 2.4% (ISRIC SoilGrids) | **Coord:** 25.07°N, 91.40°E<br>**Metrics:** 25.0658°, 91.3950° \| 1.0 ha \| 2.8% \| 150 mm \| Organic Matter: 2.4% (ISRIC SoilGrids) |

**Divergence Analysis:** Regional divergence confirmed. Parameters dynamically respond to distinct local soil hydrology, NASA POWER climate cell metrics, and parcel size.